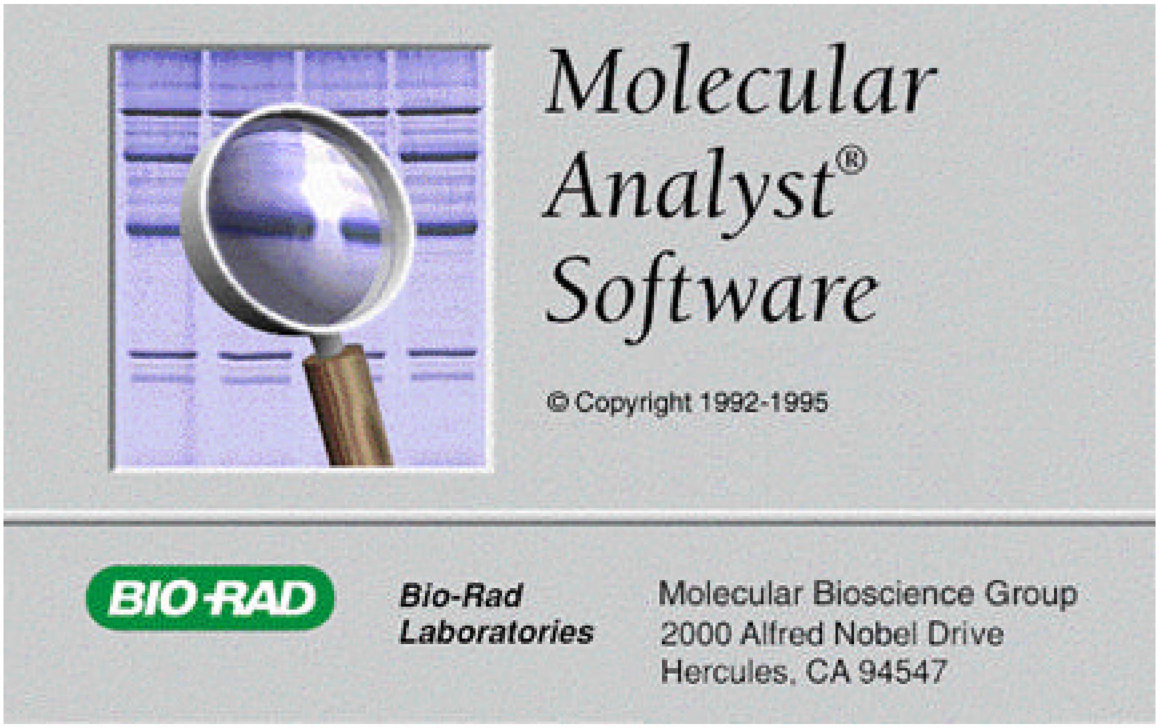


Molecular Analyst™/PC

Windows Software for Bio-Rad's Image Analysis Systems

Version 1.4

Instruction Manual

For Technical Service
Call Your Local Bio-Rad Office or
in the U.S. Call **1-800-4BIORAD**
(1-800-424-6723)

PN 4000001 Rev F

Warranty

Bio-Rad Laboratories warrants that the Molecular Analyst/PC software shall substantially conform, in all operational features, to Bio-Rad's current specifications as published in Bio-Rad's user and installation guides and that, when properly installed, it will be free of material defects which affect system performance.

The Purchaser must notify Bio-Rad in writing, within 30 days of delivery of the software (not including delivery of any subsequent modifications to the software), of any defect. If the software is found to be defective by Bio-Rad, Bio-Rad's sole obligation under this warranty is to remedy the defect in a manner consistent with Bio-Rad's regular business practices. For a defect which adversely affects the performance of the software, Bio-Rad shall use its best efforts to cure such defect as soon as reasonably practicable after receipt of Purchaser's notice. For minor defects, Bio-Rad shall use its best efforts to correct such minor defects in the next release of its software. If, however, Bio-Rad is unable to cure a major defect within 90 days of receipt of Purchaser's notice, Purchaser shall have the option to cancel this agreement, whereupon Bio-Rad shall refund only the software fees paid.

The warranties set forth in this agreement are in lieu of all other representations and warranties, expressed or implied, including warranties of merchantability and fitness for a particular purpose and any other statutory or common-law warranty. Bio-Rad on its own behalf expressly disclaims and excludes any and all such other representations and warranties. Liability of Bio-Rad to Purchaser, if any, for breach of warranty, or any other claim relating to this agreement, shall be limited to the total amount of software fees paid by purchaser to Bio-Rad. In no event shall Bio-Rad be liable for incidental or consequential damages, loss of business or profits, special or indirect damages of any nature whatsoever. No amendment, waiver, or other alteration of the warranties in this agreement may be made except by mutual agreement in writing.

Purchaser agrees that Bio-Rad's liability arising out of contract, negligence, strict liability in tort or warranty shall not exceed the amount of software license fees paid by Purchaser.

A multi-use license may be purchased to allow the software to be used on more than one computer owned by the purchaser, including a shared disk system.

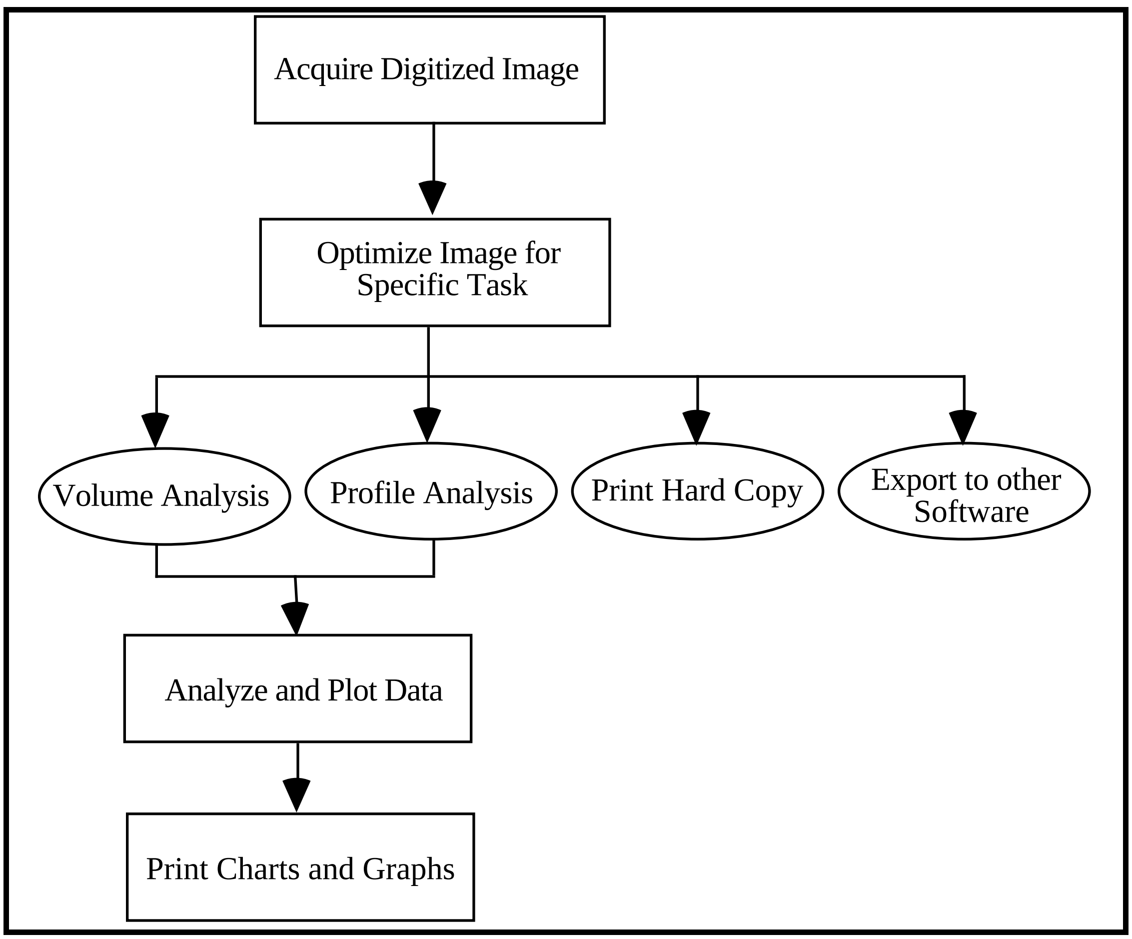
Table of Contents

Section 1 Molecular Analyst/PC Software

1.1 Introduction

The Molecular Analyst®/PC image analysis software is a powerful tool for controlling and analyzing images and data from Bio-Rad's image analysis systems. This manual contains instructions common to Bio-Rad's Molecular Imager, Densitometer, and Gel Doc systems. Please refer to the appropriate sections which are relavent to the device being used. It is recommended that the first-time user follow the tutorial section of this manual to become acquainted with some of the Molecular Analyst/PC functions and features. More detailed information about each function is contained in the sections following the tutorial.

An overview of the image analysis process is diagrammed below. After an image is acquired (either from the Densitometer, Molecular Imager, or Gel Documentation systems) and saved in a digitized form, the image can be adjusted or optimized for (1) printing, (2) performing volume analysis, (3) performing profile analysis, or (4) exporting to other software. Data obtained by volume analysis or profile analysis may be (1) used for molecular weight or conentration determination, (2) subjected to regression analysis, (3) analyzed by other functions, or (4) exported to other software. Each step of image analysis process will be described in detail in the tutorial and the sections following.



Digital data

The Gel Documentation, Densitometer, and Molecular Imager systems are light and/or radiation detectors that are capable of converting biological signals into digital data. The digital data are then displayed on the computer in a two-dimensional format using the Molecular Analyst software. The intensity of the various signals (i.e. bands or spots) is directly proportional to the intensity of the gray levels displayed on the computer monitor. The stronger the signal, the darker the image displayed on the monitor.

Currently, most image analysis software displays data in a two-dimensional format. However, the two-dimensional display does not reflect all the information stored in the digital data. A three-dimensional representation of the data gives a more realistic and clear picture (Figure 1). In a three-dimensional view, signals are displayed as mountains or hills in a landscape surrounding. The strength or intensity of the signal is represented by the volumetric capacity of the mountains or hills. This three-dimensional data is the feature that allows for more accurate quantitation of the biological signals.

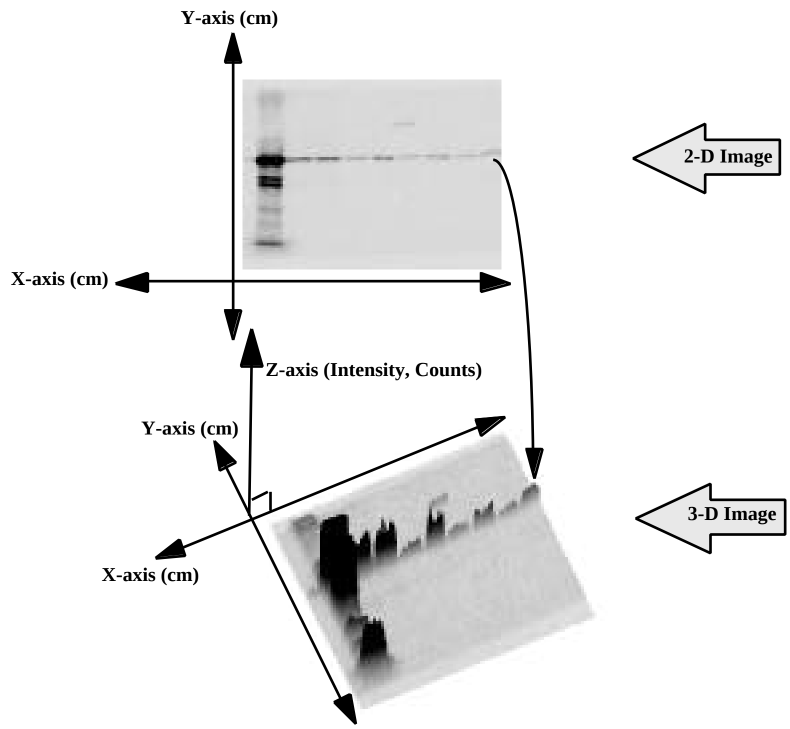


Figure 1. Two and 3-dimensional representations of a digitized image.

The digitized data provide three important pieces of information necessary for qualitative and quantitative analysis (See Figure 1):

X-coordinate which indicates the signal horizontal position
Y-coordinate which indicates the signal vertical position
Z-coordinate which indicates the signal intensity

The x- and y- coordinates define the exact location of the signal, whereas the z-coordinate establishes the intensity of the signal. The units of the x- and y- coordinates are millimeters or centimeters. The units of the z-coordinate (intensity) are “Optical Density" or O.D. when using the densitometer or the Gel Doc 1000 with the white light source, and “Counts” when using the Molecular Imager systems or the Gel Doc 1000 with the UV light source.

The signal displayed on the monitor is generated from the assembly of thousands of tiny individual screen pixels, each of which posesses its own intensity level or count (Figure 2). In order for a signal to be visible, the counts of these congregated pixels must be higher than those pixels that represent the background. The total count of a signal is determined by summing up the individual intensity or count of all the pixels (Equation 1). The mean count or intensity of a signal is the total intensity divided by the total number of pixels (Equation 2).

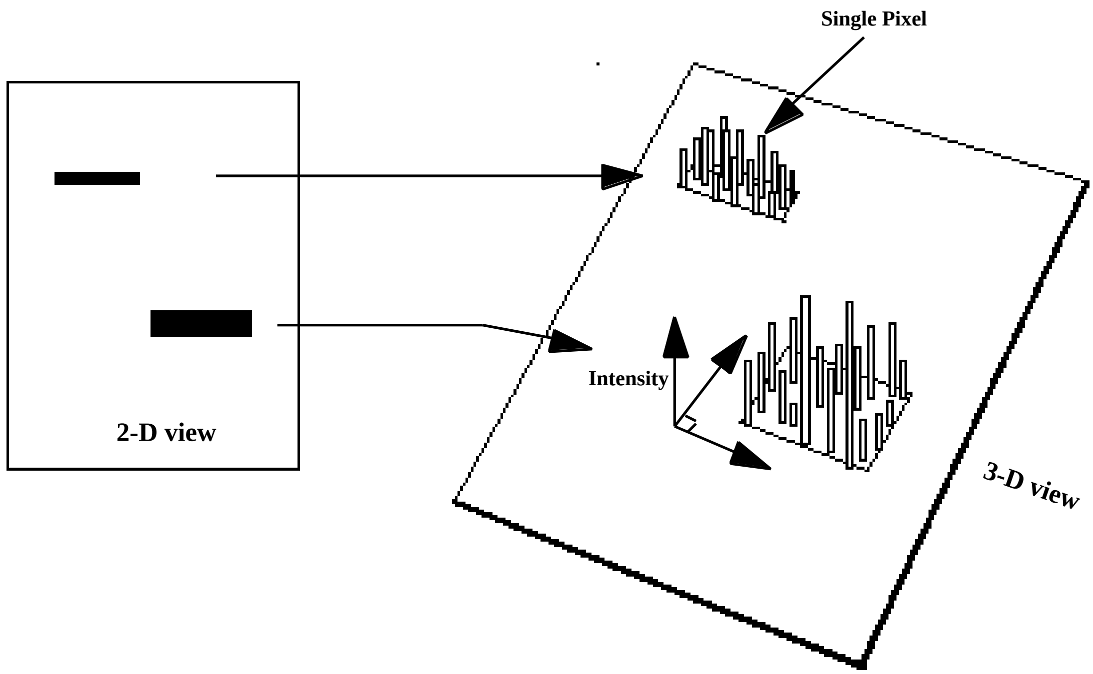


Figure 2. Pixel representation of digital signals

Equation 1

Total Count or O.D.= $\sum (I_1 + I_2 + ...)$**, where I = intensity or count of each pixel**

Equation 2

Mean Count or O.D. = Total Count or intensity / total number of pixels

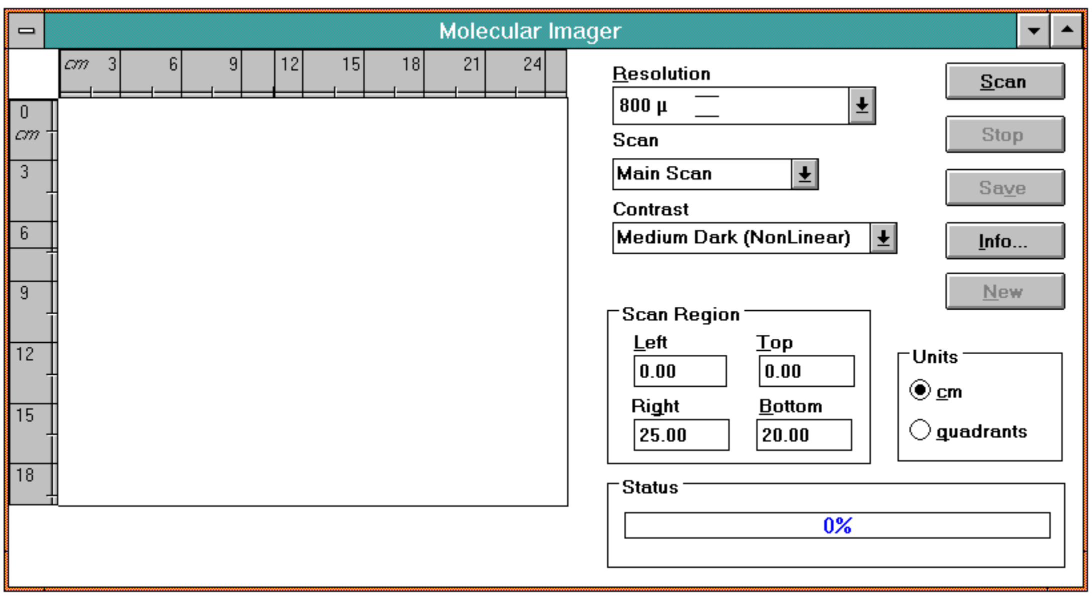
Accurate quantitation of the data is only possible if the signal lies below the maximum intensity level. This quantifiable range (which is often referred to as the linear response or linear dynamic range) varies depending on the type of detectors. Table 1 summarizes the theoretical and practical quantifiable ranges for the three types of instruments. The Gel Doc 1000, GS-700 Densitometer and the GS-363 Molecular Imager have a theoretical linear range of 0-255 pixel density units (2^8 bits), 0-4,095 OD (2^{12} bits), or 0-65,535 pixel density units (2^{16} bits), respectively. However, in practice, there are other parameters that limit the quantifiable range. For example, the linear range of densitometer samples (i.e. x-ray film) limits the quantifiable range to approximately two orders of magnitude.

Instrument	Digital Gray Scale	Theoretical Linear Range	Limiting Factor
Gel Doc	2^8 bit	256	Instrument
Densitometer	2^{12} bit	4,096	Film & Samples
Molecular Imager	2^{16} bit	65,536	Instrument

Table 1. Theoretical and practical linear dynamic range

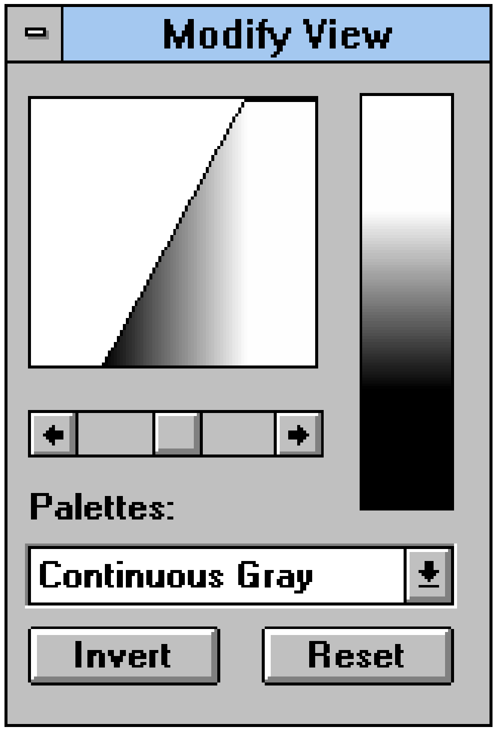
Acquiring Images

Molecular Analyst/PC software allows the acquisition of images from Bio-Rad's Densitometer (GS-670, GS-690 and GS-700), Gel Documentation (Gel Doc 1000), and Molecular Imager (GS-250, GS-363 and GS-525) systems. This integrated image analysis software provides a fast method to acquire images from radioactive, chemiluminescent, white-light, and fluorescent samples. This software will also open TIFF files.



Optimizing Images

Images from Molecular Analyst/PC can be optimized using a variety of tools. The color palette, transformation tool, and a variety of filters can adjust faint or saturated images to aid in quantitation and printing.

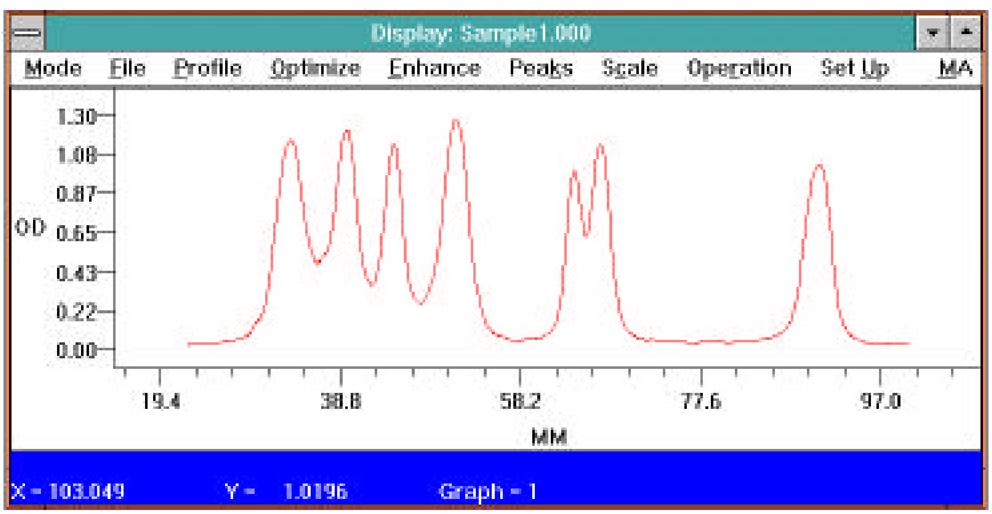


Volume Analysis

Volume analysis fully exploits the three-dimensional character of the digitized data. It allows accurate quantitation of unknown samples by extrapolation (or interpolation) from known standards.

Profile Analysis

Profile analysis determines relative differences in intensities and positions of bands or spots in an image. Using the profile tools, a two-dimensional graph or profile is generated and values such as "Percent of Total" and "Molecular Weight" can be calculated.



Printing

Molecular Analyst will print images and reports to a variety of printers. The Gel Doc 1000 will print live and saved images to a Mitsubishi P68 Thermal Printer. Please see Appendix H for configuring the Mitsubishi P68 printer.

Data Export

The results obtained from Molecular Analyst software can be exported to a variety of other software applications. For example, using simple Copy, Copy to Clipboard, and Paste commands, the data can be transferred to Excel for further analysis, and graph and chart generation.

1.2 Computer Requirements

It is recommended that Molecular Analyst/PC software use the following:

1. 80486 AT or better compatible computer with expansion slots
2. 16 MB RAM (20+ MB recommended to allow processing of large images)
3. 240 MB hard disk
4. 256 gray level color monitor
5. Windows® compatible graphics card or equivalent local bus. The Gel Doc 1000 requires that the computer or video card have video pass-through. Moreover, the video driver must support 256 gray levels and either 800 x 600, or 1024 x 768 resolutions.
6. DOS version 5.0 or better
7. Microsoft Windows version 3.1.

Due to the large potential file sizes, Bio-Rad recommends Data Analysis System PC4 to insure optimum instrument and software performance. Compatibility and performance of all other computers not purchased from Bio-Rad are the sole responsibility of the user. Due to the variability in computer hardware, please check with your local Bio-Rad representative regarding compatibility of your specific brand of computer with Bio-Rad's imaging systems.

1.3 Molecular Analyst/PC System Setup

Molecular Imager Setup

Installation of Bio-Rad's Molecular Imager Systems is completed by a field service technician.

Imaging Densitometer Setup

The PC to Densitometer connection requires that a GPIB-PCIIA adapter and GPIB software be installed in the IBM AT compatible 16 bit PC. The GPIB-PCIIA adapter and software is available when purchasing the GS-670 system from Bio-Rad.

Installing the GPIB Board

Insert the GPIB board into an available expansion slot in your PC.

Installing the GPIB Software

1. Insert the diskette labeled "National Instruments: GPIB software" in the **A:** drive.
2. Select "Run" from the "File" menu in the Application Manager.

3. Type **a:install** and press the OK button. The GPIB software installation screen is displayed.
4. Use the Down Arrow to move the check mark next to the "Windows GPIB Installation" and press Enter.
5. Use the Down Arrow to move the check mark next to the "Partial Installation" option and press Enter. The full Installation won't hurt, but will fill disk space with unnecessary software.
6. Use the Down Arrow to move the check mark next to the "Microsoft C language interface" option and press the space bar to remove the check mark and then press Enter. The extra software installed by including the "Microsoft C language interface" option won't hurt, but will fill disk space with unnecessary software.
7. Destination directories will be displayed allowing user editing. The default values should be OK for most installations. Press Enter when you are satisfied with the directory information.
8. When the message "Transfer Complete" appears, press the Enter key.
9. Press the Enter key again to select the "Win 3.1+" option.
10. Press the ESC key to exit the installation program.

The installation accomplishes two main tasks: (1) It copies two files into the windows directory - gpib.dll and gpib.ini. (2) It creates a directory called **c:\gpib-pcw** containing diagnostic software.

The diagnostic program "**wibtest.exe**" may be run to confirm correct board installation.

1. From File Manager, double click on wibtest.exe.
2. Next select the command Test GPIB0 from the Software Diagnostics menu. (Test GPIB1,2, etc. refer to other GPIB boards, if present).

If the wibtest was successful, exit, and install Molecular Analyst/PC software. If the wibtest was not successful, either the GPIB board was not present, not configured properly, or there exists a conflict with another board or device driver.

Configuring the GPIB Board if a Computer Conflict Exists

To check or to change the configuration of the GPIB board and driver:

1. Double click on wibconf.exe from the File Manager.
2. Click OK on the parameters dialog box. (If the Windows directory is located somewhere other than C:\ drive, consult your GPIB manual.
3. At the GPIB Device Map, select F8. This will display the GPIB0 configuration.
4. Next, press the <End> key on your keyboard. This will bring the cursor directly to the Interrupt Level, DMA Channel, and Base I/O Address configurations. These software configurations reflect jumper and dip switch positions on the GPIB board. Change the Interrupt level, DMA channel and Base I/O address using the right and left arrow keys to the following:

Base I/O address:	02E1H (change dip switches on the board as displayed on the screen)
DMA:	None (change board jumpers and dip switches as displayed on the screen)
Interrupt:	None (remove jumpers from the board as displayed on the screen)

Push the <esc> key to exit to the GPIB0 map, and then again to exit to File Manager. Wibtest.exe may be run again to confirm correct installation of the board. If installation is correct, load Molecular Analyst/PC software. Remember to connect all cables, start the densitometer, and then start the computer.

Gel Documentation System Setup (See Installation Notes)

The PC to Gel Documentation system requires that a frame grabber adapter be installed in your computer and that a video card with video pass-through or built-in video with video pass-through be available in your computer. The frame grabber adapter is included when purchasing the Gel Doc 1000 system from Bio-Rad. If your computer system does not have either a video card with video pass through or built-in video with video pass through, please contact your local Bio-Rad representative.

Setting the Base Address DIP Switch

The SW1 DIP switch is used to set the I/O base address of the frame grabber board. The factory default base address is 280h. The location of the SW1 on the board and diagrams for various address selections are found in Appendix G. Typically there is no address conflict requiring a change in I/O address. But if there is, the new address chosen must differ from the addresses of other boards residing in your computer. Please refer to Appendix G when changing the base address default.

Installing the Frame Grabber Board

1. Insert the frame grabber board into an available expansion slot in your PC. Due to its width, the board may require 2 ISA slots.
2. Connect the "VGA Pass Through" cable (supplied with the frame grabber board) from the frame grabber board VGA feature connector (J18) to your video board or computers built-in VGA connector. When connecting the VGA pass through cable, position the cable so that Pin 1 on the VGA cable connector (red stripe on ribbon cable) matches Pin 1 on the frame grabber board feature connector (See Appendix G for locations of Connector J18 and Pin 1 on the frame grabber board.) **Video board without a VGA pass through connection will not work. Improper video drivers will also not function (see Appendix E).**
3. Attach the supplied "Video Terminator" on the video VGA connector in the back of your computer (where the monitor normally goes.) Attach the monitor cable to the port on the frame grabber board.

Attaching The Video Cables From The Board To The Camera

1. Plug the 9 pin DB video input connector and 8 pin "MINI DIN" integration control connector of the video cable into the Frame Grabber adapter in your computer.
2. Connect the BNC video output connector and 8 pin "MINI DIN" integration control connector into the camera "Video" and "AUX" ports respectively.

Installing the Frame Grabber Board Software

The frame grabber software is installed when Molecular Analyst is installed. See section 1.6

1.4 System Software Configuration

Please refer to the included Readme Notes for the recommended AUTOEXEC.BAT and CONFIG.SYS file configuration.

1.5 Notational Conventions

Keyboard

Throughout this manual, characters enclosed in brackets <> stand for a key on the keyboard. For example, <ENTER> designates that the large key labeled **Enter** be pressed. Whenever you are supposed to type something at the keyboard, the instructions will say: "Type: ". For example, Type: EXAMPLE <ENTER> instructs you to type in the characters EXAMPLE and then press the **Enter** key.

Mouse

Unless otherwise specified, all "clicking" using the mouse is performed with the **left mouse button** (LMB). The center mouse button, if present is not functional with the Molecular Analyst/PC software. The right mouse button (RMB) is active in Profile Analyst and is only functional when the mouse cross hair is on top of a graph containing an active trace.

Objects

Throughout this manual we refer to objects within the image window. Objects can be generated and positioned on an image using either the Text, Profile or Volume tools from the tool palette. Text objects are used to annotate images or number objects. Profile objects (Line, Multi-Segment, and Averaging Rectangle) are used to create profiles for area integration and molecular weight determination. Volume objects (Circle, Ellipse, Rectangle, Square, and Grid) are used to create volume reports for volume integration and concentration, molecular weight, and pI calculation. All of the objects can be moved and manipulated within the image window.

Active Objects

Active objects are the only objects which are currently "live" and can have any function or operation performed upon them. Active objects are denoted by their color. Active objects and text will be *Fuschia* colored unless this active color is changed within the *Preferences Setup* command under the FILE menu. To make an individual object active when multiple objects are displayed, simply double click on the desired object. To make multiple objects active use the block select tool. Multiple active objects will generate multiple profiles within one graph.

Active profile

Active profiles are denoted by their color (Fuschia unless the default colors have been changed). When more than one graph is displayed on the screen, the active graph will be the graph containing the active profile. To make a different graph active, simply double click on one of the profiles within that graph.

View Modification

A view modification is a modification which affects only the display or viewing of the data, but does not actually modify the data itself.

Data Modification

A data modification is a modification which not only affects the viewing of the data, but actually transforms the data itself.

Units

The units displayed in MA/PC are related to the instrument used.
Molecular Imager - 16-bit data (0-65536 gray levels) displayed as “Counts”.
Densitometer - 8-bit data (0-255 gray levels) displayed as “O.D.”
Gel Documentation - 8-bit data (0-255 gray levels) displayed as “Counts”

1.6 Molecular Analyst Hardware Protection Key “Dongle”

A hardware protection key (HPK) is required before using Molecular Analyst software. A single-user HPK is included with each copy of Molecular Analyst software. The HPK is connected to the parallel port in the back of your computer. If a printer is connected to the parallel port, plug the HPK into the parallel port, then attach the printer cable to the key. For an additional cost, multi-user network HPKs are also available through Bio-Rad. This network key allows multiple users to use the software over the network at the same time. Please see Appendix K to install network hardware protection keys. (ALL POWER MUST BE TURNED OFF PRIOR TO ATTEMPTING ANY CONNECTION TO HPK).

1.7 Installing the Molecular Analyst/PC Software

The Molecular Analyst software is supplied on two 1.44 MB 3.5” floppy disks. One disk is labeled “Setup” and one disk is labeled “Program”.

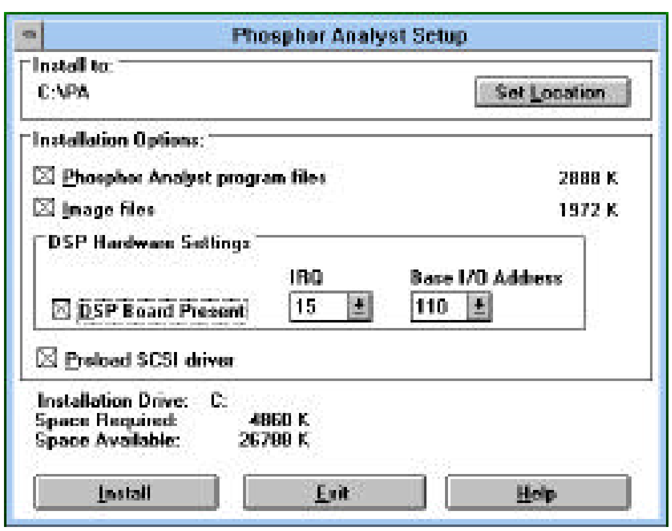
Install the Molecular Analyst/PC Program as follows:

1. Insert the Molecular Analyst software 3.5” floppy disk labeled **“Setup”** into disk drive A
2. Double click on the File Manager program icon found in the “Main” group of Program Manager.
3. Click on the disk drive icon A
4. Double click on the file: SETUP.EXE
5. At the prompt, insert the floppy disk labeled “Program” into drive A, and click **<OK>**

Install to: appoints the C:\ drive as the program destination. The Setup program also determines the necessary space required for installation of the Molecular Analyst/PC program file and image files; and the space available on Drive C. Make sure all options are set appropriately. To accept drive C for installation, use the mouse to select **<Install>**. To designate another location, select **<Set Location>** and choose an alternate path**.** For more information select **<Help>**.

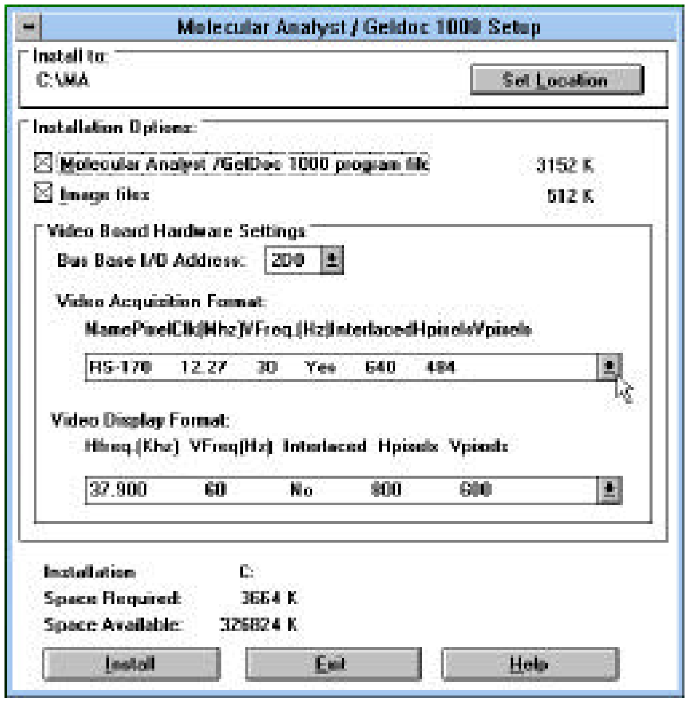
Molecular Imager Setup:

When installing Molecular Analyst for use with the Molecular Imager, there is an option for a DSP preprocessing board. The DSP board speeds all of the filtering functions. If a DSP board is present, simply click on the check box *DSP Board Present.* The default configuration values for the DSP board should work in most cases. Additionally, you should click on *Preload SCSI driver* check box. This provides for scanning under low DOS memory conditions.



Gel Doc Setup:

When installing Molecular Analyst for use with the Gel Doc 1000, you are prompted to select a Base I/O address (this must match the Base I/O address that was set in Section 1.3), Video Acquisition Format, and Video Display Format. Please see Section 2.1 for detailed description on scanning with the Gel Doc 1000. These parameters can be changed later within Molecular Analyst.



Base I/O Address

The default base address of the board is 280 hex. This configuration should work in most cases. However, if a different address is required because of a conflict within your computer, select an alternative address during the software installation. (This is selected during board installation. See Section 1.3)

Video Acquisition Format

The video acquisition for 120V/60Hz systems is RS-170. The video acquisition for 100V/50Hz, 220V/50Hz, and 240V/50Hz systems is CCIR.

Video Display Format

The video display format is determined by each computer system used. The Gel Doc 1000 functions optimally with a video frequency of 60Hz. However, refer to your computer's BIOS Setup (if your computer has built-in video) or use the configuration program (if your computer is using a video card) to check the appropriate video frequency. The Mitsubishi P68 video printers require a video display format of 800 x 600. It will not function with a video display format of 1024 x 768.

Contact your local Bio-Rad representative if you are experiencing difficulties with installation. In the U.S.A., call Bio-Rad Technical Service at 1-800-4BIORAD.

Section 2

Learning to Use the Molecular Analyst/PC Software

Using the Molecular Analyst/PC program you can quickly direct Bio-Rad's Model GS-670 Imaging Densitometer, Molecular Imager, or Gel Doc 1000 to scan, display optimize the image, perform volume or peak integration, calculate the molecular weights or concentrations of the unknown substances, and print the results. This tutorial introduces the basic skills that you will need to perform these functions. An on-line help feature is available, under the Help menu in the main menu bar, to provide additional assistance as you go through the program.

Lesson 1: Data Acquisition and Manipulation

In Lesson 1 you will learn how to acquire data from the Model GS-670 Imaging Densitometer, Molecular Imager, or Gel Doc 1000 systems.

Lesson 2: Display and Optimization

In Lesson 2 of this tutorial you will learn to display and optimize an example image provided with the software.

Lesson 3: Volume Integration

In Lesson 3 you will learn how to use the different volume tools to identify bands, perform volume integration, and generate volume reports. A standard curve will be generated and the concentrations of the unknowns calculated.

Lesson 4: Peak Integration

In Lesson 4 you will learn how to use the different tools to perform peak integration. You will learn how to extract a profile from an example image; identify and quantitate peaks from the profile; optimize the profile by subtracting background and reducing noise; and finally, integrate profile peaks.

Lesson 5: Molecular Weight Determination

In Lesson 5 you will learn how to find the molecular weight of an unknown sample. A standard curve will be generated using the standard profile data, and the resulting scatter plot will be analyzed for best fit determination. Finally, you'll determine the molecular weight of peaks identified in an unknown sample profile.

2.1 Lesson 1: Scanning

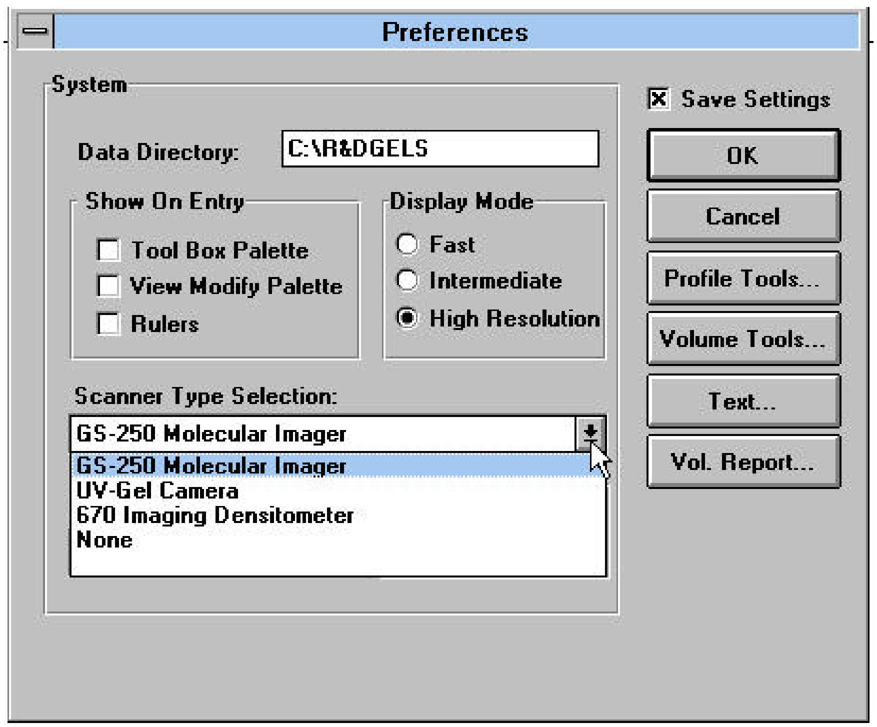
Please skip to the scanning section appropriate for your instrument.

Scanning with the Molecular Imager

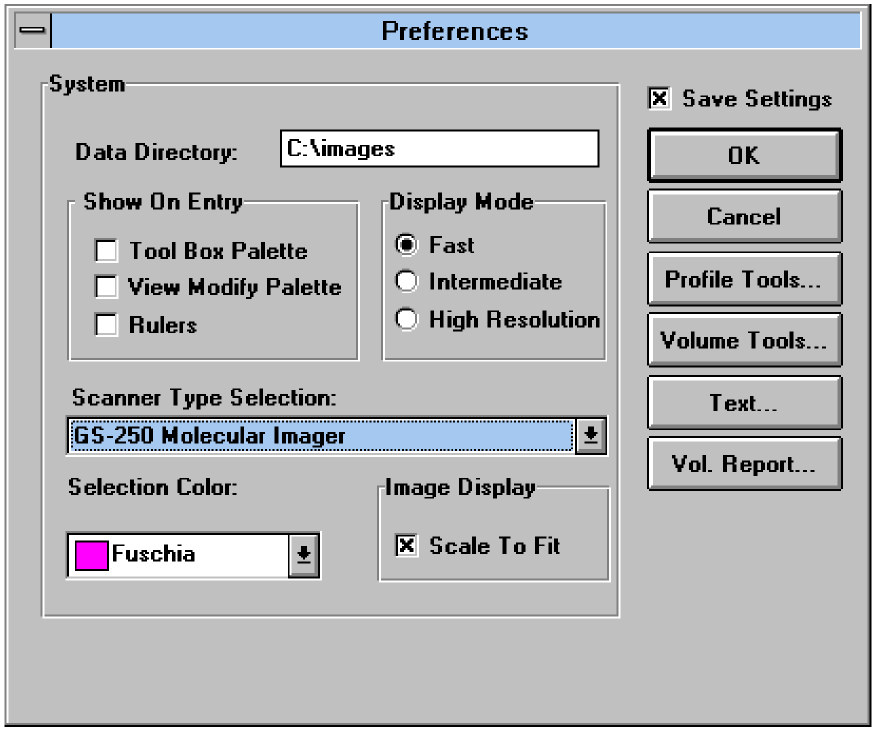
Setting Instrument Preferences

To prepare for scanning with the Molecular Imager:

1. Choose the *Preferences* command from the FILE menu.
2. Next, click on the pull-down menu "Scanner Type Selection". This will allow you to select the Molecular Imager and the defaults for profile tools, volume tools, text, and volume report.



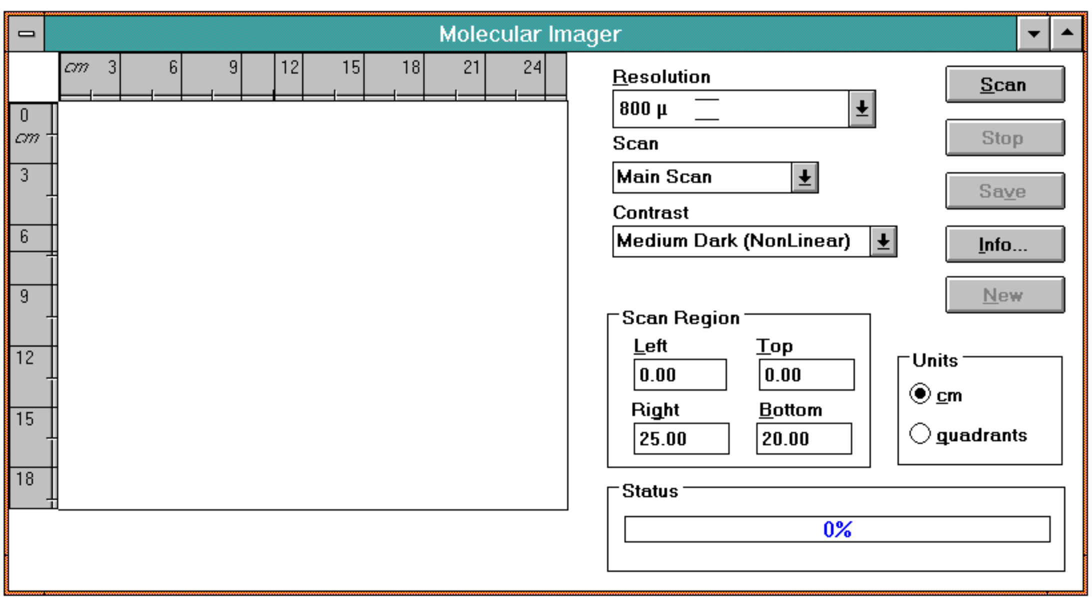
3. The Molecular Imager should automatically be highlighted. However, if using other Bio-Rad imaging systems, highlight "Molecular Imager" and click OK.



Setting Scan Parameters-

To open the scan window and to set the scanning parameters:

1. Click on the FILE menu, and drag the mouse to *New*. The scan window will appear with pull-down menus, scanning area window, control buttons, and status bar. The scan window is initially blank with its dimensions representing the full size of a phosphor screen.



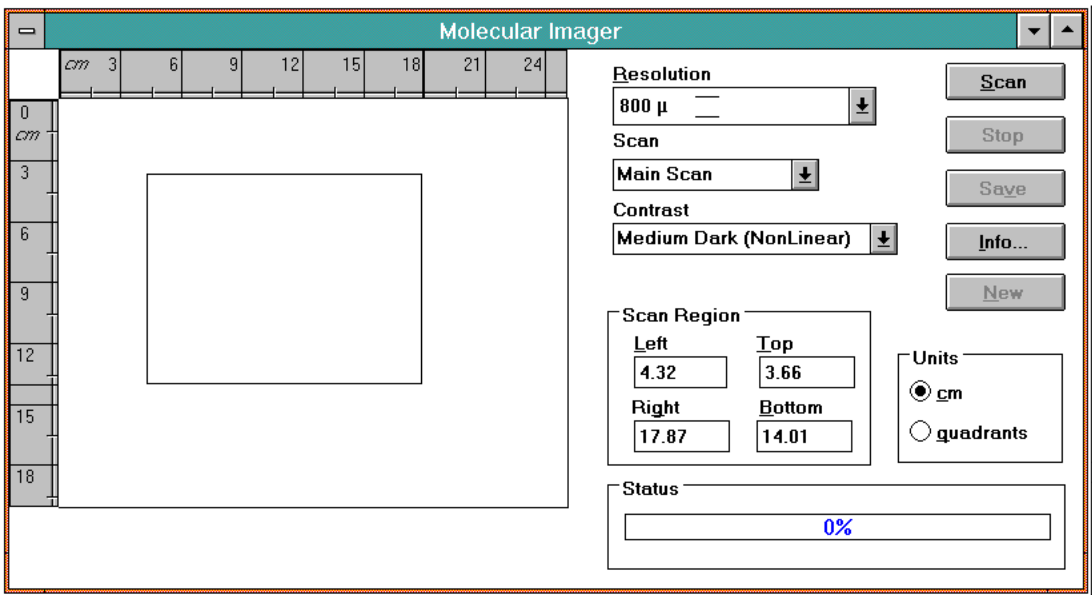
2. Using the mouse, activate the first pull-down menu titled RESOLUTION. Within this menu, select *800µ* from the list. This low resolution scanning will allow rapid preview of your sample for qualitative purposes.
3. Next, activate the CONTRAST pull-down menu and select *Medium Dark (Non Linear)*

Identifying the Area to Scan-

Next, identify the scanning area. When the *Scan Window* is opened, the default scanning area is the entire screen size, 20 cm x 25 cm. To identify a more specific scanning area on the phosphorescence imaging screen:

1. Place the mouse onto the scanning area. Notice that the pointer is replaced with a square scanning tool when placed in the scanning region. Two metric rulers are displayed in the window to aid in selecting the scanning area. These rulers are replaced with letters if quadrants is selected from the *Unit* box.
2. Place the square scanning tool at the upper left corner of the region to be scanned, click, and hold down the left mouse button, and drag down diagonally until the desired scanning area is selected.
3. Release the mouse button. As the scanning region is identified the scanning coordinates (cm or quadrants) are displayed within the *Scan Region Area* boxes designated *Left, Right, Top, and Down.*

Alternatively, the sample placement grid coordinates for the sample's upper left and lower right corner can be entered. Click on the quadrants radio button and enter the grid coordinates noted from the lid of the sample loading dock.



Starting the Preview Scan

Select the *Scan* control button to begin scanning. The status bar will indicate the percent of the scan that has been completed. The scan window will update the display as the scanning process continues. Once the image has been fully scanned, it will be displayed in the scanning area. This image in the scanning window is only a crude preview of what will be actually shown in the display window.

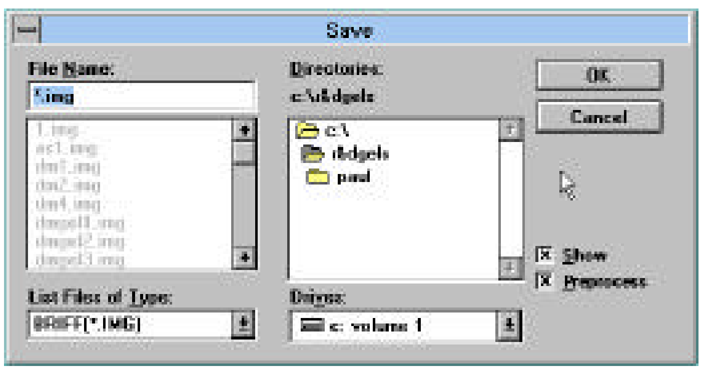
The Higher Resolution Scan

A specific area of interest can now be identified on the preview image and rescanned at medium (200µ) or high resolution (100µ) without having to re-expose the sample to the phosphorescence imaging screen. Select out the area of interest on the 800µ image and scan at a higher resolution. Each scan is sequentially named and added to the menu item, *Scan*, from which any scan can be accessed and areas identified for high resolution scanning. **Note that a high resolution scan must be completely inside or outside a preview scan area. Overlap of preview and unscanned areas will result in an error.**

To scan a new phosphorescence imaging screen and begin the process again, click on *New*.

Saving the Scan

1. Adjust the contrast of the image again by selecting the appropriate contrast to best display your image from the *Contrast* pull-down menu. To perform image optimization functions, the scan from the sequence must be saved as an image file using the *Save* control button.
2. In the *Save* dialog box, save your image as an image file with an 8 character name and a .IMG extension (TEST.IMG).
3. If a DSP board is present in your computer, you are given the option to preprocess images in the *Save* dialog box. If no DSP board is present, the *Preprocess* option is not present in the *Save* dialog box. Preprocessing automatically performs ANR and ORPA filtering on the image upon saving. Click on the preprocess check box to preprocess the image during saving.



To view the image just captured, select *Open* from the FILE menu and click on the image name in the Graphics window's title bar, and click on the *Open* command button. The size (MB) of each file saved depends upon the total area scanned and the resolution.

Deleting Images

To delete an image:

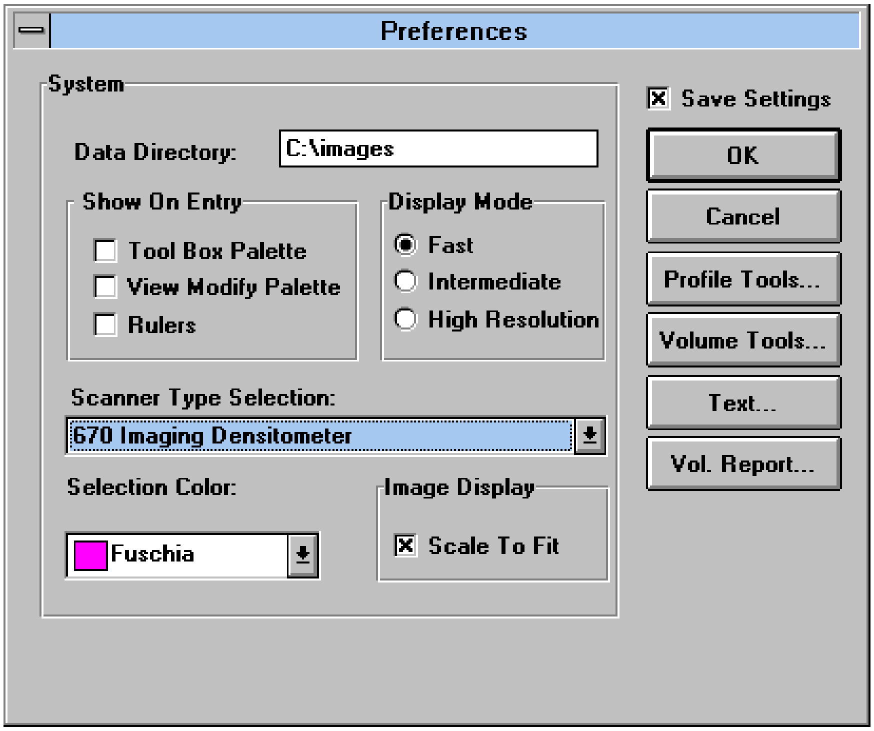
1. Quit the Molecular Analyst/PC program and open File Manager in Windows.
2. Click on the MA folder, then click on the data folder to display images and profile files.
3. Highlight the file(s) to be deleted by clicking on them with the mouse (hold the <Cntrl> key down while clicking the mouse button for deleting multiple files) and then select *Delete* from the FILE menu or simply press the <Delete> key.
4. Click OK at the prompt.

Scanning with the Model GS-670 Imaging Densitometer

Setting Instrument Preferences

To prepare for scanning with the densitometer:

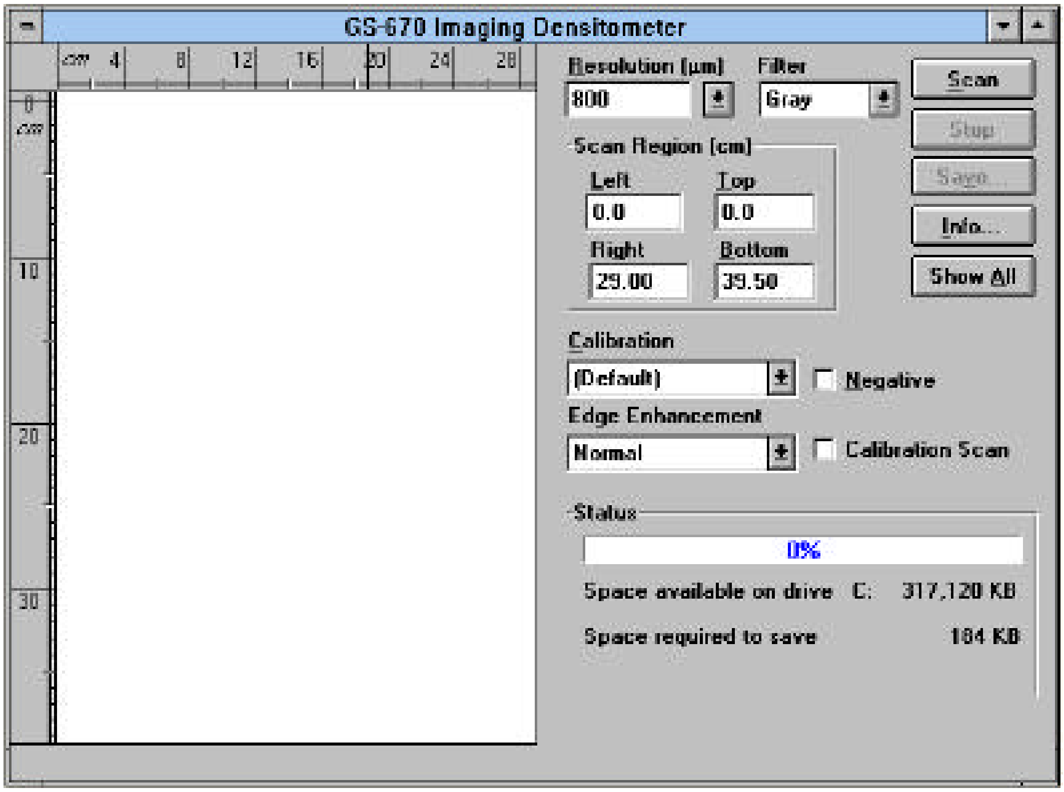
1. Choose the *Preferences* command from the FILE menu.
2. Click on the pull-down menu "Scanner Type Selection". This will allow you to select the densitometer and the defaults for profile tools, volume tools, text, and volume report.



3. The GS-670 Densitometer should automatically be highlighted. However, if using other Bio-Rad imaging systems, highlight "670 Imaging Densitometer" and click OK.

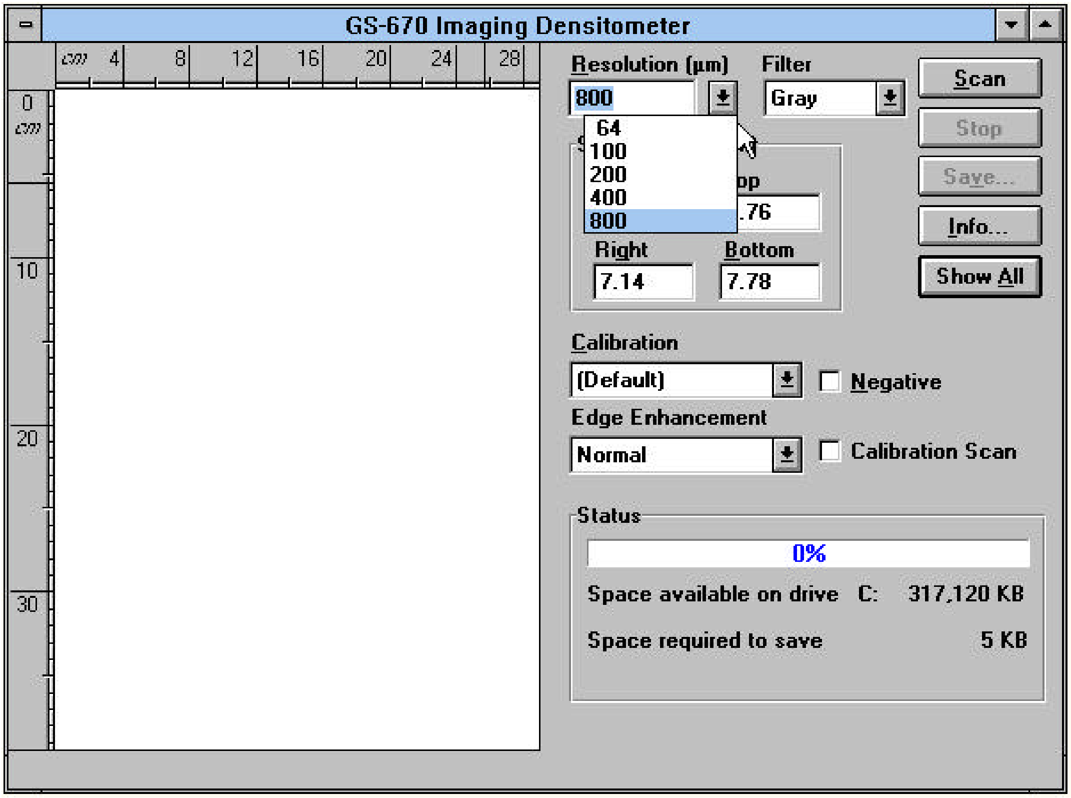
Setting Scan Parameters-

To open the scan window, click on the FILE menu, and click on *New*. The scan window will appear with pull-down menus, scanning area window, control buttons, and status bar.



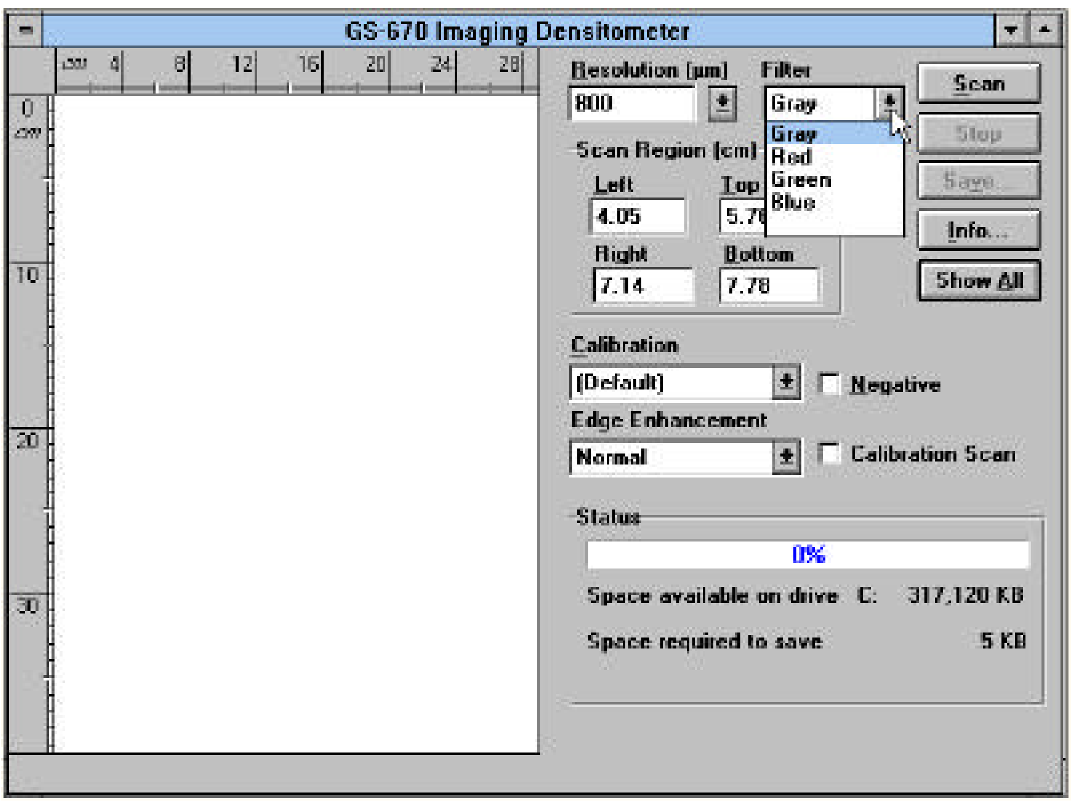
To prepare for scanning:

1. Using the mouse, activate the first pull-down menu titled RESOLUTION. Within this menu, select *800µ* from the list. This low resolution scanning allows rapid preview of your sample for qualitative purposes.
2. Next, select the type of edge enhancement desired.
3. Click on the pull-down menu titled Normal in the *Edge Enhancement* field to provide no enhancement to the image. Filters effect the raw data and may change quantitative results. For best quantitative results, use the Normal edge enhancement filter (See Appendix D).



Selecting Resolution

4. Lastly, select the filter which corresponds to a wavelength range for your sample.



Selecting Filters

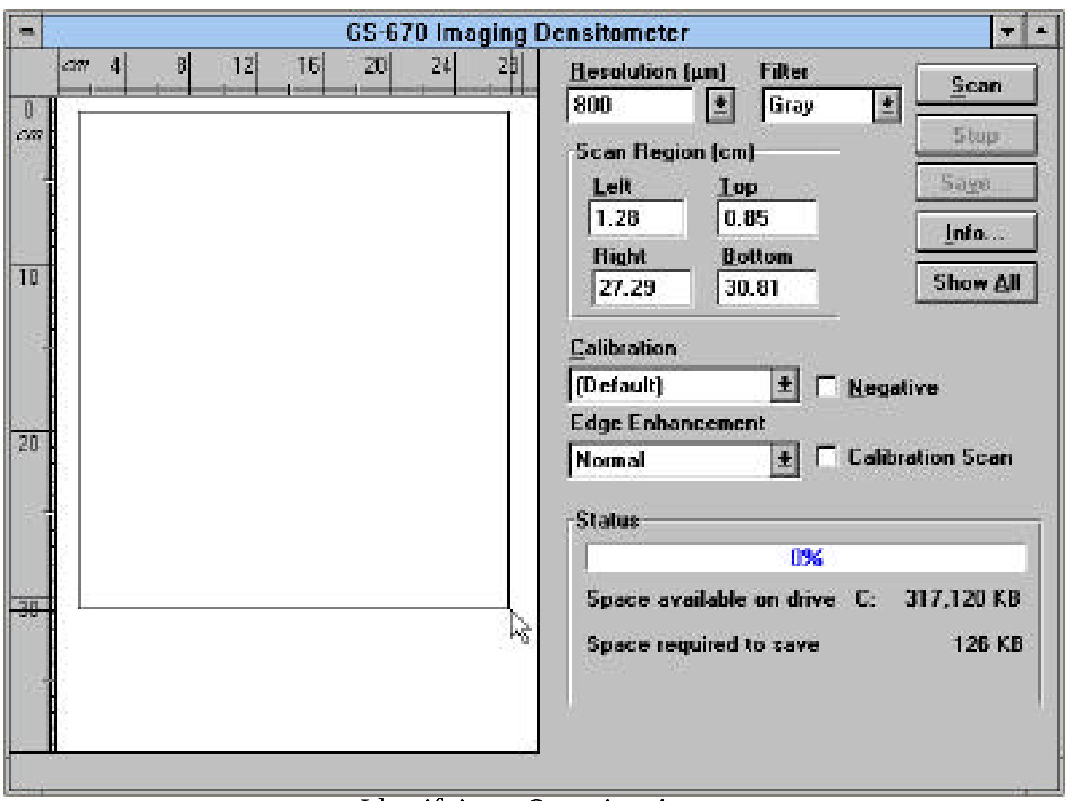
Filter Selection:

Filter Color	Wavelength	Application Examples
Red	(595nm-750nm)	Coomasie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	(520nm-570nm)	Coomasie R-250, Basic Fuchsin
Blue	(400nm-530nm)	Crocein Scarlet, Colloidal Gold
Gray	(400nm-750nm)	Silver Stain, Copper Stain, Film, Photographs

Identifying the Area to Scan-

Next, we will identify the scanning area. When the *Scan Window* is opened, the default scanning area is the entire scanning area (Transmittance: 29cm x 39.5cm, Reflectance: 29cm x 42cm). To specify a smaller scanning area:

1. Place the mouse onto the scanning area. Notice that the pointer is replaced with a square scanning tool when placed in the scanning region. Two metric rulers are displayed in the window to aid in selecting the scanning area.
2. Place the square scanning tool at the upper left corner of the region to be scanned, click, and hold down the left mouse button, and drag down diagonally until the desired scanning area is selected.
3. Release the mouse button. As the scanning region is identified the scanning coordinates (cm) are displayed within the *Scan Region Area* boxes designated *Left, Right, Top, and Down.*



Identifying a Scanning Area

Starting The Scan

Click the *Scan* control button to begin scanning. The status bar will indicate the percent of the scan that has been completed. The scan window will update the display as the scanning process continues. Once the image has been fully scanned, it will be displayed in the scanning area at 100% of the current window size. Next, with the preview image displayed, select a higher resolution, and click on the command button *Scan*. Moreover, a more defined area of interest can also be identified in the preview image using the mouse and then rescanned. Alternatively, after displaying the preview image, selecting the *Show All* button displays the area scanned in proportion to the whole scanning area. This feature is useful to allow rapid reidentification and rescanning if the desired target area was missed. Note that at any point in the scanning process, scanning can be stopped and images either saved or rescanned at a different resolution, filter, or edge enhancement.

Scanning Negatives (Ethidium Bromide Photographs)

Clicking this check box allows the scanning of negatives or samples which have light bands on dark background (i.e. ethidium bromide photographs.) This information can also be entered after the image has been saved using the *Image Info* command from the VIEW menu. Selecting the *Options* button in the Image Info dialog box allows the selection of either a positive (dark bands on light background) or negative (light bands on dark background) image.

Saving the Scan

Save your image as an image file with an 8 character name and a .IMG extension (TEST.IMG) and close the scan window. To view the image just captured, select *Open* from the FILE menu and click on the image name in the Graphics window's title bar, and click on the *Open* command button. The size (MB) of each file saved depends upon the total area scanned and the resolution.

Deleting Images

To delete an image, quit the Molecular Analyst/PC program and open File Manager in Windows. Click on the MA folder, then click on the data folder to display images and profile files. Highlight the file(s) to be deleted by clicking on them with the mouse (hold the <Cntrl> key down while clicking the mouse button for deleting multiple files) and then select *Delete* from the FILE menu or simply press the <Delete> key. Click OK at the prompt.

Calibrating the Densitometer

The MA/PC software also provides the means to calibrate densitometer linearity and eliminates any scanning variation. Once the instrument is calibrated for each of the 3 translucent plastic density strips, calibration tables are saved and are used for future scanning. The densitometer need not be calibrated before each scan.

To calibrate the densitometer in transmittance mode:

1. Place the appropriate densitometer translucent plastic density strip (Light, Medium, Dark) on the densitometer. Please refer to your Model GS-670 Densitometer Users Manual for information on selecting the proper plastic density strip.
2. Next, place a Kodak #2 or #3 calibrated step tablet on the densitometer.
3. To calibrate the denstiometer in reflectance mode, attach the reflectance lid and place a Kodak Paper Gray Scale Step Tablet on the densitometer.

(To order these step tablets, contact you local Kodak Central Domestic Customer Service Center. The Kodak catalog numbers for the calibrated #2, #3, and Paper Gray Scale tablets are 152-3406, 152-3422, and 152-2267, respectively.)

The tablet should be placed horizontally and as straight as possible in relation to the top of the densitometer (this will aid in the automatic step finding.) It is also important to surround the calibration strip with the masking template provided with the densitometer. The masking template lowers the amount of background noise caused by scattered light. **Do not cover the strip with the template.** Moreover, in transmittance mode, the tablet must be placed emulsion side (dull side) up, to reduce error due to reflection. In reflectance mode, the masking template is not necessary.

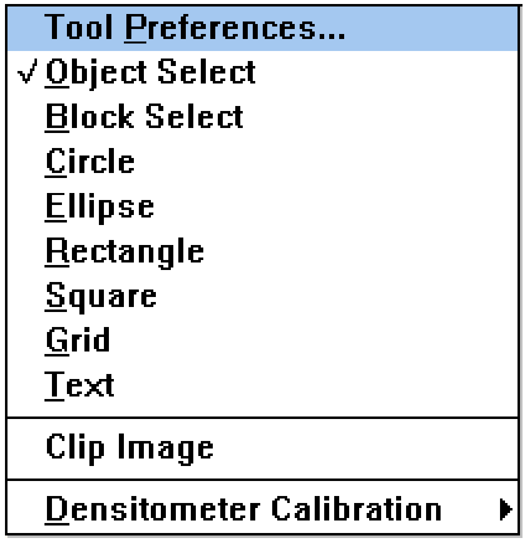
4. Within the scan window, select the lowest resolution available (*800μ*) and a *Gray* filter. 5. Click *Scan*. Stop the scan when the entire preview image is displayed. This quick scan will show you the position of the calibration strip.

5. Next, mouse out an area around the image of the calibration strip. Make sure not to cut off any area of the strip.
6. Select the highest resolution available (*64μ*),and click on the *Calibration Scan* box.
7. Click on the name of the translucent plastic strip being used. Light, Medium, or Dark should be displayed in the Calibration window.
8. Click *Scan.*

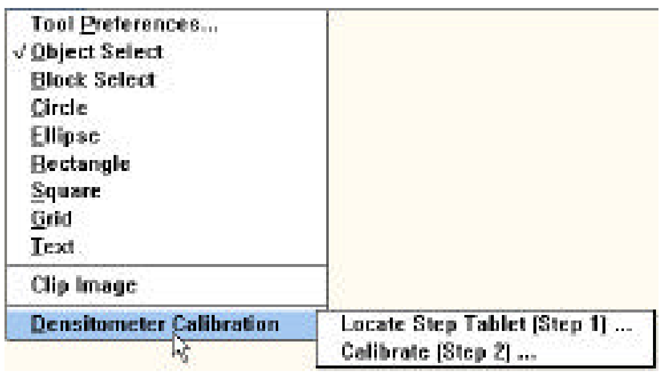
Once the scanning is completed, save the calibration image with the name of the translucent plastic density strip used (Light, Medium, Dark).

9. Close the scanning window and open the saved calibration image.

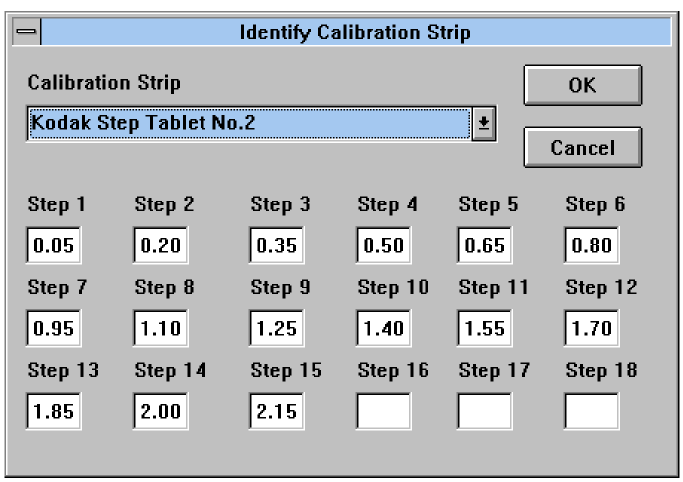
10. Next, select *Densitometer Calibration* from the TOOLS menu.



11. Select the command *Locate Step Tablet (Step 1)*.



This will display a *Identify Calibration Strip* dialog box.

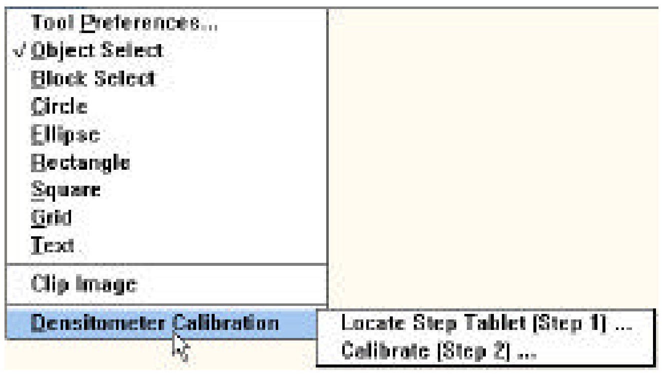


The most common O.D. values for the Kodak strip are provided as a default. However, the OD values for the calibrated step tablets can vary from tablet to tablet.

12. Enter in the values from the information sheet which accompanied the calibrated step tablet. (The maximum linear range you can expect to calibrate under the best circumstances will be approximately 0 to 2.0 ±.2 O.D.. Based on this, only use the first 15 steps of the Kodak #2 step tablet. Attempting to use more steps may reduce the linearity of the calibration.
13. Click OK.

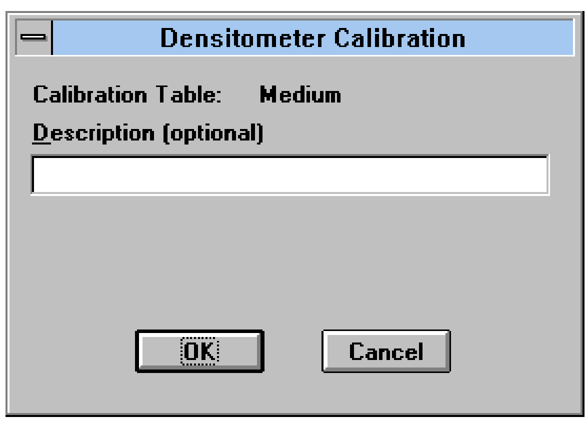
Volume rectangles should be placed onto the image in the proper density sections. If the automatic detection fails, the step rectangles are still created on the image. The user can then manually move the rectangles over the correct section of the tablet.

14. Once again select *Densitometer Calibration* from the TOOLS menu.
15. Select the command *Calibrate(Step 2)*.

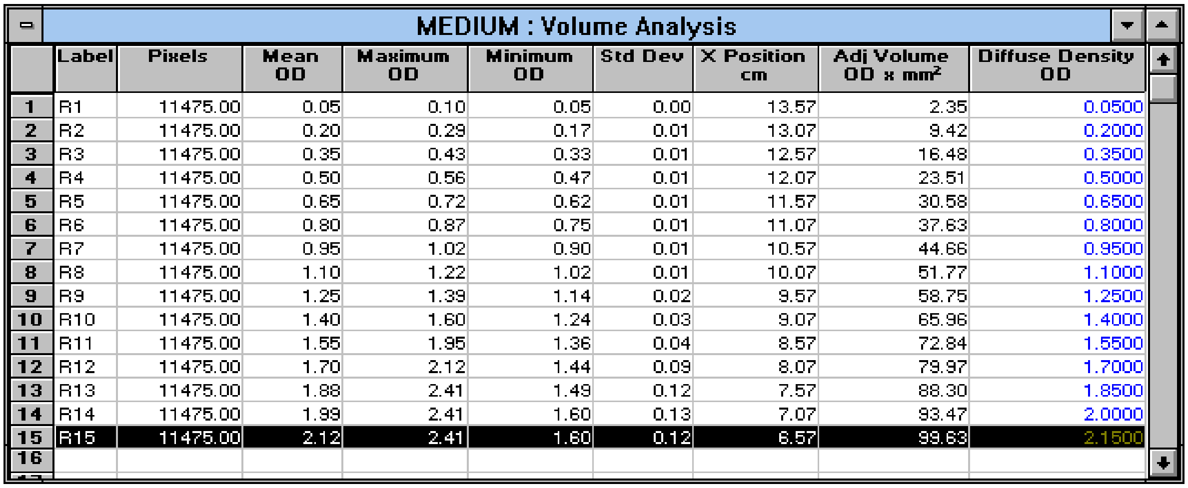


This will set volume objects onto the image and display a Densitometer Calibration dialog box.

16. Within the Densitometer Calibration Dialog box, Click *OK* to save the calibration table. Each table created (Dark, Medium, Light) is saved and is available in the Calibration pull-down menu in the scan window.



17. View the calibration report table by selecting Analyze Volume Objects from the ANALYSIS menu.

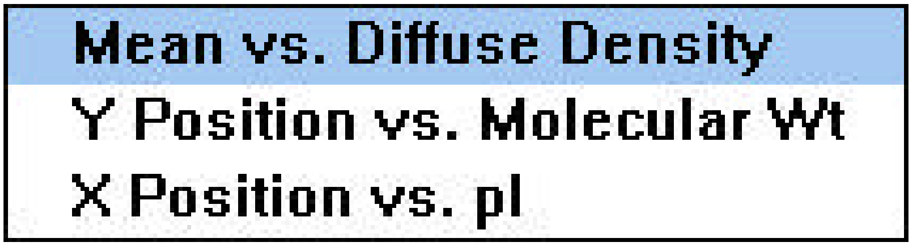


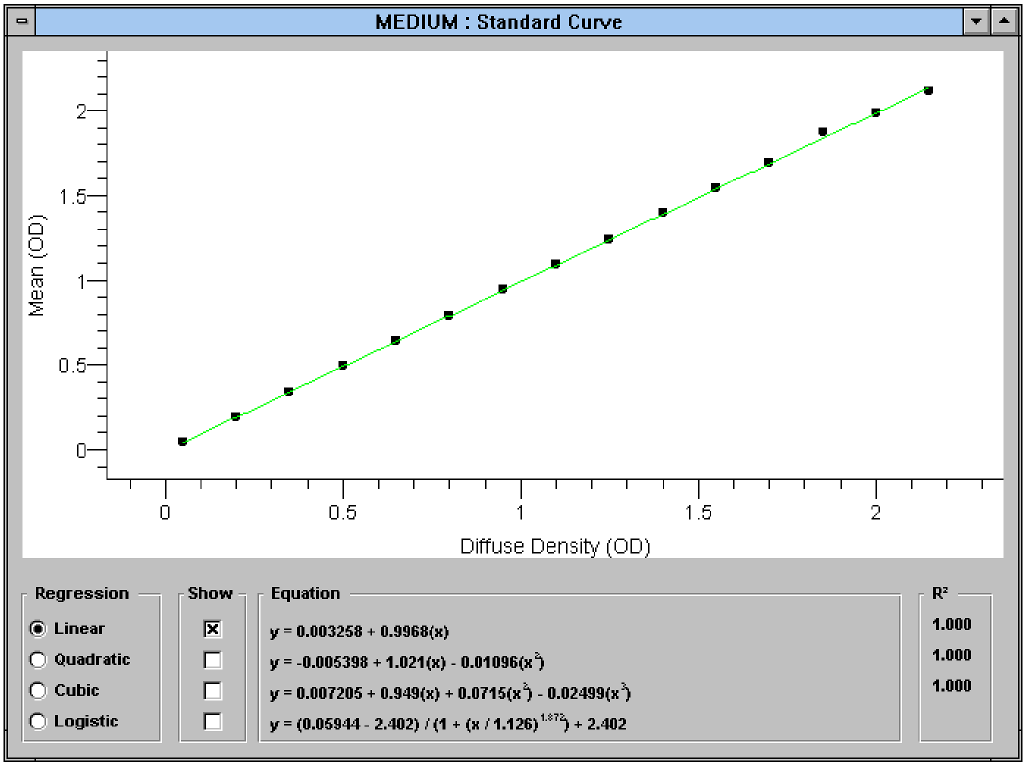
MEDIUM : Volume Analysis

	Label	Pixels	Mean OD	Maximum OD	Minimum OD	Std Dev	X Position cm	Adj Volume OD x mm²	Diffuse Density OD
1	R1	11475.00	0.05	0.10	0.05	0.00	13.57	2.35	0.0500
2	R2	11475.00	0.20	0.29	0.17	0.01	13.07	9.42	0.2000
3	R3	11475.00	0.35	0.43	0.33	0.01	12.57	16.48	0.3500
4	R4	11475.00	0.50	0.56	0.47	0.01	12.07	23.51	0.5000
5	R5	11475.00	0.65	0.72	0.62	0.01	11.57	30.58	0.6500
6	R6	11475.00	0.80	0.87	0.75	0.01	11.07	37.63	0.8000
7	R7	11475.00	0.95	1.02	0.90	0.01	10.57	44.66	0.9500
8	R8	11475.00	1.10	1.22	1.02	0.01	10.07	51.77	1.1000
9	R9	11475.00	1.25	1.39	1.14	0.02	9.57	58.75	1.2500
10	R10	11475.00	1.40	1.60	1.24	0.03	9.07	65.96	1.4000
11	R11	11475.00	1.55	1.95	1.36	0.04	8.57	72.84	1.5500
12	R12	11475.00	1.70	2.12	1.44	0.09	8.07	79.97	1.7000
13	R13	11475.00	1.88	2.41	1.49	0.12	7.57	88.30	1.8500
14	R14	11475.00	1.99	2.41	1.60	0.13	7.07	93.47	2.0000
15	R15	11475.00	2.12	2.41	1.60	0.12	6.57	99.63	2.1500
16									

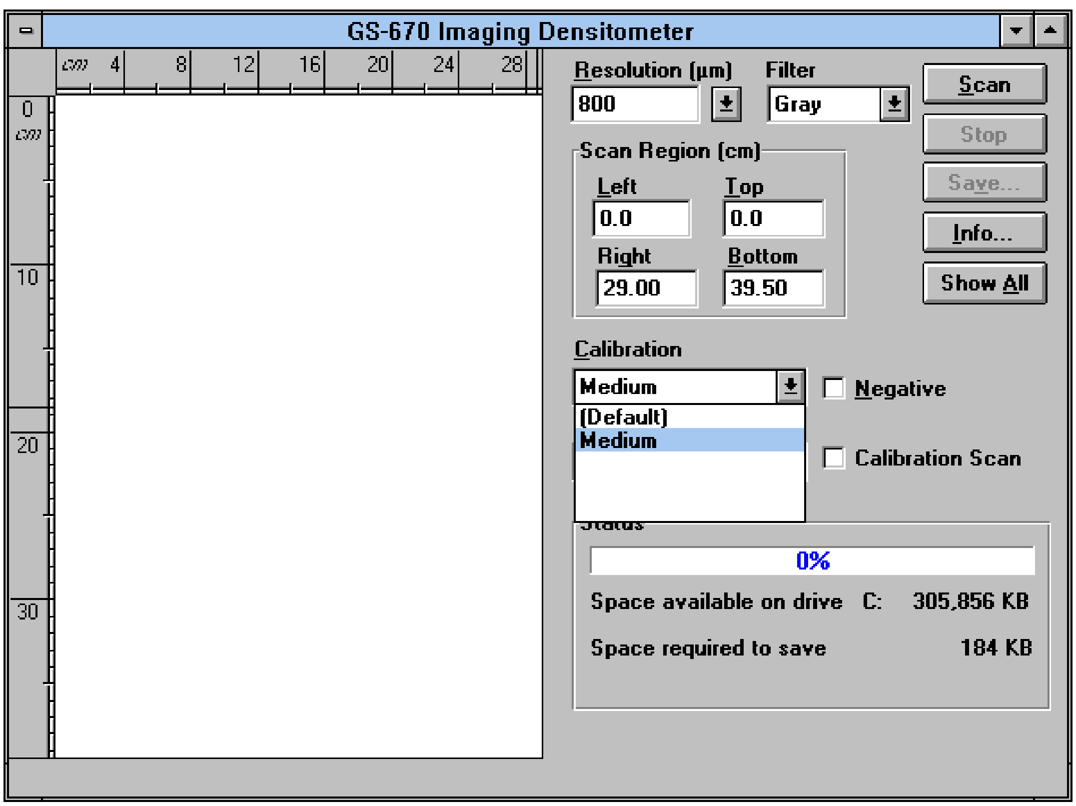
To visualize the linearity of calibration as a graph:

1. Select Format Columns from the OPTIONS menu.
2. Click on the *Next* button until "Concentration" is displayed.
3. Highlight the name "Concentration" and type in "Diffuse Density".
4. Next, click on the *Std Curve Y* Axis pull-down menu and select Mean.
5. Finally, select Mean vs. Diffuse Density from the VIEW menu.





Once the desired calibration tables are saved, the user can select the appropriate calibration table to use when scanning various samples with the different GS-670 Densitometer translucent plastic density strips.



Frequently asked questions:

Q: What happens if I do not create all the calibration tables?
A: Tables that have not been created will not be listed in the Scan Window.

Q: Do the calibration tables travel with the densitometer?
A: No, the calibration tables are stored on the host computer. The scanner must be recalibrated whenever it is attached to a different computer.

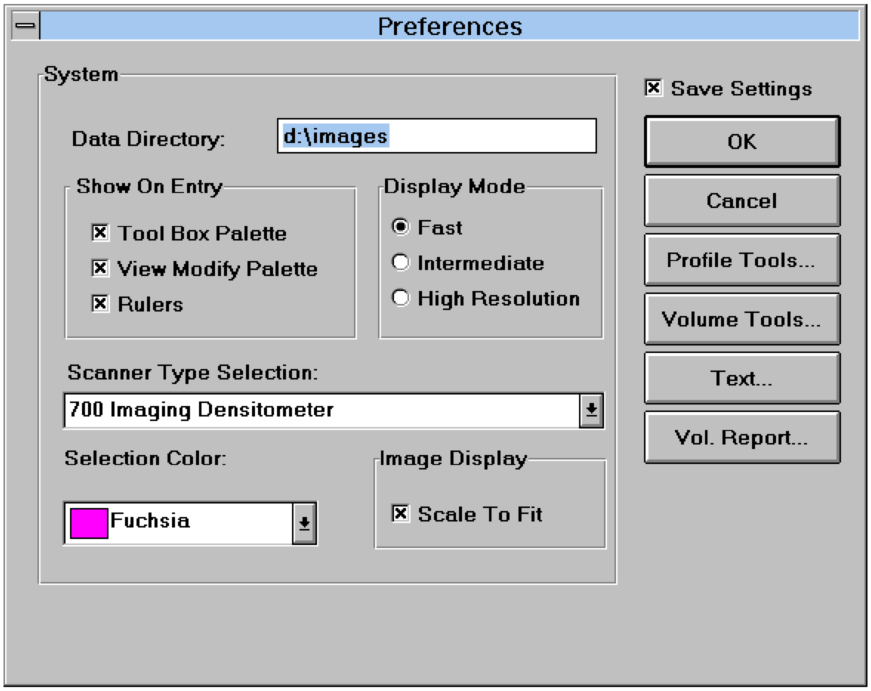
Q: What is "User Defined"?
A: The user defined calibration is any translucent plastic strip (Light, Medium, or Dark) that is used with any filter color other than GRAY. One example would be the use of the BLUE filter when creating a calibration table for Crocien Scarlet stain.

Scanning with the Model GS-700/690 Imaging Densitometers

Setting Instrument Preferences

To prepare for scanning with the densitometer:

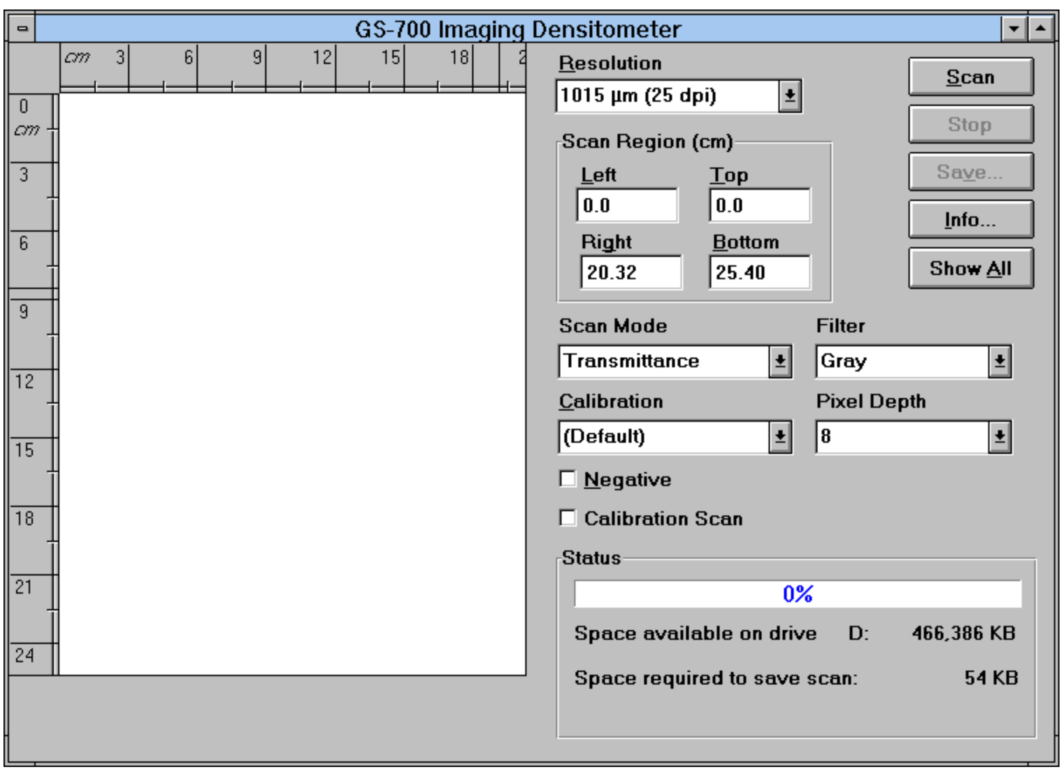
1. Choose the *Preferences* command from the FILE menu.
2. Click on the pull-down menu "Scanner Type Selection". This will allow you to select the densitometer and the defaults for profile tools, volume tools, text, and volume report.



3. Highlight the appropriate instrument and click OK.

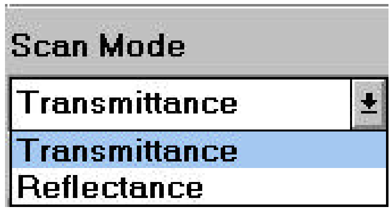
Setting Scan Parameters-

1. To open the scan window, click on the FILE menu, and click on *New*. The scan window will appear with pull-down menus, scanning area window, control buttons, and status bar.



2. Using the mouse, activate the first pull-down menu titled RESOLUTION. Within this menu, select a low resolution scan from the list (25 dpi for example). This low resolution scanning allows rapid preview of your sample for qualitative purposes.

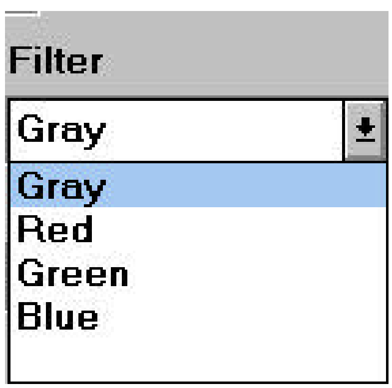
3. Next, activate the pull-down menu titled SCAN MODE. Select either Transmittance or Reflectance



Transmittance
Transparent allows the scanning of transparent samples such as films, gels, and slides.

Reflectance
Reflective allows the scanning of opaque samples such as blots, gels dried on filter paper, and photographs.

4. Next, select the filter which corresponds to a wavelength range for your sample.



Gray
This mode allows you to scan samples as 256 gray level images (8-bit) or 4096 grat level images (12-bit) and is useful with samples includeing Silver Stains, Copper Stains, Films, Negatives, etc.

Red
This mode allows you to scan samples within a specific red filter wavelength range and is useful with blue samples including Coomasie G-250, BCIP/NBT, etc.

Filter Color	*Wavelength*	*Application Examples*
Red	(595nm-750nm)	Coomasie G-250, BCIP /NBT, Fast Green FCF

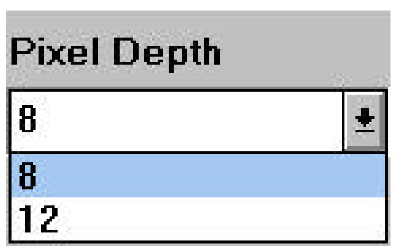
Green
This mode allows you to scan samples within a specific green filter wavelength range and is useful with samples including Coomasie R-250, etc.

Filter Color	*Wavelength*	*Application Examples*
Green	(520nm-570nm)	Coomasie R-250, Basic Fuchsin

Blue
This mode allows you to scan samples within a specific blue filter wavelength range and is useful with red samples including Crocien Scarlet, etc.

Filter Color	*Wavelength*	*Application Examples*
Blue	(400nm-530nm)	Crocein Scarlet

5. Lastly, select the Pixel Depth with which to scan your sample.



8
This mode allows you to scan samples with 8 bit precision and generates 8 bit image data files.

12
This mode allows you to scan samples with 12 bit precision and generates 12 bit image data files. These image files will be larger than 8 bit files.

6. To begin the prescan, click on SCAN. The status bar will indicate the percent of the scan that has been completed.

Identifying a Specific Area to Scan-
Next, we will identify the scanning area. When the *Scan Window* is opened, the default scanning area is the entire scanning area. To specify a smaller scanning area:

1. Place the mouse onto the scanning area. Notice that the pointer is replaced with a square scanning tool when placed in the scanning region. Two metric rulers are displayed in the window to aid in selecting the scanning area.
2. Place the square scanning tool at the upper left corner of the region to be scanned, click, and hold down the left mouse button, and drag down diagonally until the desired scanning area is selected.
3. Release the mouse button. As the scanning region is identified the scanning coordinates (cm) are displayed within the *Scan Region Area* boxes designated *Left, Right, Top, and Down.*

Starting The Scan
Select a higher scanning resolution and click the *Scan* control button to begin scanning. The status bar will indicate the percent of the scan that has been completed. The scan window will update the display as the scanning process continues. Once the image has been fully scanned, it will be displayed in the scanning area at 100% of the current window size. Alternatively, after displaying the preview image, selecting the *Show All* button displays the area scanned in proportion to the whole scanning area. This feature is useful to allow rapid reidentification and rescanning if the desired target area was missed. Note that at any point in the scanning process, scanning can be stopped and images either saved or rescanned at a different resolution, filter or pixel depth.

Scanning Negatives (Ethidium Bromide Photographs)
Clicking this check box allows the scanning of negatives or samples which have light bands on dark background (i.e. ethidium bromide photographs.) This information can also be entered after the image has been saved using the *Image Info* command from the VIEW menu. Selecting the *Options* button in the Image Info dialog box allows the selection of either a positive (dark bands on light background) or negative (light bands on dark background) image.

Saving the Scan
Save your image as an image file with an 8 character name and a .IMG extension (TEST.IMG) and close the scan window. To view the image just captured, select *Open* from the FILE menu and click on the image name in the Graphics window's title bar, and click on the *Open* command button. The size (MB) of each file saved depends upon the total area scanned, pixel depth, and the resolution.

Deleting Images
To delete an image, quit the Molecular Analyst/PC program and open File Manager in Windows. Click on the MA folder, then click on the data folder to display images and profile files. Highlight the file(s) to be deleted by clicking on them with the mouse (hold the <Cntrl> key down while clicking the mouse button for deleting multiple files) and then select *Delete* from the FILE menu or simply press the <Delete> key. Click OK at the prompt.

Calibrating the GS-700 / GS-690 Densitometers
The MA/PC software also provides the means to calibrate densitometer linearity and eliminates any scanning variation. Once the instrument is calibrated, calibration tables are saved and are used for future scanning. The densitometer need not be calibrated before each scan.

To calibrate the densitometer in transmittance mode:

1. Place a Kodak #2 or #3 calibrated step tablet on the densitometer. To calibrate the denstiometer in reflectance mode, place a Kodak Paper Gray Scale Step Tablet on the densitometer.

(To order these step tablets, contact you local Kodak Central Domestic Customer Service Center. The Kodak catalog numbers for the calibrated #2, #3, and Paper Gray Scale tablets are 152-3406, 152-3422, and 152-2267, respectively.)

The tablet should be placed horizontally and as straight as possible in relation to the top of the densitometer (this will aid in the automatic step finding.) Moreover, in transmittance mode, the tablet must be placed emulsion side (dull side) up, to reduce error due to reflection. In reflectance mode, the masking template is not necessary.

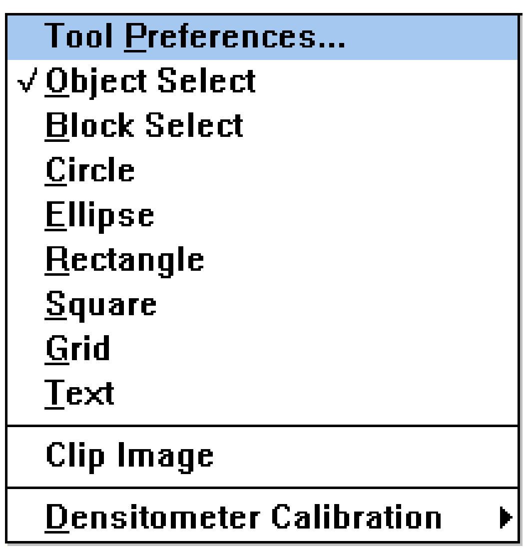
2. Within the scan window, select the lowest resolution available (25 dpi) and a *Gray* filter.
3. Click *Scan*. Stop the scan when the entire preview image is displayed. This quick scan will show you the position of the calibration strip.

4. Next, mouse out an area around the calibration strip. Make sure not to cut off any area of the strip.
5. Select a higher resolution (*64µ*)
6. Click on the *Calibration Scan* box, and click *Scan.*

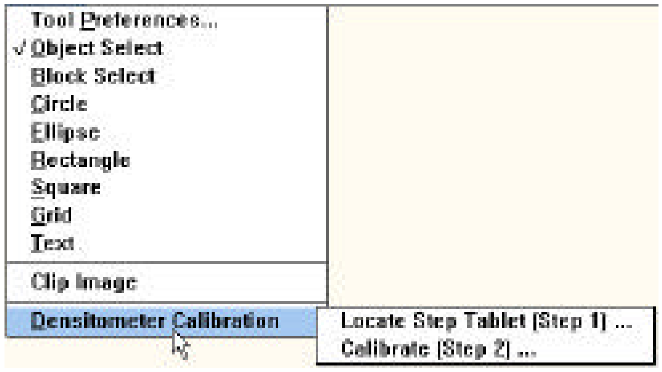
7. Once the scanning is completed, save the calibration image.

8. Next, close the scanning window and open the saved calibration image.

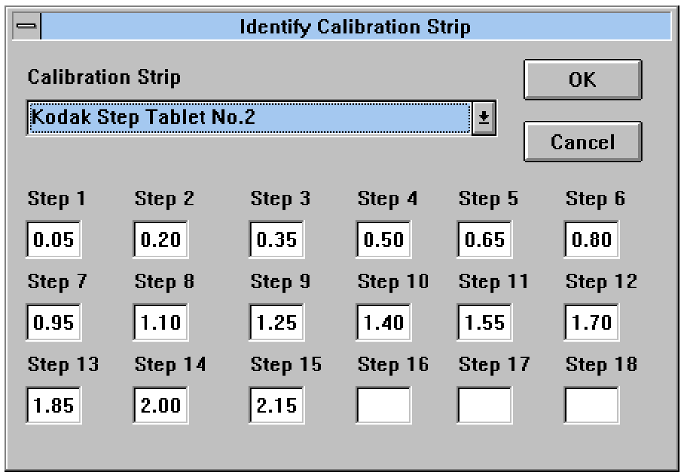
9. Next, select *Densitometer Calibration* from the TOOLS menu.



10. Select the command *Locate Step Tablet (Step 1)*.



This will display a *Identify Calibration Strip* dialog box.

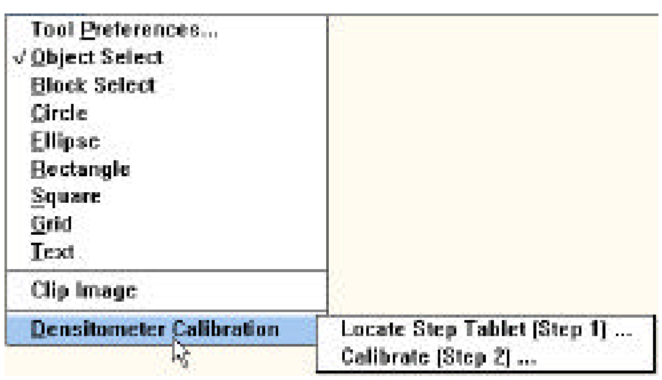


The most common O.D. values for the Kodak strip are provided as a default. However, the OD values for the calibrated step tablets can vary from tablet to tablet.

11. Enter in the values from the information sheet which accompanied the calibrated step tablet and click OK. Volume rectangles should be placed onto the image in the proper density sections. If the automatic detection fails, the step rectangles are still created on the image. The user can then manually move the rectangles over the correct section of the tablet.

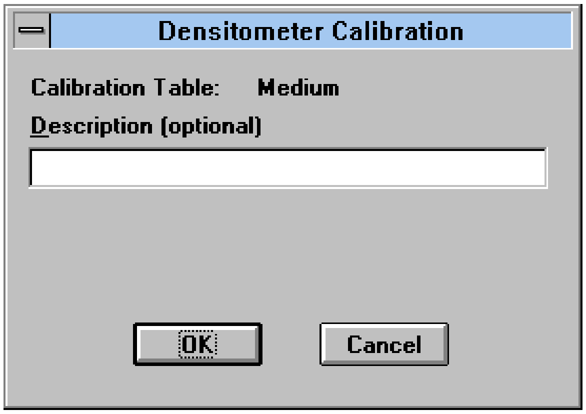
12. Once again select *Densitometer Calibration* from the TOOLS menu.

13. Select the command *Calibrate(Step 2)*.

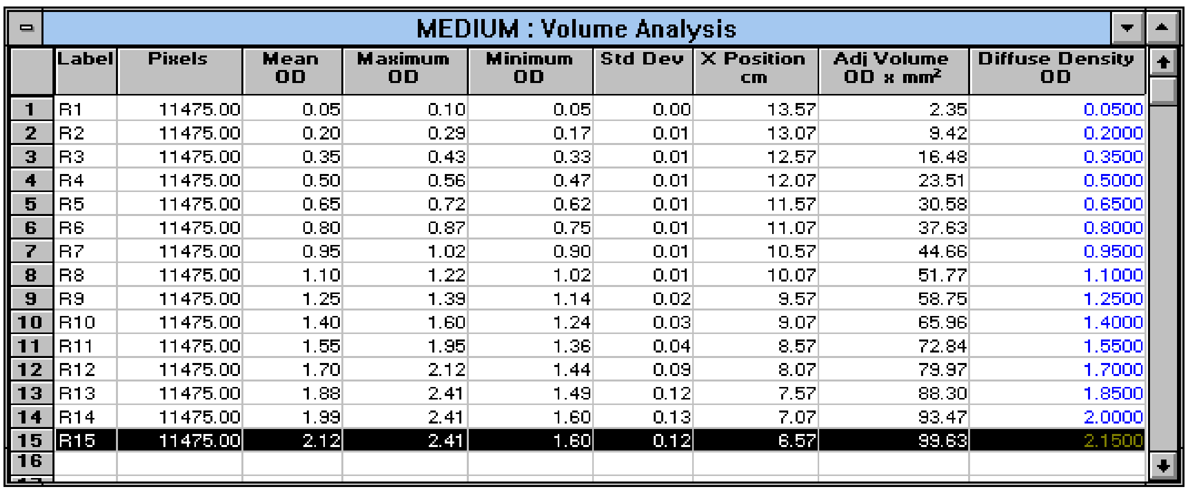


This will set volume objects onto the image and display a Densitometer Calibration dialog box.

14. Within the Densitometer Calibration Dialog box, Click *OK* to save the calibration table. Each table created is saved and is available in the Calibration pull-down menu in the scan window.



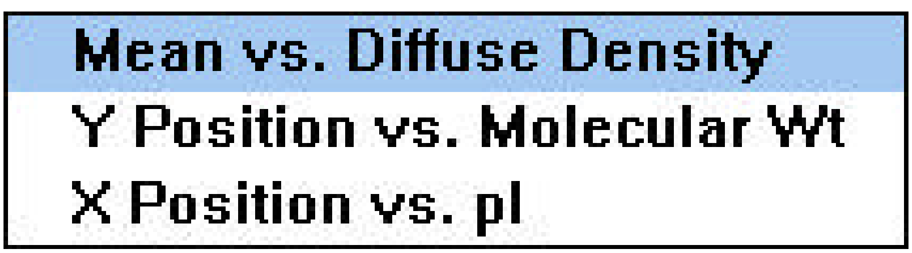
15. View the calibration report table by selecting Analyze Volume Objects from the ANALYSIS menu.

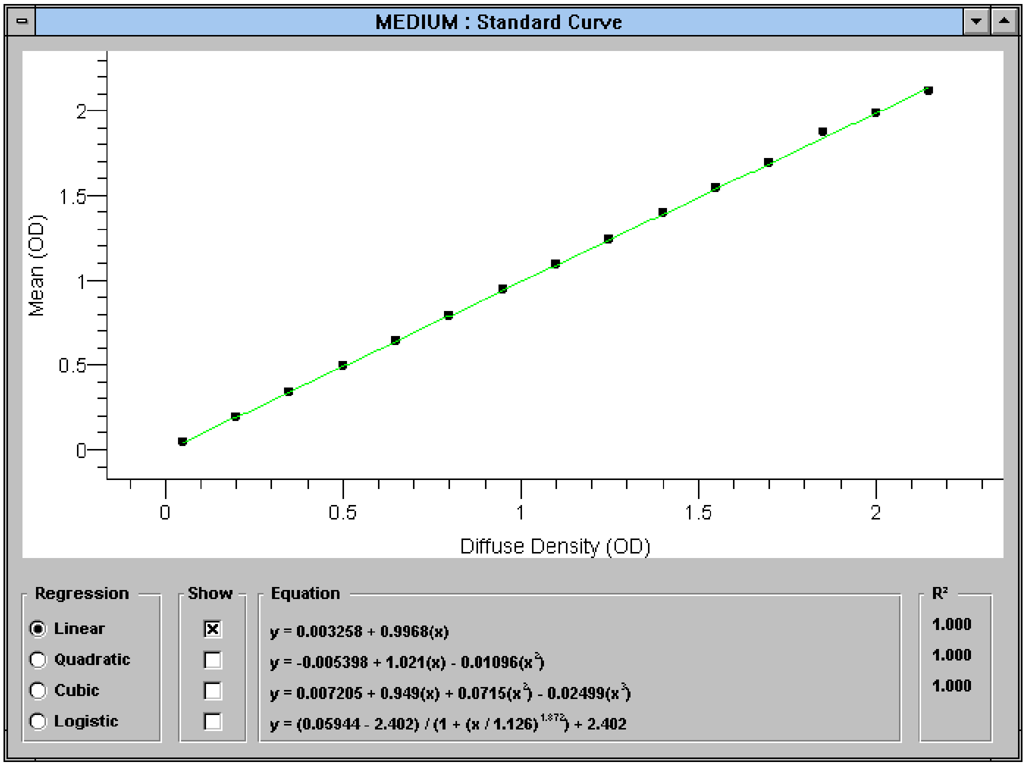


MEDIUM : Volume Analysis

	Label	Pixels	Mean OD	Maximum OD	Minimum OD	Std Dev	X Position cm	Adj Volume OD x mm²	Diffuse Density OD
1	R1	11475.00	0.05	0.10	0.05	0.00	13.57	2.35	0.0500
2	R2	11475.00	0.20	0.29	0.17	0.01	13.07	9.42	0.2000
3	R3	11475.00	0.35	0.43	0.33	0.01	12.57	16.48	0.3500
4	R4	11475.00	0.50	0.56	0.47	0.01	12.07	23.51	0.5000
5	R5	11475.00	0.65	0.72	0.62	0.01	11.57	30.58	0.6500
6	R6	11475.00	0.80	0.87	0.75	0.01	11.07	37.63	0.8000
7	R7	11475.00	0.95	1.02	0.90	0.01	10.57	44.66	0.9500
8	R8	11475.00	1.10	1.22	1.02	0.01	10.07	51.77	1.1000
9	R9	11475.00	1.25	1.39	1.14	0.02	9.57	58.75	1.2500
10	R10	11475.00	1.40	1.60	1.24	0.03	9.07	65.96	1.4000
11	R11	11475.00	1.55	1.95	1.36	0.04	8.57	72.84	1.5500
12	R12	11475.00	1.70	2.12	1.44	0.09	8.07	79.97	1.7000
13	R13	11475.00	1.88	2.41	1.49	0.12	7.57	88.30	1.8500
14	R14	11475.00	1.99	2.41	1.60	0.13	7.07	93.47	2.0000
15	R15	11475.00	2.12	2.41	1.60	0.12	6.57	99.63	2.1500
16									

16. To visualize the linearity of calibration as a graph, select Format Columns from the OPTIONS menu. Click on the *Next* button until “Concentration” is displayed.
17. Highlight the name “Concentration” and type in “Diffuse Density”.
18. Next, click on the *Std Curve Y* Axis pull-down menu and select Mean. Click on the OK button.
19. Finally, select Mean vs. Diffuse Density from the VIEW menu.



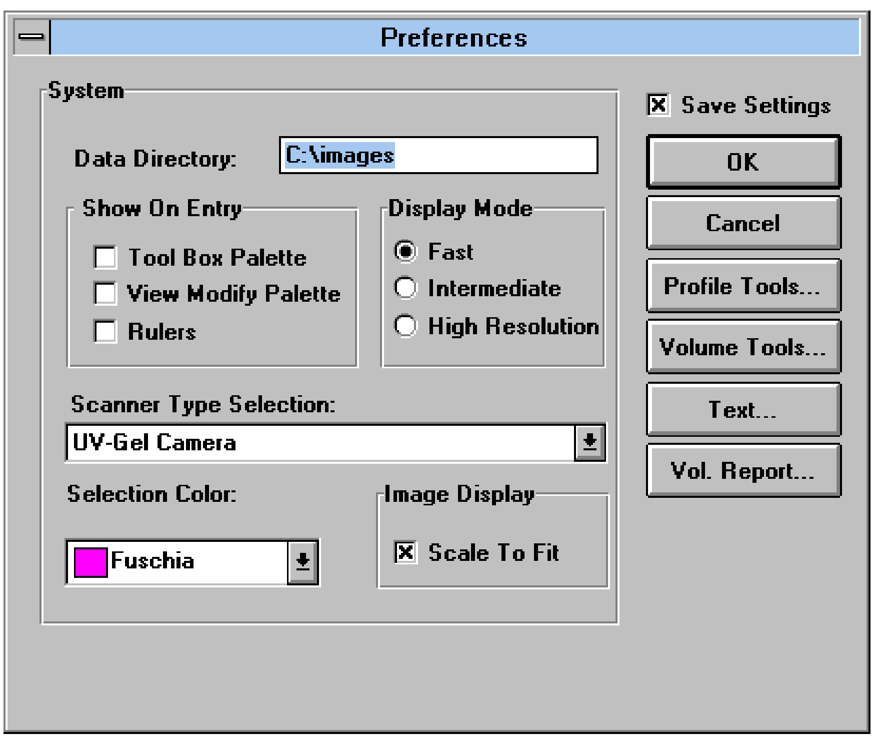


Scanning with the Gel Doc 1000

Setting Instrument Preferences

To prepare for scanning with the Gel Doc 1000:

1. Choose the *Preferences* command from the FILE menu.
2. Click on the pull-down menu “Scanner Type Selection”.

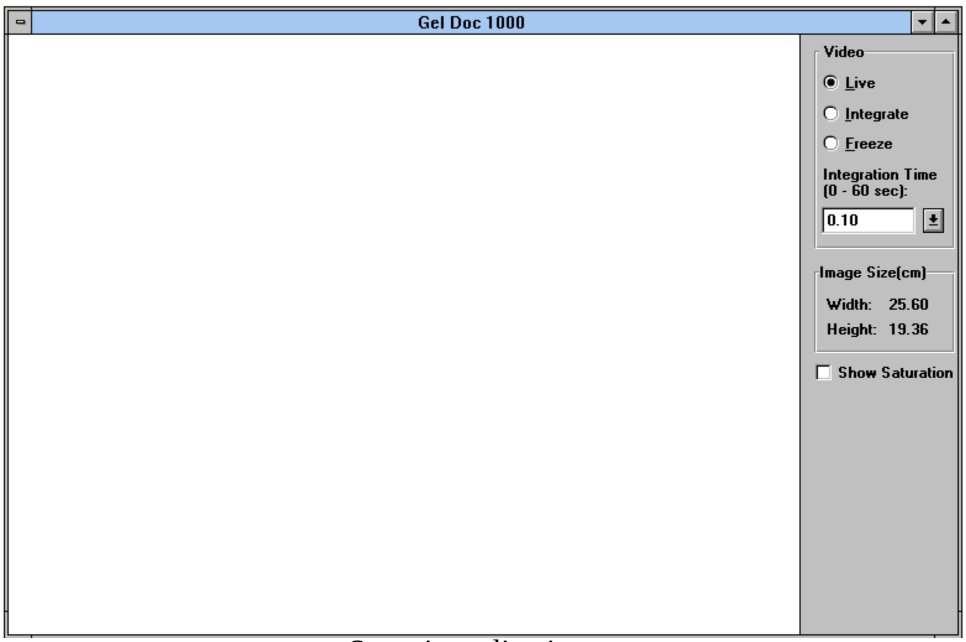


3. The UV-Gel Camera should automatically be highlighted. However, if using other Bio-Rad imaging systems, highlight “UV-Gel Camera” and click OK.

Capturing Live Images -

Click on the FILE menu, and drag the mouse to *New*. The scan window will automatically be in “LIVE” mode and will display a live video image allowing you to zoom and focus on the sample area. If *Show Saturation* is checked, all saturated values will be displayed in red. This feature is useful when quantitating bands on a gel. For accurate quantitation, reduce integration time or close down the aperature until all red saturated values are white.

If no image is visible, check cable connections from the frame grabber board to the camera and monitor. Also check that the transilluminator is on and working. The image should stay in focus while zooming from high to low magnification. If focus is not maintained, refer to Appendix J to correct the camera focus.



Scanning a live image

Setting Camera Parameters-

If no live image is displayed after checking the cables and video board, the video parameters may need to be modified. Click on the OPTIONS menu to set or to adjust the image capture and display options.

OPTIONS Menu

Live
This command displays an image in live (real-time) mode. This command accomplishes the same thing as the *Live* radio button in the scan window.

Integrate
This command displays an image in integrate mode. This command accomplishes the same thing as the *Integrate* radio button in the scan window.

Freeze
This command fixes the displayed image on the screen. This command accomplishes the same thing as the *Freeze* radio button in the scan window.

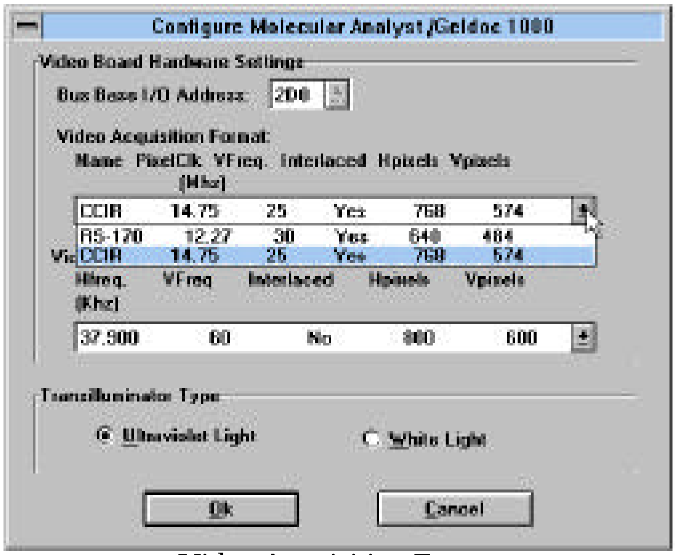
Scan Info
This command displays image information, including integration time, resolution, etc..

Video Board Configuration..
This command displays the video configuration of the system which was set when Windows® or Molecular Analyst were installed. The frame grabber supports video output of 800 x 600 or 1024 x 768, 256 colors with VGA pass through. **(Note: Any video board without a video pass through connector will not work!)** If using a Mitsubishi P68 video printer, please set the video display format at 800 x 600. The Mitsubishi P68 video printer will not function with a video display format of 1024 x 768. The settings for the Mitsubishi printer are located in Appendix H.

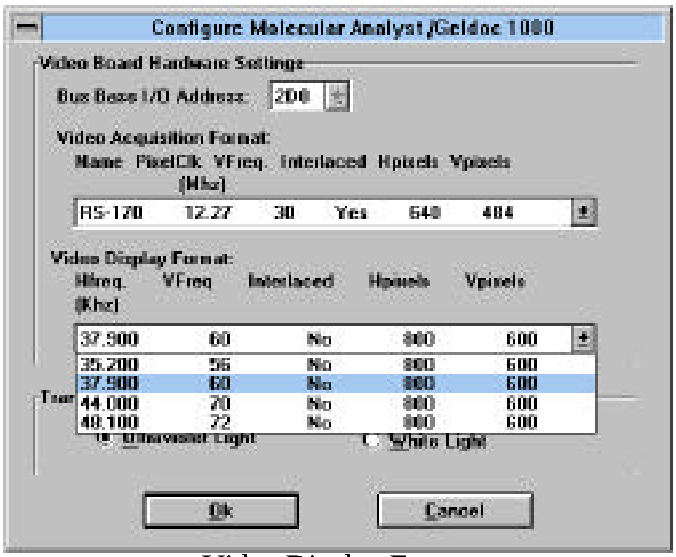
Within the Video Board Configuration dialog box, Base I/O Address (setup during installation), Video Acquisition, and Video Display Formats may be displayed and set

Video Acquisition Format describes either RS-170 (USA 120V-60Hz standard camera) or CCIR (International 100V/220V/240V-50Hz standard camera) systems.

Video Display Format describes the video frequency and display resolution of your computer system. Video frequency can be determined though your computers BIOS setup or video card configuration. The display resolution is defined within Windows setup. Resolution of 800 x 600 provides lower resolution than 1024 x 768 but displays larger fonts and menus.



Video Acquisition Format



Video Display Format

Set Range/Offset...

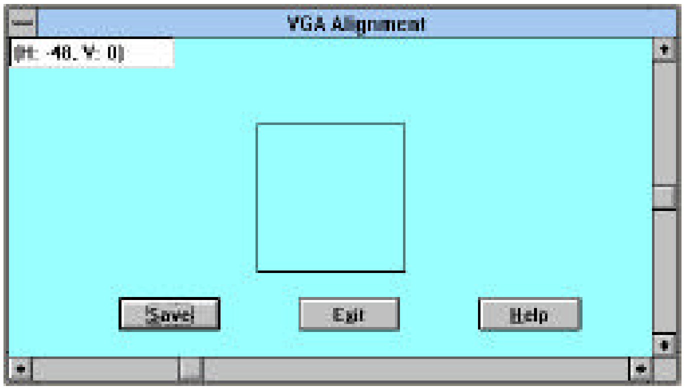
The effects of Range and Offset on the acquired image are similar to brightness and contrast controls on a video monitor. This command is rarely used because it alters the raw signal coming from the camera. Please use the default values for Range and Offset and adjust brightness and contrast with the modify view or transformation palettes.

Set VGA Alignment

This commands allows you to align the frame grabber video signal with the VGA display. If there is a large area within the scanning window (in *Live* Mode) that has no video signal, VGA alignment may be off. To set the alignment open the *VGA Alignment* window.

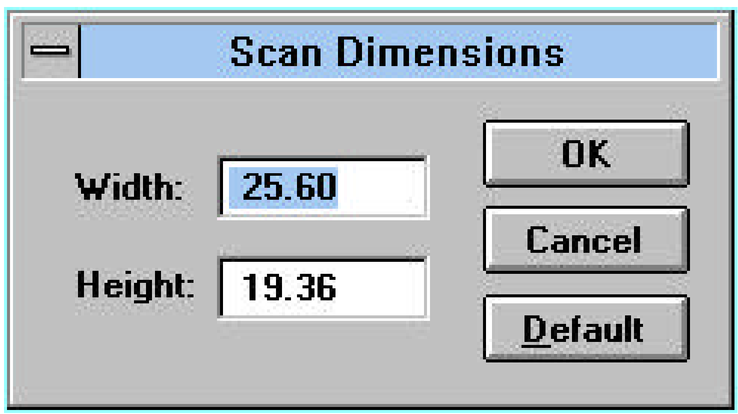
Use the scroll bars to align the two superimposed squares.

Once aligned, select *Save* to save changes for future use, and then select *Exit.*



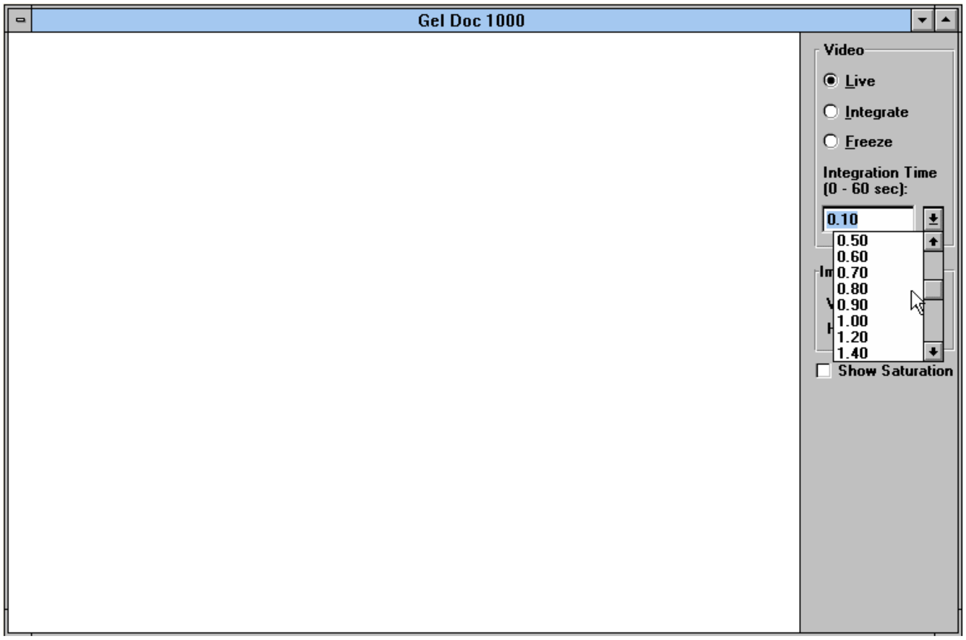
Change Image Size...

Image size always defaults to the largest possible viewing area (zoomed out.) For very accurate quantitation it is necessary to enter in the scan dimensions taken from imaging area. This is done by simply using a fluorescent ruler to measure the imaging area and entering the information. This information can also be entered after the image has been saved using the *Image Info* command from the VIEW menu. Selecting the *Options* button in the Image Info dialog box allows image size, light source (UV or White), and positive or negative image selection to be modified.



Integrating -

To identify faint areas of fluorescence, click on the “Integrate” radio button. Integration time is any number from 0.1 - 60 seconds.



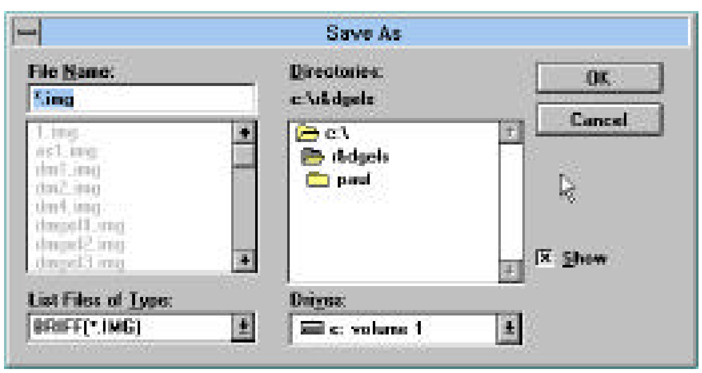
The time for integration always defaults to 0.1 second. To integrate for longer periods of time, click on the “Integration Time” pull-down menu and select the time for integration. Alternatively, to quickly change the integration time, highlight the integration number inside the *Integration Time* window (see 0.10 above) and enter another integration time using the keyboard.

Printing

Live or integrated images can be printed either to a laser or thermal printer. Select *Print* from the FILE menu.

Saving the Image

Save your image as an image file with an 8 character name and a .IMG extension (TEST.IMG) and close the scan window.



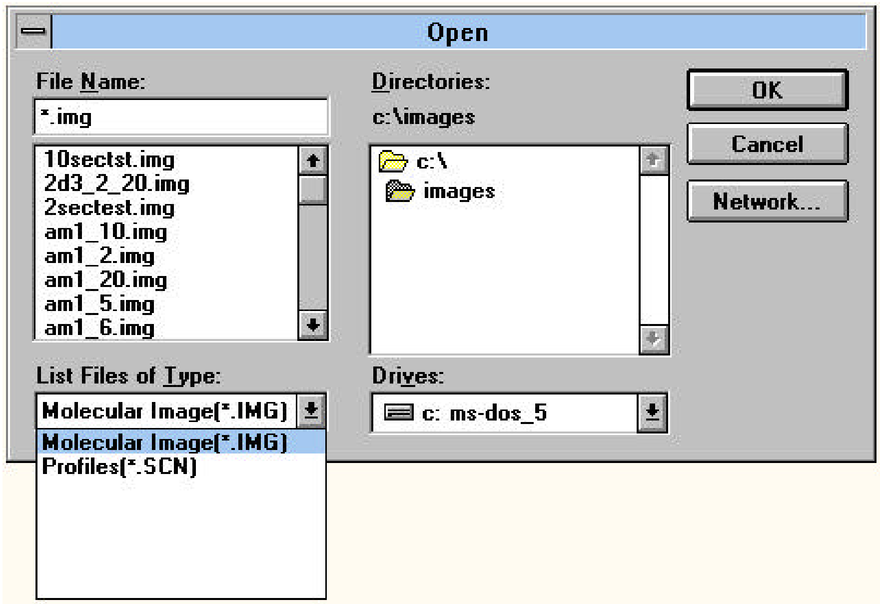
Deleting Images

To delete an image, quit the Molecular Analyst/PC program and open File Manager in Windows. Click on the MA folder, then click on the data folder to display images and profile files. Highlight the file(s) to be deleted by clicking on them with the mouse (hold the <Cntrl> key down while clicking the mouse button for deleting multiple files) and then select *Delete* from the FILE menu or simply press the <Delete> key. Click OK at the prompt.

2.2 Lesson 2: Optimizing Images

Starting the Molecular Analyst Program-
Start the Molecular Analyst/PC program by double clicking on the Molecular Analyst icon.

Displaying an Image-
To open the example image provided:
1. Click on the FILE menu, and drag the mouse to *Open*.
2. Find the directory where EXAMPLE image is located.
3. Double-click on the file name "EXAMPLE.IMG".



The image will then be displayed with metric rulers in the image window at 100% scale.

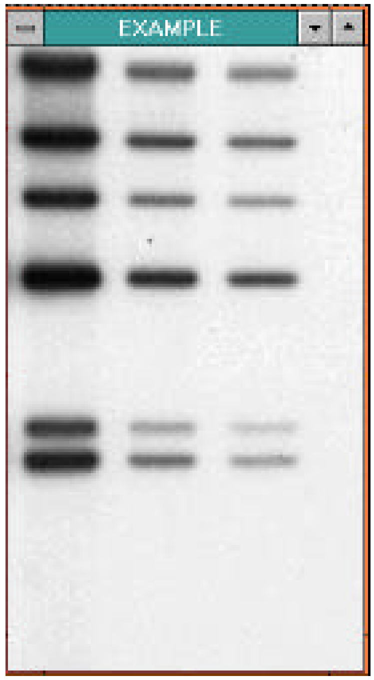
4. Hide the rulers using the *Hide Rulers* command under the VIEW menu.

5. Click on the VIEW menu and select *Scale to Fit* .

This command allows the image to always fill the entire display window as the display window is enlarged or reduced.

Magnifying the Image-

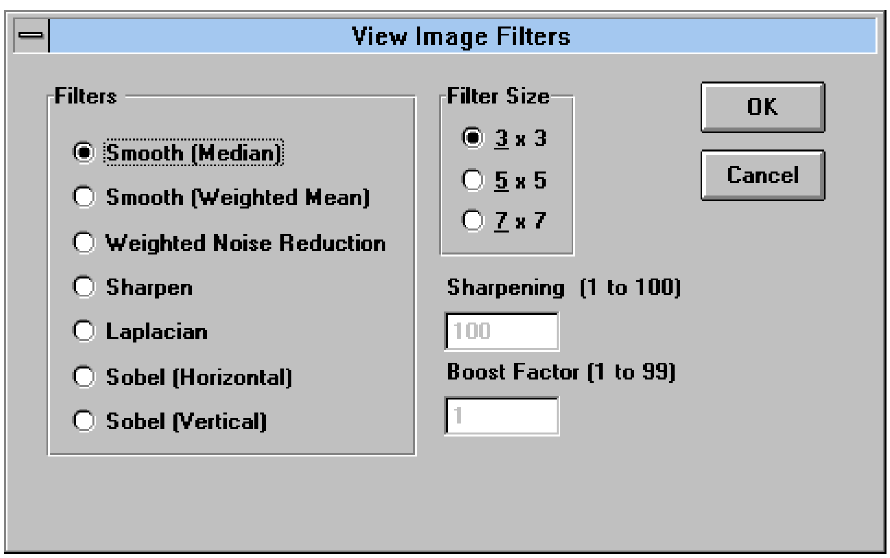
1. Click on the magnifying glass tool on the tool palette.
2. Place the magnifying glass on the image and click the left mouse button.
3. To undo magnification, hold the <Shift> key and click in the center of the image.
4. To see hidden areas of zoomed images, click on the scrolling cross icon in the tool palette.
5. While holding down the left mouse button, drag in any desired direction, and release the mouse button.
6. Return to the unmagnified image by selecting *Show All* from the VIEW menu.



Filtering background-

We will now smooth the image data to reduce background noise using one of the many filters available in Molecular Analyst.

1. Choose *View Filters* from the TRANSFORMATIONS menu.



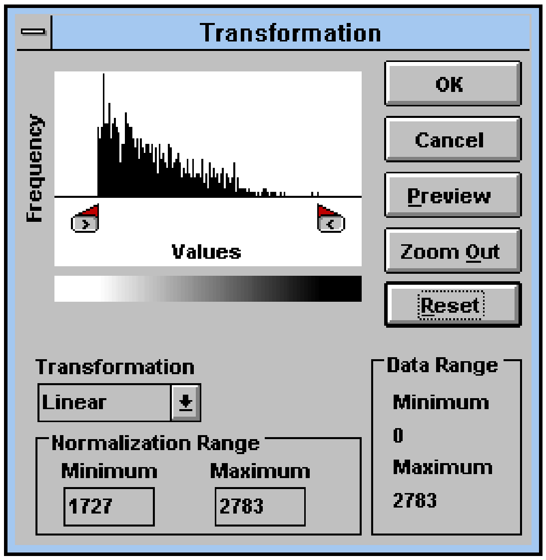
2. Select 3 x 3 kernel size for the Smooth (Weighted Mean) filter and click OK.

This is a low level filter which removes moderate background. The filtered image is then redrawn. View filters filter view data only and do not affect the quantitative nature of the image data; whereas raw filters filter raw data and will change the quantitative nature of the data (See Appendix I for additional information on enhancing image quality with filtering.)

Improving visual sensitivity- Transformation

Transformation provides a first level sensitivity adjustment to select areas of the image. Using the transformation histogram palette, it is easy to adjust gray levels or change the type of transformation to allow sensitivity adjustments to select areas of the image.

1. From the TRANSFORMATION menu, select *Transform*. This will display the transformation histogram palette.



2. Next, move the right and left tab markers to the right and left ends of the histogram plot.

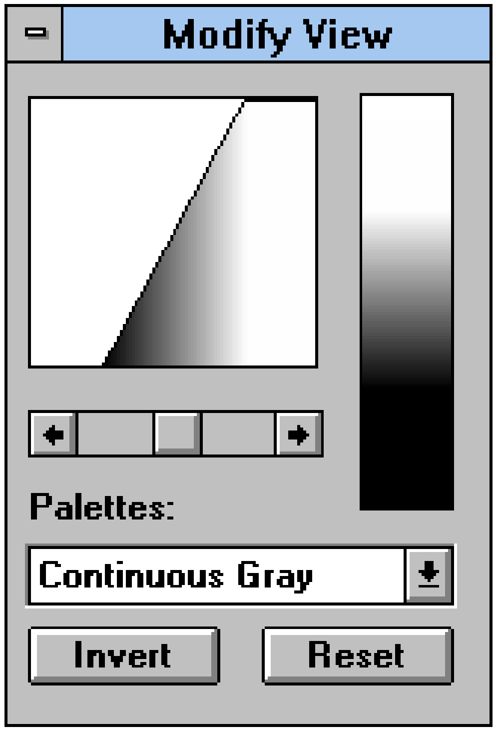
   Notice that the minimum and maximum pixel values, and the histogram plot change as the tabs are moved. Moving the tabs to the areas of the histogram plot, distribute the available gray levels to the image areas and not to background.

3. To view changes to the image, click on the *Preview* button.
4. To modify the histogram selection, select the *Zoom Out* button and move the tab markers to a new location.
5. Click on the *Preview* button to view the modifications.
6. To apply changes to the image, click on the *OK* button.

   The image will be redrawn according to the range selected.

7. To return to the original histogram plot, click on the *Reset* button and then select *Preview* and *OK*. Clicking on the *OK* button will close the histogram palette. (See Section 4.7 and Appendix I for further information on the Transformation Tool.)

Improving visual sensitivity- Modify View Palette

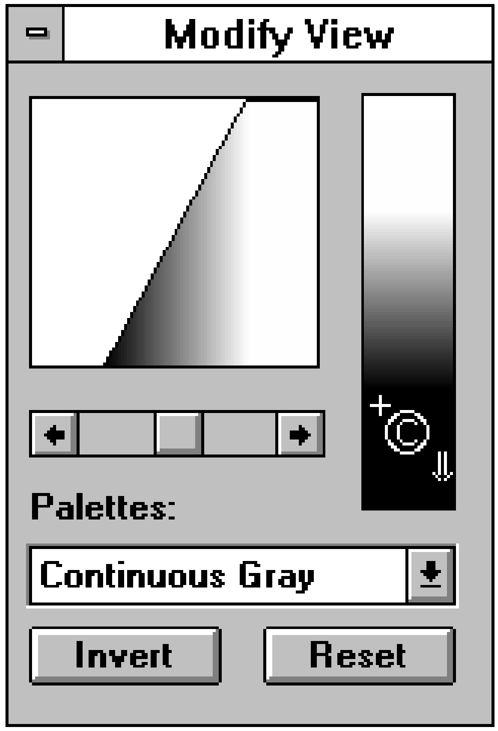


Using the Modify View palette’s equalization scroll bar, we can further enhance specific bands within the image increasing the distinction between the bands and the background.

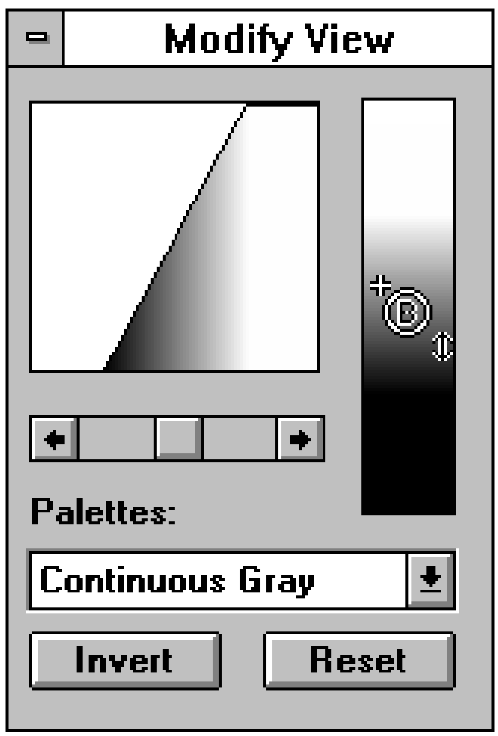
1. Use the scroll bar to visually enhance most of the bands of lane 1 and 2 on EXAMPLE.IMG.

We will now adjust the contrast of the image using brightness and contrast tools. The brightness and contrast tools allow even further image enhancement and band to background distinction by compressing (Contrast) and shifting (Brightness) the 256 gray levels.

2. Place the mouse cursor at either the top or bottom end of the gray scale window until the cursor changes to a *Circled C.*
3. Hold the mouse button down and move this contrast tool up vertically, adjusting the contrast of the image to the point were all bands in lanes 1 and 2 are clear and distinct.



4. To adjust brightness, place the cursor inside the continuous gray or color scale. The cursor will change to a *Circled B.*
5. While holding down the left mouse button, slide the brightness tool vertically until the desired brightness is achieved.



Note that the brightness tool is only functional after image contrast has been modified and gray scale LUT compressed.

6. To undo all modifications made with the Modify View palette, use the mouse to select *Reset.*

Saving An Image-

1. Save this image by selecting *Save As* from the FILE menu.
2. Type in EXAMPLE1.IMG and click <OK>.

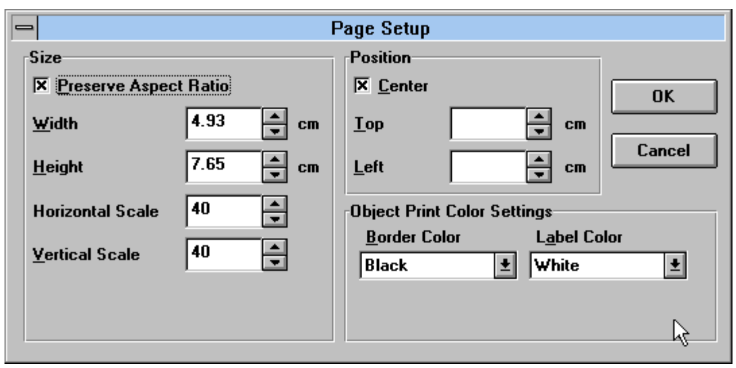
All of the modifications made to the image will be saved as a (*.IMG) file. This Bio-Rad Image File Format (BRIFF) is the default. Save all images to be analyzed by Molecular Analyst/PC in the BRIFF format.

Printing An Image-

1. Hard copy output of the optimized image can be obtained by selecting *Print* from the FILE menu, or by pressing *Print* on your video printers front panel.

Laser printer output will only display transformation changes. Modify view palette changes are not printed.

2. The *Page Setup* command within the FILE menu allows the size and position of the image to be changed.



Additionally, print outs to thermal printers can be optimized by:

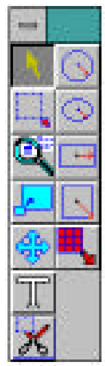
1. Selecting *Show Full Screen* from the VIEW menu.
2. The display is further modified when the command *Scale to Fit* is off.

The type and quality of the output will depend upon the particular printer system. Molecular Analyst/PC will support most Windows 3.1 compatible printers. If you are using a film recorder to generate prints or slides, please refer to your film recorder instruction manual.

2.3 Lesson 3: Volume integration

Creating Volume Integrating Objects-

1. Call up the image, EXAMPLE1.IMG.
2. From the ANALYSIS menu, select Volume Analysis. The volume tools will appear on the right side of the tool palette.
3. Click on the rectangle volume tool.



Notice that when you click on the rectangle volume tool, and place the cursor on the image window, the pointer disappears and is replaced with a rectangle with a cross-hair in the upper left corner

4. Using the mouse, move the cross-hair cursor to a point above and to the left of the first prominent band of the first lane of data in the image.
5. Click the left mouse button, and drag the mouse downward and to the right. A rectangle will begin to appear on the graphics screen. Moving the mouse adjusts the size of the rectangle so that it completely encloses the band.
6. Release the mouse button and a Fuschia rectangle will appear with a center dot enclosing the band. (Active square and rectangle objects have center dots. Active circle and ellipse objects will be surrounded by squares).
7. To adjust the size of the rectangle, place the cursor on a side of the rectangle you wish to adjust. Notice that the cursor now becomes a sizing square tool.
8. Using this sizing square tool, adjust the size of the rectangle appropriately so that the band is properly enclosed.
9. Next, place the cursor on a point inside the rectangle until the cursor changes to a curved arrow.
10. Click and hold down the mouse button. Using this positioning arrow, move the rectangle if necessary to enclose all of the band.

Duplicating Volume Integrating Objects-

1. Now duplicate the rectangle object by placing the cursor within the rectangle (but not on the center dot) changing the cursor to the movement arrow.
2. While holding the <CTRL> key down, click and hold the mouse button to duplicate the rectangle.
3. With the movement cursor still present, click and drag the duplicated rectangle to the next band in the lane and release.
4. Continue these steps until the first four bands are enclosed.

Identifying Background Objects-

Identifying Global Background:

Ideally, a background object should be placed above and below the band or spot you wish to quantitate. This type of background subtraction is called general or global background subtraction and is useful when background is uniform.

Note that when using global background subtraction, results are more accurate when objects are drawn close around image bands or spots.

1. Move the cross-hair of the rectangle tool on an area of background.
2. Click the left mouse button, and drag the mouse downward and to the right to create another rectangle to be used as background.

Many objects on an image can be used to determine an average background value. The size of a global background object is not critical because background volume is calculated as the mean pixel density within the object regardless of shape or size.

Identifying Local Background:

Alternatively, local background subtraction of each individual band is also available. This type of background subtraction is useful when background is non-uniform. No background object is necessary for this type of background subtraction. Local background subtraction is completed after the results table is generated.

The command *Local Background Subtraction* is active from within the OPTIONS menu after the Volume Integration Report is generated and displayed. The *Local Background Subtraction* command subtracts the mean pixel value around the perimeter of each volume object. When using local background subtraction, results are more accurate when objects are drawn farther from image bands or spots. This command affects all volume objects displayed on the image.

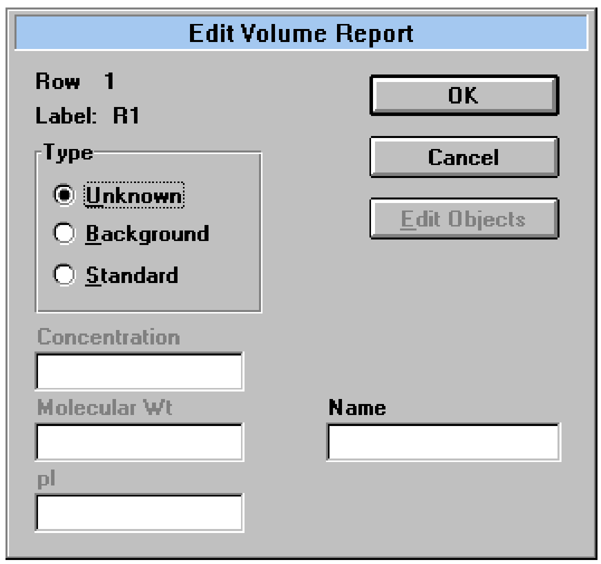
Generating a Volume Integration Report-

1. Select *Analyze Volume Object List* from the ANALYSIS menu to display the volume report for the objects you created.
2. Next, click on the OPTIONS menu and select *Sort by Creation Order*.
3. Select *Format Columns* from the Options menu. This will display the *Volume Report Preferences* dialog box. Display all columns of interest.
4. To hide columns, click on the *Hide* check box. Check the *Save Settings* box and click OK to return to the volume report.

Editing Volume Integrating Objects-

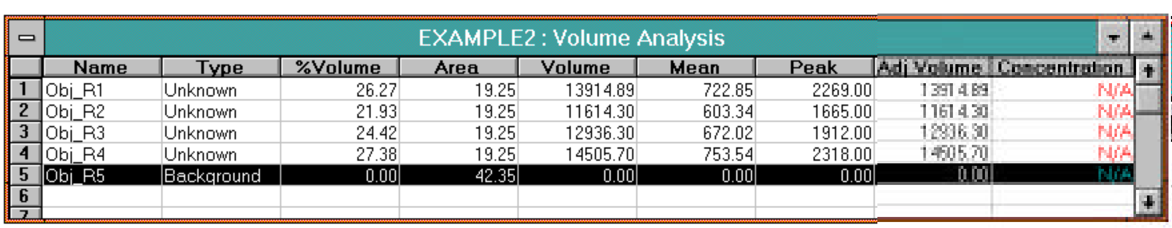
Within the volume report are rows corresponding to Object 1 through Object 5.

1. Click on any object in the image and notice that the corresponding row within the volume integration report becomes highlighted. **Active objects in the image correspond to highlighted rows in the volume integration report.**
2. Double click on the background object. This will open an *Edit Volume Report* dialog box.



3. Click on the radio button labeled *Background*.
4. Click OK to return to the volume report.

Notice that the name in the *Type* column for Object 5 has been changed from *Unknown* (the default) to *Background* within the volume report. Background values will be automatically subtracted from the unknowns and standards to provide *Adjusted Volume* data. This is global background subtraction.


EXAMPLE2 : Volume Analysis

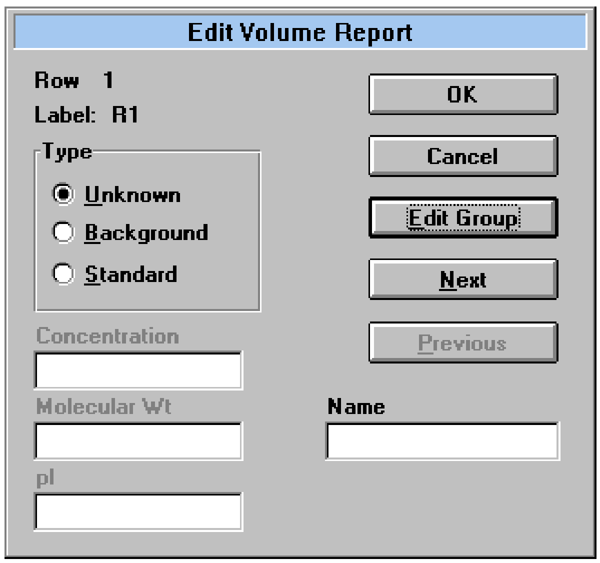
	Name	Type	%Volume	Area	Volume	Mean	Peak	Adj Volume	Concentration
1	Obj_R1	Unknown	26.27	19.25	13914.89	722.85	2269.00	13914.89	N/A
2	Obj_R2	Unknown	21.93	19.25	11614.30	603.34	1665.00	11614.30	N/A
3	Obj_R3	Unknown	24.42	19.25	12936.30	672.02	1912.00	12936.30	N/A
4	Obj_R4	Unknown	27.38	19.25	14505.70	753.54	2318.00	14505.70	N/A
5	Obj_R5	Background	0.00	42.35	0.00	0.00	0.00	0.00	N/A
6									

When using multiple background objects to average background values,

1. First activate all of the objects surrounding background by holding down the <Shift> key and clicking on each background object.
2. With all of the background objects selected, double click on one of the background objects to display the *Edit Volume Report* dialog box.
3. Clicking on the radio button labeled *Background* will change all of the objects to background.

Assigning Standards-

1. Activate all of the objects surrounding standards by holding down the <CTRL> key and clicking on each object R1, R2, and R4.
2. With all of the standard objects highlighted, select <ENTER> or select *Modify* from the OPTIONS menu to display the *Edit Volume Report* dialog box.
3. Click on the *Edit Volume Report* command button. The Edit Volume Report dialog box will be modified to the following



This command will now let us enter in the standard values for each of the standard values sequentually.

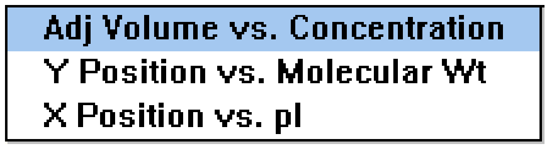
4. Click on the radio button labeled Standard and enter in the value .90 in the *Concentration* box. (Concentration values can not be entered unless the corresponding concentration column is visible. This also applies for molecular weight and pI determination).
5. Click on the *Next* command button.
6. Repeat this procedure for Object 2 and Object 4, entering concentration values of .70 and .60 respectively.
7. Click on OK.

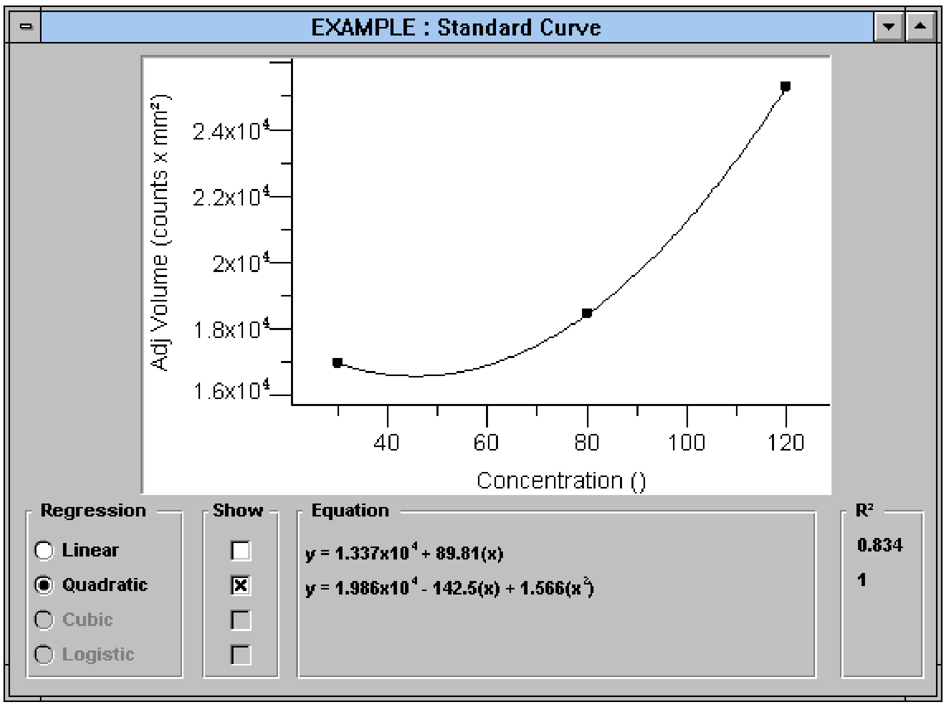
When finished, return to the volume report. Notice that within the Type column of the volume report is one unknown, one background and three standards. Additionally, notice that the background volume has been subtracted from the unknown and standard volumes, generating *Adjusted Volume.*

Generating a Standard Curve and Calculating Concentration-

To generate the standard curve(s) and calculate the value of the unknowns:

1. Click on the VIEW menu and select the command *Adj Volume vs. Concentration.* This will display the plot of the standard curve(s)



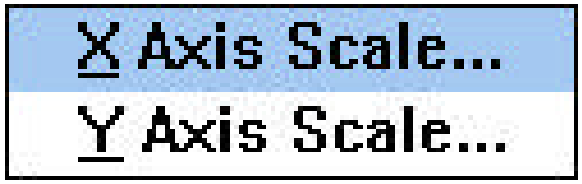


Notice within the standard curve is a legend with linear, quadratic, cubic, and logistic curve boxes. Linear Regression requires a minimum of two (2) standard data points, Quadratic Regression requires a minimum of three (3) standard data points, and Cubic Regression analysis requires a minimum of four (4) standard data points. Four Parameter Logistic Regression analysis requires a minimum of five (5) standard data points. Given 3 standard data points, Linear and Quadratic curves will be displayed. Clicking on each box will display or hide the curves fit.

2. Click on the box to select the curve that best fits the data and to calculate the concentration of the unknown for a specific regression curve. This will also display the regression formula and standard error for the standard curve.
3. Position the standard curve display and volume report so that both are visible.

The X and Y axis also allow for linear and logarithmic functions.

4. With the Standard Curve window displayed, click on OPTIONS menu.
5. Select the X and Y axis for display by clicking on the appropriate box.



Exporting Volume Data to Other Applications-

The data on the volume report can be exported to any number of spreadsheet or desktop applications which accept data from Windows clipboard. To export data from the volume report to the Windows clipboard:

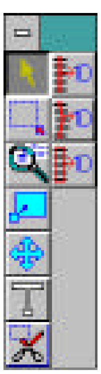
1. Simply highlight the data to export
2. Click and drag the cursor down the volume report table to highlight more than one sequential row.
3. Use the <Ctrl> key and click to highlight non-sequential groups of cells
4. Use the *Copy to Clipboard* command or the *Copy* command in the EDIT menu. This will copy the volume report values to clipboard.
5. Next, open up the application you wish to export to.
6. Use the Paste or Paste Special function to transfer the volume results from clipboard to the new application.

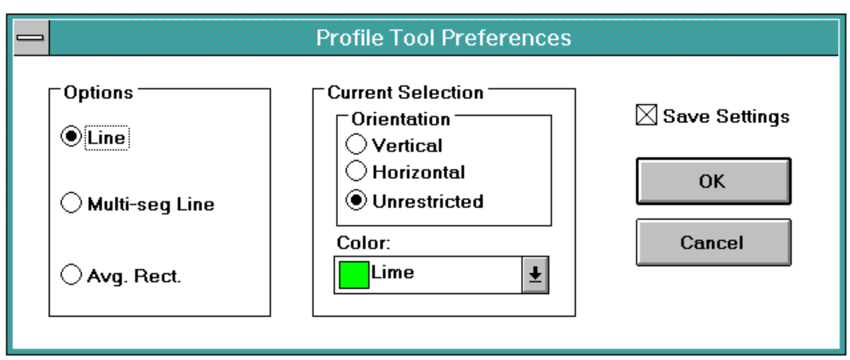
Using the <Alt>+<Tab> keys will allow you to rapidly move between Molecular Analyst and other open windows such as Windows Program Manager. Note that the volume report results are transferred in ASCII tab delimited format.

2.4 Lesson 4: Peak integration

Creating A Profile Object-

1. If not already displayed, open the image “EXAMPLE1.IMG”. **Note that profiles are generated from the raw or filtered data.**
2. From the ANALYSIS menu, select *Profile Analysis.* The profile tools will appear on the right side of the tool palette.
3. Double click on the first profile tool (single line tool) to bring up the *Profile Tool Preferences* dialog box.
4. Choose the following preferences: Line Option with Unrestricted Orientation.
5. Click on *Save Settings* box and then click on OK.

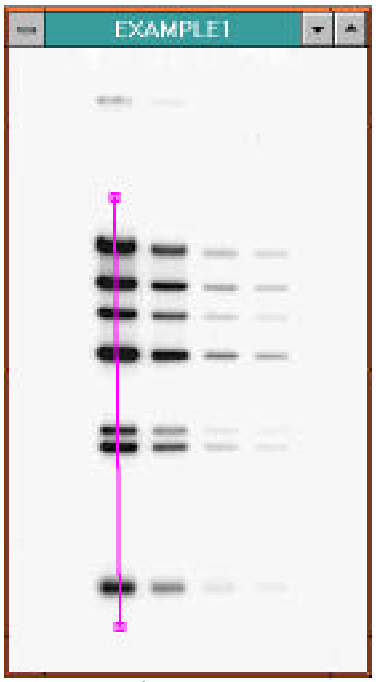




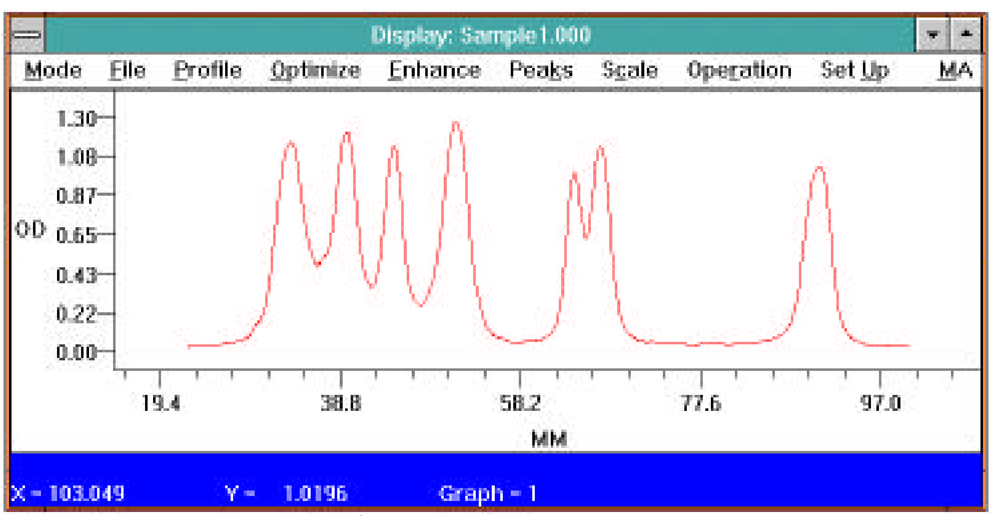
Tool Palette with Profile Tools. Profile Tool Preferences

6. To create a profile object for area analysis, place the line cursor above the first band over the center of the first lane of data.
7. Click the mouse button and drag the mouse down until the line has completely passed through each band on the lane, and release the mouse button. A Fuschia colored line with squares on each end will appear. This is the active object and will generate the active profile.

8. Next, select the pointer tool from the tool palette and then double click on the line object. A *Profile Name* dialog box will appear.
9. Name the profile SAMPLE1.
10. To display the new profile, go to the ANALYSIS menu and select *Analyze Profile(s).* A display screen will appear with the profile name SAMPLE1.000 and the menu commands for Profile Analyst



Example 1 with active profile line passing through lanes



Profile Analyst display screen

Duplicating a Profile Object-

You will begin to optimize the profile by subtracting out sample background data. First, generate a background profile that is the same length as the sample profile, by duplicating the profile object used to create the sample profile.

1. To do this, return to Molecular Analyst by clicking on the MA menu located in the upper right corner of the Profile Analyst window.
2. Now place the cursor on the center of the line used to create the sample profile. Notice that the cursor has changed to the curved movement arrow.
3. Next, while pressing the <Ctrl> key, click and hold the mouse button and drag the duplicated object to a blank area next to the sample lane. This new object is now the active object.
4. Double click on this new line and enter the name BACKGROUND in the *Profile Name* dialog box.
5. Display the new profile by selecting *Analyze Profile(s*) from the ANALYSIS menu. Another profile graph will appear with the new profile labeled BACKGROUND1.000 underneath the original SAMPLE1.000 profile. Note that the background profile is the active profile as indicated by its Fuschia color.

Setting and Saving Profile Type

Before the background profile can be subtracted from the unknown profile, the profile type must be changed and then saved.

1. With the background profile active, select the PROFILE menu and click on the *Set Type* command.
2. Within this menu, identify the profile as *background* by clicking on the background radio button.
3. Click on the *Set* command.
4. Next, from the PROFILE menu, select *Save.*

Subtracting Background

You will now subtract the background profile from the unknown profile.

1. First, make the unknown profile labeled SAMPLE1.000 active by double clicking on the profile bottom.
2. Click on the *OPTIMIZE* menu and select the *Sub Background* command from the pull-down menu. A background profile information dialog box will appear containing a list box of the background profiles contained within the EXAMPLE1.SCN file.
3. Click on BACKGROUND1.000-a. Notice that the "Processed" and "Background" radio buttons are highlighted signifying that the profile is a processed (ie. saved) background profile.
4. Now point and click on the *SUBTRACT* command button. The program will display the resulting profile with the background removed.

Note that the *Sub Background* command is only one of three commands that eliminate background data, each of which has advantages.

Optimizing a Profile

You will now trim the profile at the origin (where loaded sample meets resolving interface) and the front (location where dye front stopped) to determine the Rf value.

1. Click on the *OPTIMIZE* menu and select the *Origin* command. The cursor vanishes and a pair-of-scissors cursor appears on screen.

2. Using the mouse, move the scissors to just right of the origin at approximately X = 24 and click the mouse button to clip the profile at the origin.

3. **Make a note of the exact X coordinate position of the origin at the bottom left corner of the screen to be used later in Lesson 3: Molecular Weight Determination.**
4. Click on the *OPTIMIZE* menu and select the *Front* command.
5. Move the scissors to just left of the right last peak in the profile at and click the mouse button to clip the profile at the front line.
6. **Note the exact X coordinate position of the front at the bottom left corner of the screen to be used later in Lesson 3.**

You will now zoom in on the section of the profile of interest using the *Zoom* command.

7. To *Zoom*, press and hold-down the **right** mouse button. A small, pull-down menu appears.
8. Drag to select the *Zoom* command from this menu and then release the right mouse button. Notice that the cursor has become a magnifying glass.
9. Use the *Zoom* command by positioning the magnifying glass above and to the left of a portion of the profile that interests you.
10. Press the left mouse button and drag to box the desired area; release the mouse button. The boxed area now appears at higher magnification. The zoom function is completely variable allowing you to zoom in as much as you wish.

To return to the previous, unmagnified profile:

11. Press the right mouse button and select the *Previous* command from the pull-down menu that appears.
12. Again, press the right mouse button and select *Zoom*. This time we will magnify the whole of the profile so that a baseline can be readily constructed.
13. Position the magnifying glass cursor slightly above the highest peak, and to the left of the left most end of the profile.
14. Press the left mouse button and box the entire profile.
15. Release the mouse button to execute the *Zoom.*

By zooming we have performed a "view modification". A view modification is a modification which affects only the display, or viewing of the data, but does not actually modify the data itself. When a view modification has been done, any operation performed at this point will still affect the whole profile, not just the section which is currently in view. In order to just work with the zoomed section of the profile we must perform a "data modification". A data modification is a modification which not only affects the viewing of the data, but actually transforms the data itself. All operations performed after the data modification will only affect the portion of the profile currently in view.

16. To work with only the zoomed portion of the graph, point and click on the *OPTIMIZE* menu and select *Snip at View*.

Smoothing the Profile

The profile has been optimized, and now we will reduce high frequency background noise in the profile using the *Smooth* command. Eliminating high frequency noise is required so that the peak detection algorithm can properly identify the peaks of interest.

1. Set the smooth criteria by clicking on the *ENHANCE* menu and selecting the *Criteria* command. The Enhance/Smooth Criteria dialog box appears; the cursor prompting for window size.

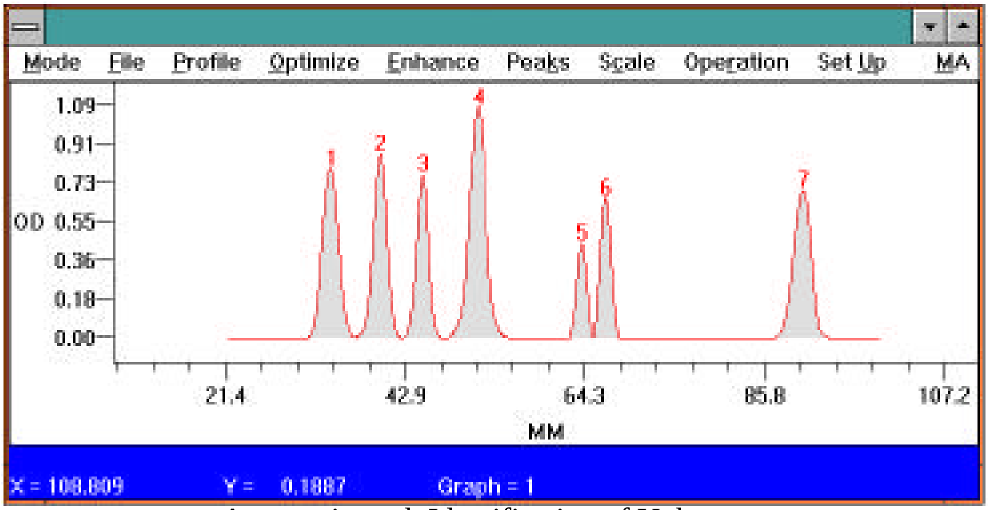
2. Select a large (10-99) window size, if you believe that your data contains a lot of noise and, therefore, needs a lot of smoothing.
3. Select a small (3-10) window size, if you believe that your data needs minimal smoothing,
4. Enter a window size of 9 in the window size edit field.
5. Next, <Tab> or move with the mouse to the Iterations Value field. The iteration value specifies how many times the smoothing will be executed. For the purposes of this tutorial, enter 3.
6. The enhancement boost factor (*En Boost*) is a constant which is multiplied by high spatial frequencies extracted from the data. Enter a value of 3. *En Boost* is discussed in further detail in Section 5, ENHANCE menu.
7. Click on the Display Each check box so that each smoothing will be displayed on screen.
8. Set these criteria by pointing and clicking on the *Set* command button.
9. Finally, point and click on the *ENHANCE* menu and select the *Smooth* command. The profile will be smoothed and re drawn after each iteration.

Identifying And Integrating Peaks

Now you will identify the peaks.

1. Using the cursor and the X and Y coordinates displayed at the bottom of the profile graph, determine the amplitude or height of the smallest peak.
2. Next, click on the *PEAKS* menu and select the *Peak Criteria . . .* command. The Peak Criteria dialog box appears. In it, you establish the criteria by which you define peaks.
3. In the Amplitude field, enter in the value below that of the smallest peak of interest you identified with the cursor. The other criteria, Peak Width (the minimum width for a peak) and Area (the minimum area for a peak) can be used to further find peaks.
4. To set these parameters, click on the *Set* command button. The other criteria fields are discussed in Section 5, PEAK menu.
5. Now, to identify and integrate the peaks allowed by your criteria, click on the *PEAK* menu and select the *Auto Identify* command. The area under the curve for the peaks which meet the specified criteria is shaded, and drop lines identify the beginning and end of each peak. The peaks are numbered from left to right starting with peak number 1.

Note that if a peak at either end of the profile was not identified, the front and/or origin of the profile was probably clipped to close to the peak. If the end peaks are not identified, return to the original profile by selecting the *Original* command from the PROFILE menu and repeat the above procedures (subtract background, clip front and origin, smooth, and identify peaks). If peaks are still not identified, lower the amplitude value set in the *Peak Criteria . . .* command.



Automatic peak Identification of Unknowns

Information on Identified Peaks

For quick information about **identified** peaks:

1. Go to the *PEAK* menu, and select the *Peak Information* command. A hand-shaped-cursor appears.
2. Move the hand to the area under any identified peak and click the mouse button. Read that peak's information, including area, from the table that appears.
3. To close the *Peak Information* box, click on the *OPERATION* menu and select the *Cancel* command.

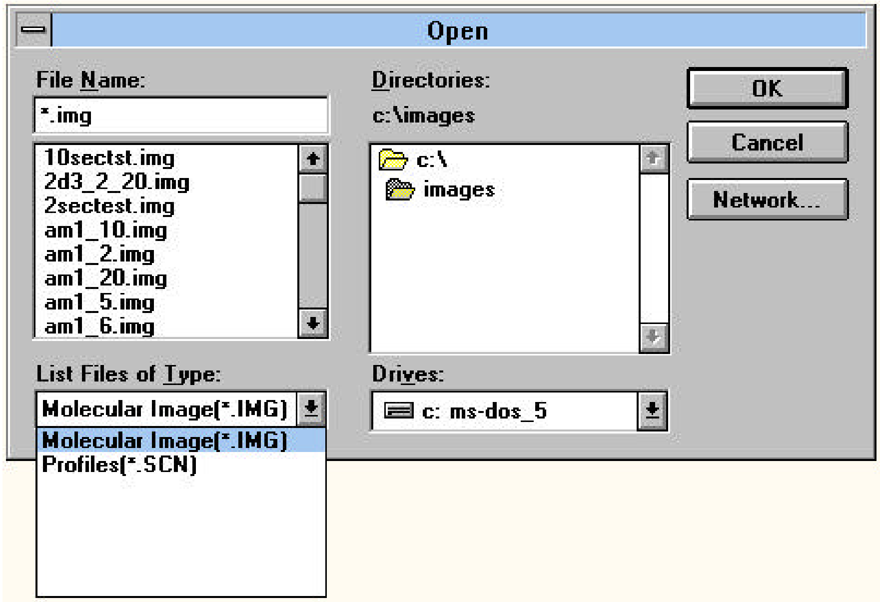
Information on Unidentified Peaks

For quick information about peaks that have **not been identified**:

1. Click on the *PEAKS* menu.
2. Select the *Integrate Interactive* command. The cursor vanishes, replaced by an integral-sign-cursor.
3. Move the integral-sign to the beginning of a significant peak.
4. Press the left mouse button and drag to the end of the peak. Note that the area of the peak appears in the lower right hand corner of the screen. These area values are volatile and will disappear as soon as another peak is integrated, or the command is quit.
5. To quit *Integrate Interactive*, click on the *OPERATION* menu, and select the *Cancel* command.

Saving The Profile

All of the profiles from the image, EXAMPLE1.IMG, will be saved to a file named EXAMPLE1.SCN. Profile file names have the image name and the suffix (.SCN) It is important to save profiles after peak identification or the modifications will be lost. To do so, point and click on the PROFILE menu and select the *Save* command. *Save* stores the active profile, with all modifications for future use. *Save* also tags the profile name, to differentiate the processed profile from the original. For example, SAMPLE1.000 will be saved as SAMPLE1.000-a. A list of previously saved profiles may be opened within Molecular Analyst by selecting the *Open* command from the FILE menu; and selecting Profile Files(*SCN) from the *List Files Type* pull-down menu.



Double clicking on the profile file name will generate a list of profiles within that file. Clicking on the profile name and then clicking on the Add button will generate the profile in the DISPLAY function of Profile Analyst. The same saved profiles can be similarly opened when in the Profile Analyst module by selecting the *New* command from the PROFILE menu.

Labeling Profiles-

Text can be added to profiles

1. Select the *Profile Label* command from the PROFILE menu.
2. Type “Bio-Rad Tutorial Sample1” in the profile label dialog box and click on the *Set* command.
3. Place the pointer where you want to place the text and click the mouse button.

Printing Profiles and Reports-

The profiles and the results of your analysis can be printed. You can print a report that lists the profile peaks areas and/or a report that lists the molecular weights. **Paper orientation (Profile or Landscape) is set under Windows control panels**

Printing Profiles:

To print the profile:

1. Press the **right mouse button** and select the *Print* command.

Printing Reports:

To print a peak report:

1. First click on the *MODE* menu and select the *Report* function. The REPORTS window appears.
2. Point and click on the *FUNCTIONS* menu and select the *Contents* command. A dialog box appears in which the type(s) of report to be printed may be specified.
3. Click in the Profile Data, Peak Report, and Molecular Weight Report check boxes.
4. Click on the *Set* command button to set these choices.
5. Next, view the reports to be printed by clicking again on the *FUNCTIONS* menu and selecting the *View* command. A dialog box appears, prompting for a filename.

6. Select EXAMPLE1.SCN and click on the *Print* command button under the *FUNCTIONS* menu. A second dialog box appears prompting for profile name.
7. Select SAMPLE1.000; click once on the *Add* command button.
8. To print the reports, select the *Print* command from the FUNCTIONS menu. Two profile reports can be viewed at one time to allow profile report comparison.

Saving Reports and Returning to the Profile-

To save the reports:

1. Use the *Save as* command from the FUNCTIONS menu.
2. Enter in the appropriate report file name followed by the extension .rpt. Reports are saved in a ASCII tab delimited format.
3. To return to the profile, first select *Display* from the MODE menu.
4. Next, select *New* from the PROFILE menu. A dialog box appears, prompting for a filename.
5. Select EXAMPLE1.SCN and click on the *Open* command button. A second dialog box appears prompting for profile name.
6. Select SAMPLE1.000; click once on the *Add* command button.
7. Click on the MA menu located in the upper right corner of the profile to return to the image and to Molecular Analyst.

2.5 Lesson 5: Molecular Weight Determination

Displaying And Optimizing The Standard Profile-

With the image EXAMPLE2.IMG displayed:

1. Duplicate the active profile object used to create the sample profile by clicking on the active profile object with the movement arrow while holding the <Ctrl> key down.
2. Place the new object on the second lane to the right of the sample lane. This new object is now the active object to generate a standard profile.
3. Double click on this new line and enter the name BRDNASTD in the *Profile Name* dialog box.
4. Display the new profile by selecting *Analyze Profile(s*) from the ANALYSIS menu. A display screen will appear with the new profile labeled BRDNASTD.000 underneath the original SAMPLE1.000 profile, and the background profile, BACKGROUND.000. Note that the standard profile is the active profile and is denoted by its Fuschia color.
5. Next, prepare BRDNASTD.000 for analysis by subtracting background, identifying the front and origin, (using the same X coordinates used for the unknown scan in Lesson 2) and by smoothing the profile.

Optimizing the Standard Profile

1. Click on the *OPTIMIZE* menu and select the *Origin* command.
2. Use the scissors-cursor to clip the profile at close to the same X coordinate (X = 24) as for the unknown profile.
3. Click on the *OPTIMIZE* menu and select the *Front* command. Use the scissors-cursor to clip the profile at close to the same X coordinate as for the unknown profile.
4. Next, smooth the profile using the criteria and technique outlined prior.

Subtracting Background from the Standard Profile

Now we will subtract background data. The *Sub Background* command has been introduced. *Sub Horizontal Baseline* and *Construct Baseline*, the two additional commands for eliminating background data, are presented below.

Horizontal Baseline:

The *Sub Horizontal Baseline* command allows you to set a horizontal baseline and subtract it from the profile.

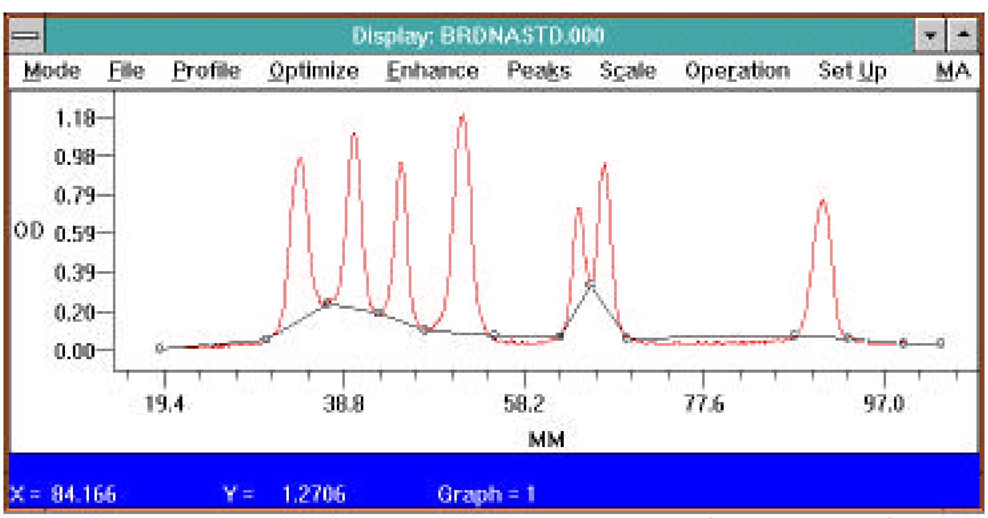
1. Click on the *OPTIMIZE* menu and select *Sub Horizontal Baseline*. The cursor changes to a horizontal line extending across the graph which can be adjusted in the vertical direction.
2. Move the horizontal baseline down the graph until the Y (O.D. or Counts) value in the lower left corner of the graph reads approximately 0 or is close to the profile background.
3. Press the left mouse button and the horizontal line will now become the new baseline (O.D. or Counts) eliminating all area below the line.
4. Now, so that the third and final command for background elimination can be introduced, click on the *PROFILE* menu and select the *Previous* command to restore the unmodified profile.

Construct Custom Baseline:

The *Construct Baseline* command lets you draw a valley to valley baseline which is useful for profiles containing a background envelope, or a drifting baseline. To use this command:

1. Click on the *OPTIMIZE* menu and select *Construct Baseline*. Notice that the cursor becomes a pointer.

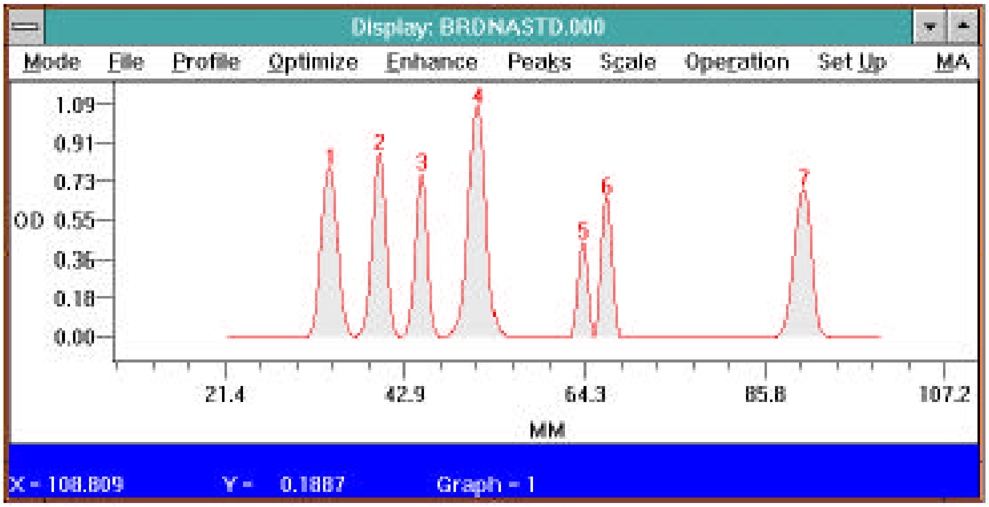
2. Use the mouse to move the pointer to a spot to the left end and **off** of the profile.
3. Click the left mouse button.
4. Follow the contour of the profile, moving from valley to valley pointing and clicking. See below for an example of baseline construction.
5. Finally, use the mouse to click on a spot to the right and **off** of the profile. Be sure that you always extend the constructed baseline beyond the left and right ends of the profile.
6. Once the baseline is constructed, point and click on the *OPERATION* menu.
7. Select the *Complete* command to draw the baseline. (*Subtract Baseline* from the *Graph Setup* from the SET UP menu should be checked). Background area below the constructed line will be eliminated, and the profile redrawn.



Construct baseline operation (sample and background profiles removed for clarity)

Identifying Peaks On The Standard Profile-

1. From the *PEAK* menu, select the *Auto Identify* command. A total of 7 peaks are found.



Automatic identification of standard peaks.

Editing Peaks On The Standard Profile-

In some cases, you may choose to edit the peaks integrated by the *Auto Identify* command. The *Peak Edit* function lets you add, delete, or re integrate the peaks on your profile. To use the *Peak Edit* function:

1. Point and click on the *PEAKS* menu and drag to select *Peak Edit*. The display window closes, and the EDIT window opens.
2. Select *DELETE* from the menu bar; notice that the cursor becomes a hand grenade. 3. Position the arrow of the grenade on the filled area of peak 1 and click. The peak is deleted, and filled area removed.
3. To restore the peak, select *CREATE* from the menu bar; notice that the cursor becomes a wand cursor.
4. Position the wand at the beginning of the peak, click the mouse button, and drag the wand until the entire peak is shaded. Note: peaks will not be automatically renumbered until you exit the edit function.
5. To adjust the width of peaks, click on *ADJUST* in the menu bar. Notice that the cursor becomes a tuning-fork.
6. Using the mouse, move the tuning fork into the area under a peak.
7. Click and the tuning-fork-cursor vanishes, replaced by a wand cursor.
8. To adjust a peak, position the wand cursor at the beginning of the peak and drag.
9. Release the wand and the width is adjusted. The wand will now change back to a tuning fork for adjusting more peaks.
10. Delete the adjusted peak using the *DELETE* command and then use the *CREATE* command to recreate the proper peak.
11. Click on the *EDIT* menu and select the *Exit* command to exit the EDIT window.
12. Next, save the modified standard profile. To do so, click on the *PROFILE* menu and select the *Set Type* command.

13. Within this menu, identify the profile as a standard profile by clicking on the standard radio button.
14. Click on the *Set* command.
15. Next, from the PROFILE menu, select the *Save* command.

We will conclude this part of the lesson by printing the active profile which is currently displayed. Simply press the right mouse button and select *Print*. Profile Analyst will automatically send the graph to the Windows spooler for printing.

Assigning Molecular Weights-

From within Profile Analyst:

1. Click on the *MODE* menu and select the *WEIGHT* function. The display window vanishes and the MOLECULAR WEIGHTS window appears.
2. Click on the *FUNCTIONS* menu and select the *Size Standard Editor* command. A dialog box appears; in it, we will tabulate names and molecular weights for the standards. Subsequently, the table will be used to associate standard names and molecular weights with the peaks within the standard profile.
3. Enter the names and molecular weight values for the Bio-Rad DNA Size Standards as follows:

Fragment 1 <Tab> 23130 <Enter>
Fragment 2 <Tab> 9416 <Enter>
Fragment 3 <Tab> 6557 <Enter>
Fragment 4 <Tab> 4361 <Enter>
Fragment 5 <Tab> 2322 <Enter>
Fragment 6 <Tab> 2027 <Enter>
Fragment 7 <Tab> 546 <Enter>

Note: Enter in the complete molecular weight values. Do not abbreviate the molecular weights of the standards (i.e. kb.)

4. Click on the *Save* command button. A dialog box appears prompting for a filename. The table will be stored in the named file.
5. Type: BRDNASTD and click on the *Done* command button.
6. Return to the *FUNCTIONS* menu and select *Calculate Standard* command. This will calculate log molecular weight vs. relative mobility curve for the standard profile.

A full screen dialog box appears with interior fields: the size standard field, the peaks field, and the standard field.

7. Click the *Open* command button in the size standards field. A dialog box appears, prompting for a filename.
8. Double click on the file that holds the table of names and molecular weights for your standards. In this case, select BRDNASTD.MW. The names and molecular weights just tabulated appear in the standard field file box.
9. Next, click the *Open* command button in the peaks field. Again, a dialog box appears, this one prompting for the name of the file in which the standard profile is stored.
10. Double click on EXAMPLE.SCN. A second dialog box appears prompting for the standard molecular weight profile name, select BRDNASTD.000-a.
11. Click on the *Add* command button to open BRDNASTD.000-a. It appears on the lower part of the screen.

Now we will associate the standard names and molecular weights from the table with the peaks on the standard profile. To do so,

1. Highlight (point and click the left mouse button) both the molecular weight size standard listed first (23130 bp) and the first listed peak (1).
2. Click on the *Associate* command button.
3. Continue associating standards and peaks down the tables by simply continuing to press the *Associate* command button.

Note that the size standards are listed in descending order of molecular weight and that the peaks are listed in descending order of relative mobility. In this case, because there are seven standards and seven peaks, a 1:1 correspondence between molecular weight and peaks is possible.

After the 7 molecular weight standards and peaks have been associated, we are ready to calculate a standard curve.

Calculating a Standard Curve-

1. In the standard field, click once on the *Calculate* command button. A dialog box appears; it contains check boxes in which curve regression type may be specified.
2. Click on all four check boxes, Linear, Quadratic, Cubic, and Logistic.
3. Finally, click on the *OK* command button.
4. Now, click on the *Curve* command button. A scatter plot graph of relative mobility vs. log (MW) appears.
5. So that you can evaluate the fit of each curve: click in the Linear check box to see the linear curve drawn; click in the Quadratic check box to see the quadratic curve drawn; click in the Cubic check box to see the cubic curve drawn; and click in the Logistic check box to see the logistic curve drawn.
6. To remove any of the curves, simply click again on the appropriate check box.

Due to the sigmoidal shape of the standard curve, the third or fourth order regression will often give the best fit of the curve. The user is however cautioned when using higher orders of regression (Appendix C).

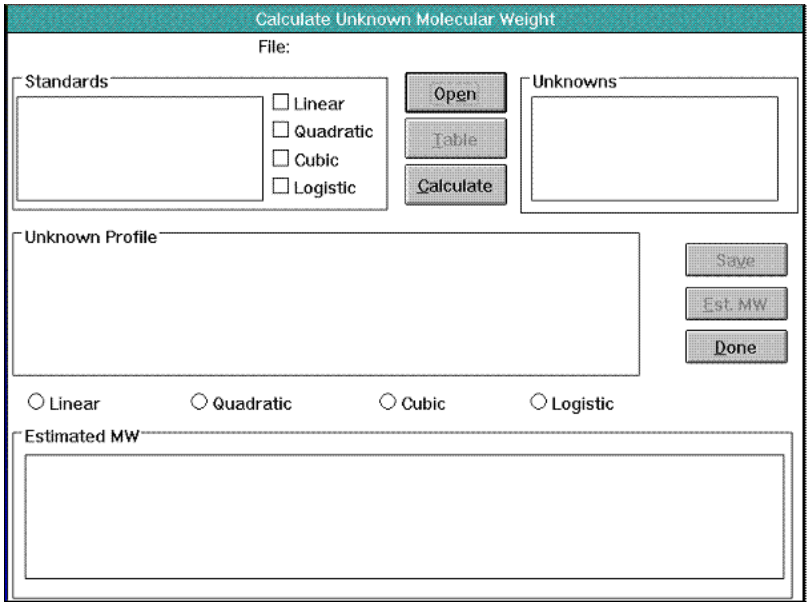
7. After you've seen all curves, close the graph by double clicking on the small menu bar in the upper left hand corner of the box. Select *Close* from the menu that appears.

You will conclude this part of the lesson by storing the curves with the standard profile. To do so, click on the *Save* command button. Note: To see a list of the standards, relative mobilities, and molecular weights, click on the *Table* command button. To quickly close the table double click in the upper left hand corner on the menu bar.

When finished click on the *Done* command button.

Calculating The Molecular Weight Of An Unknown-

1. Click on the *FUNCTIONS* menu and select the *Calculate Unknown* command. A full screen dialog box appears.



2. Click on the *Open* command button.
3. Double click on the EXAMPLE1.SCN file. Notice that all unknown and standard profiles stored in this file are listed in the appropriate file boxes.
4. Next, highlight the standard profile that you would like to compare to the unknown by clicking once on it. In this case, select BRDNASTD.000-a; recall that curves of relative mobility vs. log molecular weight have been stored with this profile.
5. Then, click on the unknown. In this case select SAMPLE1.000-a.
6. Finally, click on the *Calculate* command button. The profile of the unknown appears on screen as well as four estimates of the unknown's molecular weight. Notice that the molecular weight is calculated using linear, quadratic, cubic, and logistic regressions.
7. Move the cursor into the graph of the unknown and notice how the cursor changes to a vertical bar. As the bar is moved across the profile, the relative mobility and molecular weights values are continuously displayed on the bottom of the graph.
8. By clicking on the regression type radio buttons below the graph, the read out will change to reflect that order of regression.

9. Click on the *Save* command button to save this information for later reports.
10. Click on the *Done* command button to exit the Calculate Unknown dialog box.

Selecting the MODE menu will allow you to return to the DISPLAY function, REPORTS function or to EXIT the Profile Analyst module.

11. Select EXIT from the MODE menu.
12. Next, open the FILE menu and choose EXIT. This will return you to the Molecular Analyst program.

Exporting Profile Reports and Data Points To Other Applications-

The data on the profile reports can be exported to any number of spreadsheet or desktop applications which accept data from Windows clipboard.

To export data from the profile peak report to any Clipboard receiving application, first view the data to be exported.

1. To view the reports, click on the *FUNCTIONS* menu and select the *View* command.

2. When the dialog box appears, enter the filename and click on the *Print* command button.
3. When the second dialog box appears enter the profile name and click once on the *Add* command button.
4. With the profile data displayed, simply highlight the data to export (click and drag the cursor down the profile report table to highlight more than one row) and use the *Clipboard* command in the FUNCTIONS menu. This will copy the profile report values to clipboard.
5. Next, open up the application you wish to export to and use the Paste function to transfer the data from clipboard to the new application. Using the <Alt>+<Tab> keys will allow you to rapidly move between Molecular Analyst and other open windows such as Windows Program Manager.

Note that the profile report results are ASCII tab delimited values and will transfer as such.

The data points from a profile may also be exported to any number of spreadsheet or desktop applications which accept data from Windows clipboard. Using the command *Copy to Clipboard* from the FILE menu of the Profile Analyst module, data points (x-Distance and Y-O.D. or Counts) from a displayed profile can be transferred to the Microsoft Windows Clipboard to be later pasted into other clipboard receiving applications for additional analysis or publication.

2.6 Tutorial Summary

To start the Molecular Analyst/PC program:

- Double click on the Molecular Analyst/PC program icon

To display an image:

- From the File menu, select Open
- Double click on the image file to be displayed

To optimize an image:

- From the View menu, select *Scale to Fit*
- Using the magnifying glass tool, magnify the image
- From the TRANSFORMATION menu, select *View Filter*
- From the Raw Image Filter dialog box, select *Smooth(Median)*, then click OK
- Use the Modify View palette to enhance image brightness and contrast

To generate volume integration report

- From the EDIT menu, select *Clear All* to remove previous profile objects
- Select Volume Analysis from the ANALYSIS menu to display volume analysis tools
- Use the volume tools to select the desired spots and a background
- Select *Analyze Volume Object List* from the ANALYSIS menu

To create a standard curve and calculate concentration, molecular weight, or pI

- Identify objects and enter standard value in the *Edit Volume Report* command box.
- Identify a background object for global background subtraction, or select the *Local Background Subtraction* command from the OPTIONS menu for local background subtraction
- From the OPTIONS menu select the *Format Columns* command. Display the appropriate columns, choose the regression type
- Select *Adj. Volume vs. Concentration* from the VIEW menu

To generate a profile

- Select Profile Analysis from the ANALYSIS menu to display profile tools
- Highlight the *line* tool and draw a line through the image lanes
- Select *Analyze Profile(s)* from the ANALYSIS menu

To display a previously saved profile from Molecular Analyst:

- From the FILE menu, select the Open ... command
- Open the pull-down menu *List Files of Type*:. and select Profiles(*SCN)
- Double click on the File name from the Document Type Box
- Click on a profile name in the profile title bar in the Graphics window
- Click on the profile file name and click Add

To display a previously saved profile from within Profile Analyst :

- From the *MODE* menu, select the *Display* command
- From the *PROFILE* menu, select the *New* command
- In the Filename dialog box, identify the file where the profile is stored and click OK
- In the Profile name dialog box, click on the name; click on the *Add* command button.

To optimize a profile:

- From the *OPTIMIZE* menu, select Sub *Horizontal Baseline, Construct Baseline, or Subtract Background* command
- From the *OPTIMIZE* menu, select the *Origin* command; Clip profile at origin
- From the *OPTIMIZE* menu, select the *Front* command; Clip the profile at front
- In the DISPLAY window, press the right mouse button, select the *Zoom* command
- Box the area of interest
- From the *OPTIMIZE* menu, select the *Snip at View* command

To smooth a profile:

- From the *ENHANCE* menu, select *Criteria*
- In the Set Criteria dialog box, enter Window Size, and Iterations Value
- From the *ENHANCE* menu, select the *Smooth* command

To Identify and integrate peaks:

- From the *PEAKS* menu, select the *Peak Criteria* command
- In the Peak Criteria dialog box, enter minimum Amplitude, Width and Area.
- From the *PEAKS* menu, select *Auto Identify*
- From the *PEAKS* menu, select the *Peak Info* command
- From the *PEAKS* menu, select the *Integrate Interactive* command
- Drag the cursor from origin to front of the peak of interest

To edit peaks:
- From the *PEAKS* menu, select the *Edit. . .* command
- From the *EDIT* menu bar select *ADJUST*, *CREATE*, OR *DELETE*

To save:
- From the *PROFILE* menu, select the *Save* command

To Print a profile:
- With the mouse pointer within the graph, press the **right** mouse button and select *Print.*

To calculate a standard curve:
- From the *MODE* menu bar, select the *Weight* command
- If your standard is not a Bio-Rad standard, from the FUNCTIONS menu, select *Standard Editor;* enter the names and molecular weights for peaks on the standard profile
- From the *FUNCTIONS* menu, select *Calculate Standard*; in the dialog box, select the file holding the table of names and weights.
- In the dialog box, select the standard profile; associate names/molecular weights with peaks.
- In the dialog box, click on the *Calculate* command button; check regression type; click on the *Curve* command button; click on the *Done* command button.

To calculate the molecular weight of an unknown:
- From the *MODE* menu, select the *Weight* command.
- From the *FUNCTIONS* menu, select the *Calculate Unknown* command: open the file in which the unknown and standard profiles are stored; select standard profile for which curves have been generated, select unknown standard profile, click on the *Calculate* command button

To view or print reports:
- From the *MODE* menu, select the *Report* command
- From the *FUNCTIONS* menu select the *Contents* command; select the type of report desired
- From the *FUNCTIONS* menu, select the View command
- From the *FUNCTIONS* menu, select the *Print* command

To export profile or volume report data to Window's clipboard
- Highlight the data to export and select *Copy to Clipboard* or *Copy* command from the EDIT menu
- Use <Alt>+<Tab> to go to Program Manager
- Open the application and use the Paste or Paste Special function to transfer the ASCII tab delimited formatted data

To delete profiles:

- From File Manager menu, highlight profiles for deletion
- Select *Delete Profiles ...*

To print images or reports

- Select Page Setup command from the FILE menu to size and position images on Windows compatible printers
- Select Print command from the FILE menu to print reports to Windows compatible printers
- Press the Print button on the Video Printer to print images

Section 3 Description of Palettes

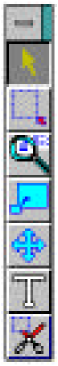
Molecular Analyst/PC provides “floating” palettes which will aid in image enhancement and optimization. The Tool Palette provides access to all of the commonly used tools such as zooming and scrolling. The Modify View Palette provides access to all of the image display manipulation tools such as brightness, contrast, and color slider controls. Every palette has a title bar that can be dragged to reposition the palette. Each title bar also has a close box for putting the palette away until it is needed again. Both tool palettes can be displayed automatically

3.1 Image Tool Palette

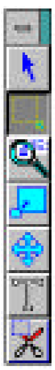
Choosing *Show Tools* from the TOOLS menu displays the Tool Palette. Alternatively, selecting *Preferences Setup* under the FILE menu, and clicking on the Tool palette *Show on Entry* box, will display the Tool palette when an image is opened.

The Tool palette consists of a number of boxes, each of which displays an icon that represents a specific tool or effect. When one of the tools is selected, that specific tool icon will be highlighted and the mouse cursor changes to the tool when placed on the image.

The Tool palette consists of the following:



Pointer (Object Select)
The pointer is the default tool as it is used to select all other tools. It is also used to select windows, window sizes, and menu functions.



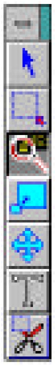
Block Select (Multi-object Select)
The block select tool is used to select multiple objects as an active group on an image.

• Click on the block select icon.

• Place the block select tool at one corner of the group of objects to make active.

• While holding down the left mouse button, drag diagonally until the desired objects are selected.

Deselecting objects in a group - While holding down the <Shift> key, block select on the group of objects to deselect. Alternatively, to deselect a single object from a group, place the pointer tool on the object and while holding down the <Shift> key, click the mouse button.



Magnifying Glass (Magnify)
The magnifying glass tool is used to rapidly magnify a specific point within the image by a factor of two. The maximum magnification is 1600% from the actual image size.

• Click on the magnifying glass icon.

• Magnify the image two fold from the point of clicking.

• Magnify the image another two fold from the point of clicking.

Undo magnification - Holding down the <Shift> key and clicking the mouse button de-magnifies the image two fold from the point of clicking. Alternatively, selecting *Show All* from the VIEW menu will display the original, unmodified image.

Zooming Box (Zoom)
The zooming box tool is used to quickly magnify a selected area of the image by a factor of two. The maximum zoom is 1600% from the actual image size.

• Click on the zooming box icon.

• Place the zooming box tool at one corner of the area to magnify.

• While holding down the left mouse button, drag diagonally until the desired area is selected and release the mouse button.

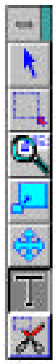
Undo zoom - To undo a zoom function, select *Show All* from the VIEW menu to display the original, unmodified image.

Scrolling Cross (Pan)
The scrolling cross tool is used to move an image that has been zoomed or magnified around the image window in horizontal, vertical, and diagonal directions.

• Zoom or magnify a region of interest.

• Click on the scrolling cross icon.

• While holding down the left mouse button, drag horizontally, vertically, or diagonally, and release the mouse button.



T (Text)
The text tool is used to label images or profiles. It also has an auto-numbering feature for rapid sequential numbering of objects.

• Click on the text icon.

• Move the text cursor to the location where you want the left edge of the label to appear, and click the mouse button.

• Backspace to remove the auto number if desired.

• Type in the text entry box and click on the OK button.

• To alter text on an image, double click on the text. When the text entry box appears, retype the text or change text parameters. To relocate the text, first click on the text to make it active. Then, click and hold on the text and drag the text to the new location.

• To delete text, click on the text to be deleted and use the Edit Command: *Delete* or press the <Alt> + <Del> keys.

Scissors (Clip Image)
The scissors tool is used to clip and display selected areas from images.

• Click on the scissors icon.

• Place the clip image tool at the upper left corner of the region to be clipped.

• While holding down the left mouse button, drag diagonally until the desired area is selected, and release the mouse button.

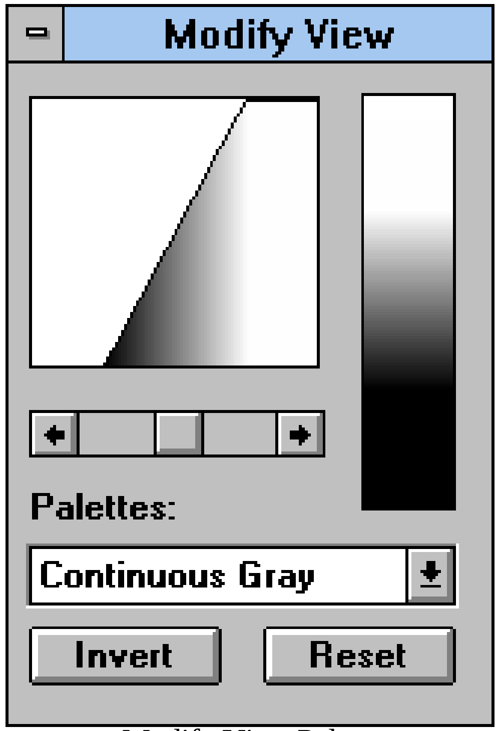
• Select the yes button from the Image Clip dialog box.

Undo clip - To undo a clip function, select *Undo* from the EDIT menu or display the original, unmodified image. To close the clipped image, select *Close* from the FILE menu. At the dialog box, do not save the changes. Reopen the original image by choosing the *Open* command from the FILE menu.

3.2 Modify View palette

Choosing Show Palette from the View menu displays and hides the Modify View palette. Alternatively, selecting *Preferences Setup* under the FILE menu, and clicking on the Modify View palette *Show on Entry* box, will display the Modify View palette when an image is opened.

The Modify View palette consists of horizontal equalization scroll bar, brightness and contrast look-up table (LUT) window slider controls, pull-down palette menus, and control buttons. These palette features control brightness and contrast, invert or negative display, and image reset.

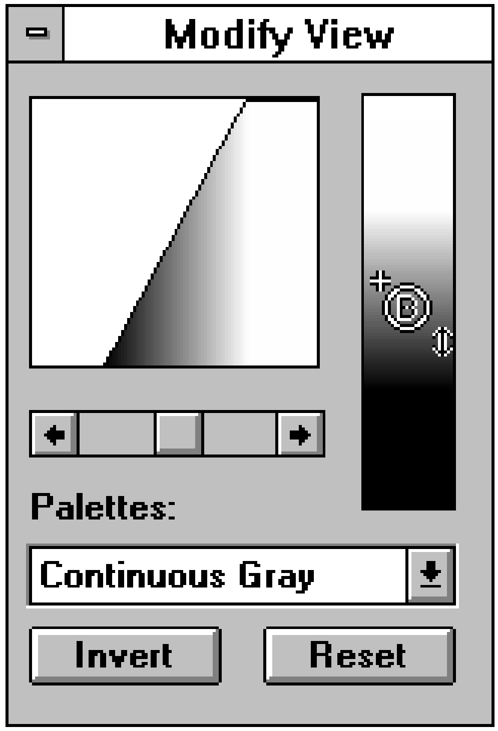


Modify View Palette

Pull-Down Palette Menus
•*Monochrome:*: Distributes black and white throughout the 256 gray level display.
•*8 Grays*: Distributes 8 gray levels throughout the 256 gray level display.
•*16 Grays* : Distributes 16 gray levels throughout the 256 gray level display.
•*32 Grays*: Distributes 32 gray levels throughout the 256 gray level display.
•*64 Grays*: Distributes 64 gray levels throughout the 256 gray level display.
•*8 Colors:* Distributes 8 colors throughout the 256 gray level display.
•*16 Colors*: Distributes 16 colors throughout the 256 gray level display.
•*32 Colors* : Distributes 32 colors throughout the 256 gray level display.
•*Cont. Grays:* Distributes the entire gray scale throughout the 256 gray level display.
•*Cont. R,G,B:* Distributes Red, Green, and Blue throughout the 256 gray level display.
•*Spectrum:* Distributes entire spectrum throughout the 256 gray level display.

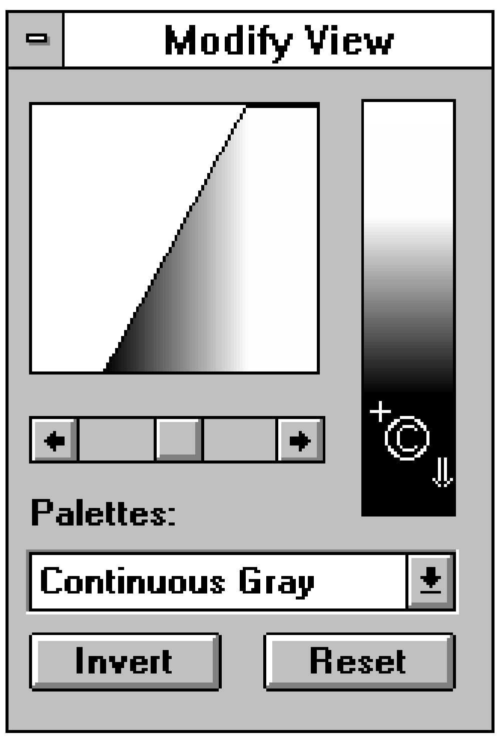
Equalization 256 Gray Level/Color Scroll Bar
This scroll bar option controls the gray/color level and equalization curve (box at top left) of the image as it passes through the 256 gray level or color scale.

Brightness and Contrast Sliding Controls
Brightness - Brightness of the image is controlled by a sliding gray/color display tool. Placing the pointer on a point other than black and white on the continuous gray or color scale (rectangle on the right), changes the pointer to a circled B.



While holding down the left mouse button, slide the brightness tool vertically until the desired brightness is achieved. Altering the brightness shifts the curve in a vertical direction. **The brightness of an image cannot be adjusted until the contrast has been modified.**

Contrast - Contrast of the image is also controlled by a sliding gray/color display tool. Placing the pointer at the top or bottom of the continuos gray or color scale changes the pointer to a circled C.



While holding down the left mouse button, slide the contrast tool vertically until the desired contrast is achieved. Altering the contrast compresses the 256 gray levels and shifts the equalization curve toward horizontal.

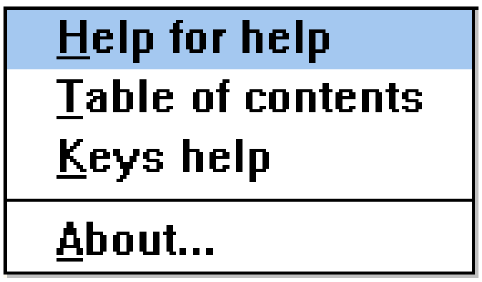
Invert Button
Selecting this option inverts the image and displays the negative of the image.

Reset Button
This option returns the display to its original, unmodified display.

Section 4 Description of Menus

All of the Molecular Analyst/PC menus can be accessed from the main menu bar once an image is displayed or scanned.

4.1 Help Menu



The Molecular Analyst/PC program has on-line, context sensitive help available. To access this feature, select Help from the menu bar. You will be provided with a list of topics regarding the use of the program. Alternatively, to access help on a particular menu function or command, highlight the desired topic and either select help or press the <F1> key.

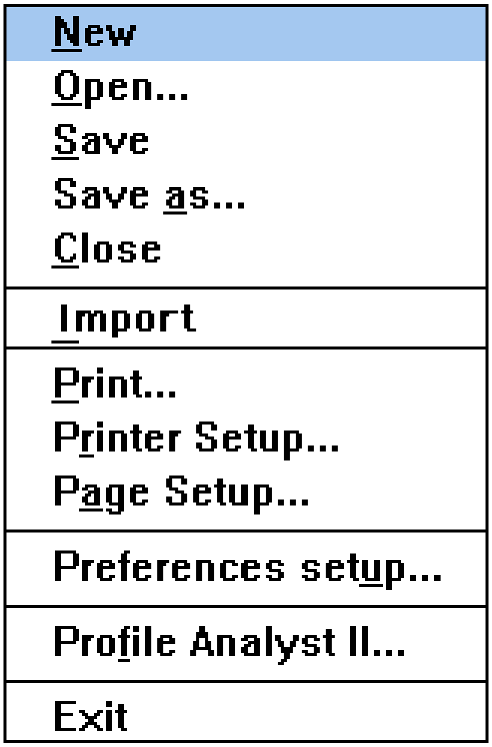
HELP COMMAND: *Help for help*
This command provides a table of contents and information on how to use Help.

HELP COMMAND: *Table of Contents*
This command provides a table of contents and information on the relevant topics of Molecular Analyst/PC .

HELP COMMAND: *Keys help*
This command provides information on all of the keys on the keyboard.

HELP COMMAND: *About*
This command displays the name, version, and copyright date for the Molecular Analyst software program.

4.2 FILE Menu



FILE COMMAND: *New*
This command opens the scanning window for a new scanning session.

FILE COMMAND: *Open...*
Selecting this command allows you to open images or profiles. The opened file will replace any current information on the display window. Up to 4 images may be opened at one time.

FILE COMMAND: *Save*
This command allows you to save data as a file. If the data has not been saved previously, the save dialog window will appear. Select the file type and enter a file name up to 8 characters long. Files are saved in the BRIFF format (Bio-Rad Image File Format), to be used exclusively within Molecular Analyst/PC. If the file had been saved, the data will be saved to the previous file.

FILE COMMAND: *Save as...*
This command allows you to save data under a different name or file type. To save a file to a different disk or directory, save the file the first time using the *Save* command; the second time, use the *Save As...* command to specify an alternate destination. Different file types can be saved by selecting it from the *Save File as Type* pull-down menu. The default file type is a BRIFF format (Bio-Rad Image File Format). Alternatively, images can also be saved in the 8-bit or 16-bit (16-bit TIFF are Molecular Imager images only) TIFF format to be exported to many image analysis software programs.

FILE COMMAND: *Close*
Selecting this command will close the active window displayed in the foreground and the tool palettes associated with it.

FILE COMMAND: *Import*
This command imports 8-bit and 16-bit raw data TIFF files, 8-bit display data with a table of palette colors, and 8-bit and 12-bit data with calibration table.

FILE COMMAND: *Print...*
This command will send the image and its associated reports to the printer.

FILE COMMAND: *Printer Setup*
This command is used to select the desired characteristics and functions for the current printer. The current printer is set using Print Manager, which is available in Window's Program Manager window.

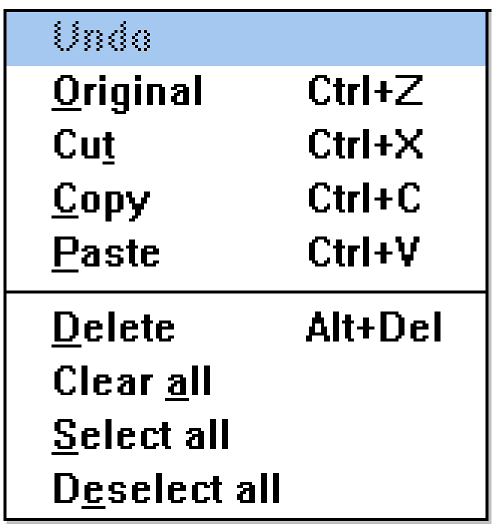
FILE COMMAND: *Page Setup...*
This command is used to size and position images to be printed. This command does not effect 1D profiles or reports.

FILE COMMAND: *Preferences setup ...*
This command is used to determine the default directory, tool palette display options, fonts, rulers, filters, color selection of the profile and volume tools, and the column preferences for the volume analysis report. Within this command, the display mode may be changed from fast, intermediate, and slow. This effects image display speed and scanning resolution.

FILE COMMAND: *Profile Analyst II*
This command allows you to quickly move to the display window of the Profile Analyst module.

FILE COMMAND: *Exit*
This is used to exit from the Molecular Analyst/PC program.

4.3 EDIT Menu



EDIT COMMAND: *Undo*
This function reverses the last image operation (filter, rotate, or clip) performed.

EDIT COMMAND: *Original*
This function displays the original, unmodified image.

EDIT COMMAND: *Cut*
Selecting this command will cut any selected object and place it into the clipboard. These objects can be pasted into most clipboard compatible programs.

EDIT COMMAND: *Copy*

Copy will duplicate any selected object(s) into the clipboard

EDIT COMMAND: *Paste*
This function will paste any object(s) cut or copied into an active image.

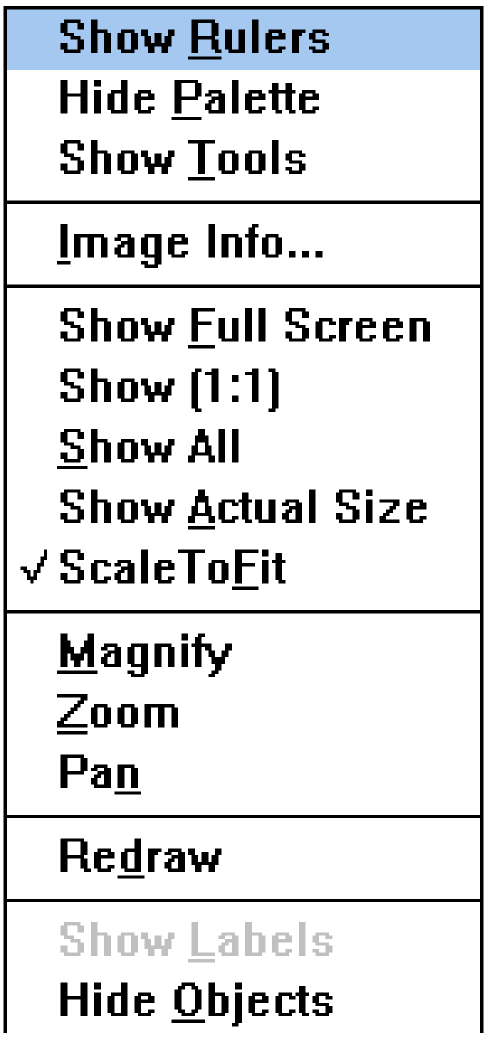
EDIT COMMAND: *Delete*
This function will delete any active object(s) or text within the displayed image.

EDIT COMMAND: *Clear all*
Selecting this command will delete all objects and text within the displayed image.

EDIT COMMAND: *Select All*
This command selects all objects and text and makes them active.

EDIT COMMAND: *Deselect All*
This command deselects all objects and text and makes them inactive.

4.4 VIEW Menu



VIEW COMMAND: *Show Rulers / Hide Rulers*
Selecting this command allows you to display or hide the metric rulers. The rulers automatically change from (cm) to (mm) when the image is zoomed.

VIEW COMMAND: *Show Palette / Hide Palette*
Selecting this command allows you to display or hide the Modify View palette.

VIEW COMMAND: *Show Tools / Hide Tools*
Selecting this command allows you to display or hide the Image, Profile, and Volume tools.

VIEW COMMAND: *Image Info...*
This command allows you to review image information such as image creation date, image resolution, image and image file size, and to add comments, exposure times, and patient information.

VIEW COMMAND: *Show Full Screen*
This command will magnify or zoom the active image to fill the window. This command is useful when printing to a video printer and the command *Scale to Fit* is off

VIEW COMMAND: *Show [1:1]*
This command will maintain a 1:1 aspect ratio in the displayed image. This command is useful when visualizing saved images that seem to have rough edges or have more pixels displayed caused by an uneven magnification.

VIEW COMMAND: *Show All*
This command will reverse a magnification or zoom of the active image to show all portions of the image in the window.

VIEW COMMAND: *Show Actual Size*
Using this command displays the actual size of the image.

VIEW COMMAND: *Scale To Fit*
This command maintains the image to window aspect ratio as the display window is enlarged or reduced.

VIEW COMMAND: *Magnify*
When this command is selected the cursor is changed to a magnifying glass when placed on the image. This command magnifies a specific point within the image by a factor of 2 and then centers that point in the display window. Holding the <Shift> key and clicking will reduce the magnified image by a factor of 2. Alternatively, selecting *Show All* from the VIEW menu will display the original, unmodified image.

VIEW COMMAND: *Zoom*
When this command is selected the cursor is changed to a zooming box when placed on the image. This command zooms a designated area within the image and fills the current display window. To undo a zoom function, select *Show All* from the VIEW menu to display the original, unmodified image.

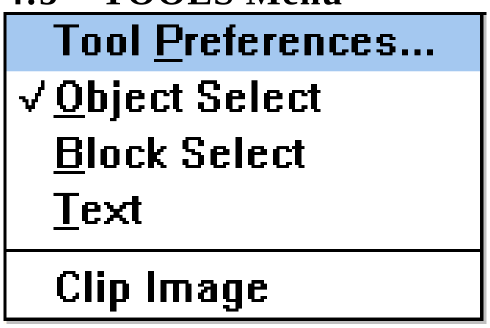
VIEW COMMAND: *Pan*
When this command is selected the cursor is changed to a pointed cross when placed on the image. By holding down the mouse button and dragging, this command scrolls the image across the window in the designated direction. This command is only functional after zooming or magnifying the image.

VIEW COMMAND: *Redraw*
This command redraws the image and removes any object or window artifact which may have been left behind from a previous function.

VIEW COMMAND: *Show Labels / Hide Labels*
Selecting this command allows you to show or hide the object labels on an image when the volume analysis report is present.

VIEW COMMAND: *Show Objects / Hide Objects*
This command allows you to show or hide text, profile, and volume objects. To view hidden objects, either select *Show Objects* from the VIEW menu or click on any profile or volume object icon displayed on the right side of the tool palette.

4.5 TOOLS Menu



TOOLS COMMAND: *Tool preferences...*
Through this command, you can select the preferences for the available tools.

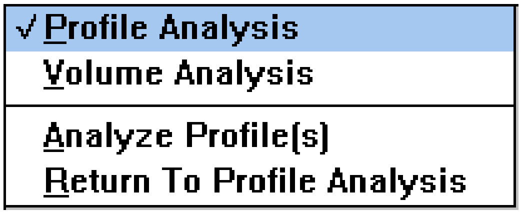
TOOLS COMMAND: *Object Select*
This command allows you to use the pointer from the tool palette. It is used to select all other tools, windows, menu functions and adjusting image window size.

TOOLS COMMAND: *Block Select*
This command allows you to use the block select function from the tool palette. The block select tool is used to make multiple objects active, thereby grouping them together on an image.

TOOLS COMMAND: *Text*
Selecting this command allows you to use the text function from the tool palette. The text tool is used to rapidly annotate and number objects.

TOOLS COMMAND: *Clip Image*
This command allows you to use the clip image function from the tool palette. The clip image tool is used to clip and display selected areas from images. To undo a clip function, close the clipped image by selecting *Close* from the FILE menu. At the dialog box, do not save the changes. Reopen the original image by choosing the *Open* command from the FILE menu.

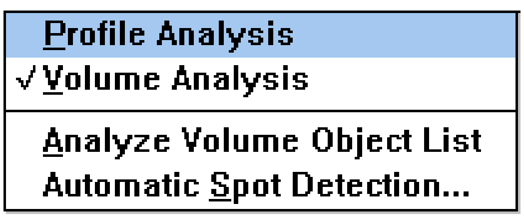
4.6 ANALYSIS Menu



ANALYSIS COMMAND: *Profile Analysis*
Selecting this command activates the Profile Analysis module and incorporates the profile tools onto the right side of the tool palette and into the TOOLS menu. **All volume objects must be removed from the image before profile tools can be displayed or used.**

ANALYSIS COMMAND: *Analyze Profile(s)*
Once a profile tool has been used and an object created, this command is used to transfer the object data to the Profile Analysis module and to create a profile. Each time this command is selected, a profile is created from all active profile objects and displayed. A total of four profile graphs can be displayed at one time within the display function of the Profile Analysis module. Please refer to Section 5 for further information on the Profile Analysis module.

ANALYSIS COMMAND: *Return to Profile Analysis*
Selecting this command allows you to return to the Profile Analysis module. No additional profile is created.

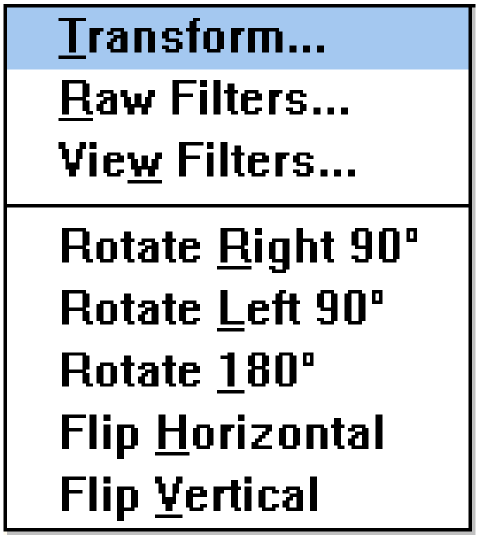


ANALYSIS COMMAND: *Volume Analysis*
Selecting this command activates the Automatic Spot Detection and Volume Analysis module and incorporates the volume tools onto the right side of the tool palette and into the TOOLS menu. All profile objects must be removed from the image before volume tools can be displayed or used.

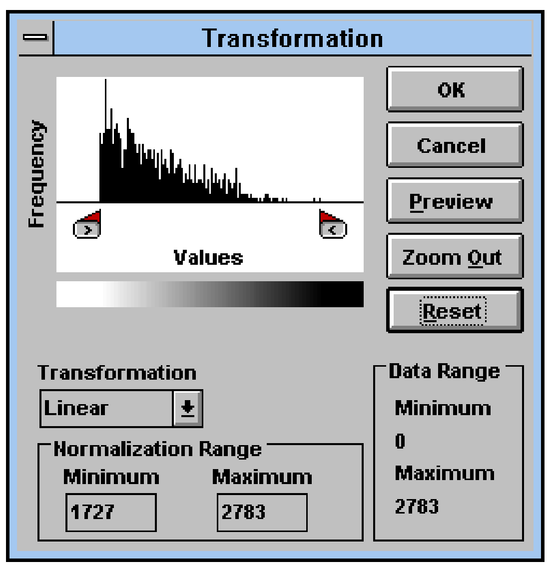
ANALYSIS COMMAND: *Analyze Volume Object List*
Once a volume tool has been used and an object created, selecting this command generates a volume report. Please refer to Section 6 for further information on the Volume Analysis.

ANALYSIS COMMAND: *Automatic Spot Detection*
Selecting this command displays the spot finding criteria dialog box and allows image region, spot size, spot criteria, and artifact exclusion criteria to be entered. The number of spots found is heavily dependent upon the set criteria. Please refer to Section 6 for further information on the *Automatic Spot Detection* command.

4.7 TRANSFORMATIONS Menu



TRANSFORMATIONS COMMAND: *Transform*
Selecting this command displays the transformation histogram. The histogram displays a distribution plot of the gray levels found within the image which is currently displayed (or the active window when two images are displayed at the same time).



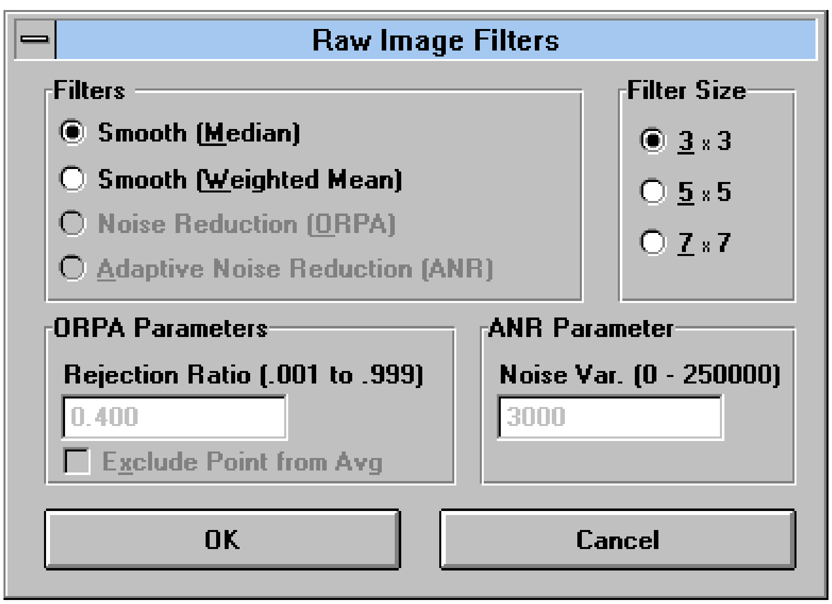
You can use the Transformation Histogram to examine how grays are distributed in your image. This information may help determine how successfully an image can be enhanced. The histogram's horizontal axis (Grays) represents the 256 (densitometer and Gel Doc) or 65,536 (Molecular Imager) possible shades of gray, from white at the far left to black at the far right. The histograms vertical or y-axis (O.D. or Counts) shows the number of pixels found in the image for each possible gray shown on the horizontal axis. For example, if Molecular Analyst searches the image and finds many pixels for gray shade 200, it draws a relatively tall bar at position 200 on the histogram's horizontal axis. If Molecular Analyst finds only a few pixels for gray shade 200, it draws a relatively short bar. If no pixels are found, no bar is drawn and a gap is shown at that position. You should examine the histogram palette before optimizing that particular image.

Adjusting the Histogram Range
Dragging the tab markers changes both the histogram plot and the maximum or minimum gray level values. To view changes to the image, click on the *Preview* button. To apply changes to the image, click on the *OK* button. The image will be redrawn according to the range selected.

The transformation palette also provides linear, non-linear and equalization transformations. Non-linear transformation maps the image data in a tight distribution toward the lower or upper-end of the 0-255 gray level display. Selecting "Nonlinear" from the Transformation pull-down menu activates both the non-linear slider controls, most useful for Molecular Imager images. These tools allow you to preview and control the gray level distribution to each section of the image. Note that the color/gray level distribution bar aligns directly under the histogram plot. Any changes to the gray level/color distribution bar (using the slider controls) will correspond directly to the histogram underneath, and subsequently to the image when the *Preview* and *OK* button is selected. Histogram equalization is the third type of transformation available. This type of transformation takes the specified histogram area and equalizes the gray level values according to frequency. Therefore, high signal areas will be given a large number of gray levels, while background will be given fewer gray or color levels. This type of transformation can provide greater contrast and detail to some images.

To reset the histogram, click on the *Zoom Out* button. This will show you how your selected range corresponds to the entire dynamic range of the image. Clicking on the *Reset* button will move the range controls to their left and right limits. Clicking on the *Preview* or *OK* button will then apply the histogram selection to the image. **(See Appendix I for additional information on enhancing image quality with the Transformation Tool.)**

TRANSFORMATIONS COMMAND: *Raw Filters*
Selecting this command allows you to smooth high frequency noise from the **raw** data with a variety of filters. This type of filtering suppresses high frequency noise which is often the result of imperfections of the sample or image digitization and **will change the quantitative nature of the image data**. Once an image has been filtered, it can be undone using the *Undo* or *Original* command under the EDIT menu. **(See Appendix I for additional information on enhancing image quality with filtering.)**



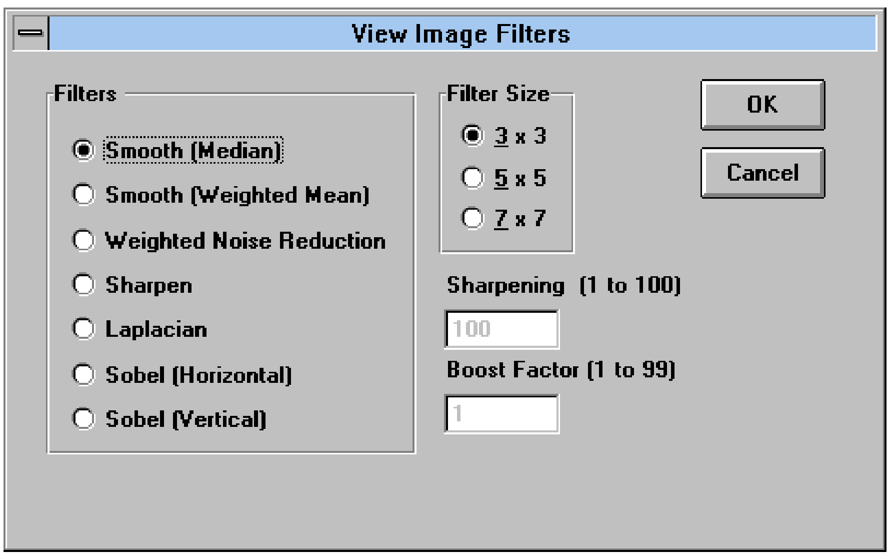
Smooth (Median)
This filter replaces each pixel with the median value of its neighboring pixel. This often blurs or softens the selected area.

Smooth (Weighted Median)
This filter replaces each pixel with the weighted mean value of its neighboring pixel. This filter also blurs or softens the selected area.

Out-of-Range-Pixel-Averaging (ORPA)
ORPA or Out-Of-Range-Pixel-Averaging reduces the overall "grain" in images. This filter is only active with images from the Molecular Imager.

Adaptive Noise Reduction(ANR)
This filter actively removes additive noise in images. It does not soften or blur the edges and is useful when visualizing images with poor signal-to-noise ratios. This filter is only active with images from the Molecular Imager.

TRANSFORMATIONS COMMAND: *View Filters*
Selecting this command allows you to smooth high frequency noise from the **view** data with a variety of filters. By passing a 3 x 3, 5 x 5, or 7 x 7 size filter grid through the data points, view filtering reduces noise but **will not change the raw data of the image data.** Once an image has been filtered, it can be undone using the *Undo* or *Original* command under the EDIT menu. Once a filtered view image has been saved, it may be undone using the transformation histogram.



Smooth (Median)
This filter replaces each pixel with the median value of its neighboring pixel. This filter also blurs or softens the selected area.

Smooth (Weighted Mean)
This filter replaces each pixel with the weighted mean value of its neighboring pixel. This often blurs or softens the selected area.

Weighted-Noise Reduction
This filter is useful for bringing out details in extremely noisy images, and can be used for good publication printing. A boost factor, selectable from .001 to .999 is used for bringing out low values and saturating high values. Each pixel is replaced with the median value of its 3x3 neighborhood. This is a time-consuming operation since each of the nine pixels in the 3x3 neighborhood must be sorted and the center pixel replaced with the median value. This filter preserves the edges of objects (bands and dots) in the images.

Sharpen
This filter increases contrast and accentuates detail in the selection. It may, however, accentuate noise.

Laplacian
This filter can be used as a preliminary for spot detections. Its effect is to enhance boundaries.

Sobel (Horizontal or Vertical)
This filter can be used to enhance electrophoretic images. The effect is to bring out edges in either horizontal or vertical directions.

TRANSFORMATIONS COMMAND: *Rotate Right 90*
This command allows you to rotate the image 90° in a clockwise direction.

TRANSFORMATIONS COMMAND: *Rotate Left 90*
This command allows you to rotate image 90° in a counter-clockwise direction.

TRANSFORMATIONS COMMAND: *Rotate 180*
This command allows you to rotate image data points 180°.

TRANSFORMATIONS COMMAND: *Flip Horizontal*
Selecting this command allows you to flip the image relative to its vertical (left to right) center to provide a mirror image of itself.

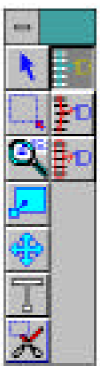
TRANSFORMATIONS COMMAND: *Flip Vertical*
Selecting this command allows you to flip the image relative to its horizontal (top to bottom) center to provide a mirror image of itself.

Section 5 Profile Analyst - Peak Integration

Profile Tool Palette

Choosing *Profile Analysis* from the ANALYSIS menu displays the profile tools on the right side of the tool palette. When displayed, the profile tools consists of three boxes, each of which represents a specific object which can be used to generate a profile from an image. When a tool is selected (clicked on) and used, the specified active object will be displayed on the image. A maximum of eight (8) objects can be displayed at one time. Conversely, profile objects can be deleted from an image using the EDIT menu command *Delete* or *Clear all*.

All of the objects (lines, multi-segments, and column averaging rectangles) can be used in vertical, horizontal, and unrestricted positions by using the profile *Tool Preferences* command which can be accessed from the FILE menu. Alternatively, double clicking on any profile tool icon in the tool palette will also display the *Tool Preferences* dialog box. Additionally, all of the profile tools may be modified by using the altering tools which are activated when the cursor is placed on different parts of the object. These cursor tools (movement arrow, activating hand, sizing square, rotating arrow, and duplication tool) are active when properly placed on parts of an object. Once a profile tool has been used, the command *Analyze Profile(s)* from the ANALYSIS menu will send the profile to the Profile Analyst module where it will be displayed in a graph for further analysis.



Single Line Tool

The single line profile tool is used to identify lanes of data in a straight vertical, horizontal, or unrestricted line. This tool uses a **single** pixel cross section of the image data to display the image profile.

• Double click on the single line profile tool icon. Select either a vertical, horizontal or unrestricted line.

• Move the pointer to a location over the data, normally above the first band in the lane of interest.

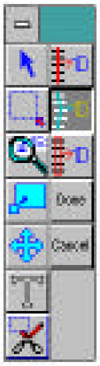
• Click the mouse button and drag the mouse until the line has completely passed through the data, and release the mouse button. A single line profile object will appear in the active color.

• *Object Altering tools*- When placed on the line, these tools allow the single line object to be modified to better fit the image.

Movement arrow- Allows the line to be moved to another location. Place the cursor directly on the line, click and drag the line to the new location.

<Ctrl> +Movement arrow- Clicking the movement arrow while holding the <Ctrl> key down, duplicates the line.

Sizing square- Allows the line to be lengthened or shortened. Place the cursor on the square ends of the line. Notice that the cursor has changed to the sizing square. Adjust the length of a vertical or horizontal line, or the position of an unrestricted line.



Multi-Segment Tool
The multi-segment profile tool is used to identify lanes of data that are skewed or are not in a straight vertical line. This tool uses controlled line segments to identify single pixel cross sections of the data to display the image profile.

• Double click on the multi-segment profile tool icon. Select either a vertical, horizontal or unrestricted multi-segment.

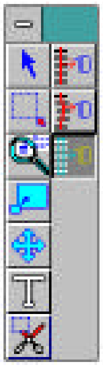
• Move the pointer to a location over a skewed lane of data

• Click and hold the mouse button and move the cursor until one line segment has completely passed through one set of data. Click the mouse button. Move the mouse again and draw the another line segment through the second data point and click the mouse. Continue until a line has passed through all of the data points to analyze.

• To complete the multi-segment line, select *Done* from the tool palette or simply press the <Esc> key.

• *Object Altering tools*- When placed on the multi-segments, these tools allow the segments to be modified to better fit the data.

> *Movement arrow*- Allows the line segments to be moved to another location. Place the cursor directly on a segment of the line (not on an anchor point), click and drag the line to a new location.
>
> *<Ctrl> +Movement arrow*- Clicking the movement arrow while holding the <Ctrl> key down, duplicates the line segments
>
> *Sizing square*- Allows each segment to be angled, lengthened, or shortened. Place the cursor on any square anchor point of the multi-segment line. Notice that the cursor has changed to the sizing square. Adjust the length or position of each segment of the line as desired.



Averaging Rectangle Tool
The averaging rectangle tool uses a user-defined rectangle to identify and average all of the data pixels in the selected area to display a profile. When used across a uniform rectangle shaped band, this tool generates the most accurate profile.

• Double click on the rectangle tool icon. Select either a vertical or horizontal rectangle tool. Note that using the rectangle tool in the wrong orientation will result in an improper profile calculation.

• Move the pointer to a location over the data.

• Click the mouse button and drag the mouse in the orientation as selected in the tool preferences dialog box until a rectangle has completely passed through the data, and release the mouse button. A rectangle will appear. Note: Do not make the rectangle larger than the band, as this will average low background values with high sample values resulting in erroneous calculation.

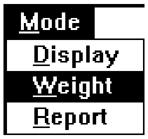
• *Object Altering tools*- When placed on the rectangle, these tools allow the object to be modified to better fit the data.

> *Movement arrow*- Allows the rectangle to be moved to another location. Place the cursor directly inside the rectangle (not on the center dot), click and drag the rectangle to a new location.
>
> *<Ctrl> +Movement arrow*- Clicking the movement arrow while holding the <Ctrl> key down, duplicates the rectangle.
>
> *Sizing square*- Allows the height or width of the rectangle to be lengthened or shortened. Place the cursor on any square anchor point of the multi-segment line. Notice that the cursor has changed to the sizing square. Adjust the length or position of each segment of the line as desired.
>
> *Rotating arrow*- Placing the cursor on the center dot of the rectangle object, clicking and moving the mouse vertically rotates the rectangle. Note: For fine adjustments of the angle of rotation, click on the center dot, hold the mouse button down, and move the cursor farther away from the rotating point.

5.2 Profile Menus

In this section, each of the Profile Analyst module functions, menus and commands will be described in detail.

Display Function



MODE Menu

The mode menu is continuous throughout the profile analysis module, allowing the user to move rapidly from one function (Display, Weight, and Report) to another without having to go through multiple menu levels. All the Profile Analyst module functions can be accessed from the *Mode* Menu throughout the module. Also available in the *MODE* Menu is the *Exit* command, which allows the user to exit to the image module instead of to another function.

MODE COMMAND: *Display*
These commands are used to display, optimize, and enhance profile data; as well as, perform advanced algorithms for the identification of peaks. The *Display* commands appear when a profile object is used and a profile is created using the *Analyze Profile(s)* command from the image module.

MODE COMMAND: *Weight*
These commands allow for the entering of standard molecular weight values into a table, calculation of a relative mobility vs. log molecular weight standard curve, and calculation of unknown standard molecular weights.

MODE COMMAND: *Report*
These commands allow you to view or print a report of any profile contained within any file. Reports include: information about the profile itself, an area report if peaks have been identified, and a molecular weight report if molecular weights have been calculated. Note: to print the profile in a graph, the profile must be displayed from the DISPLAY window and printed by pressing the right mouse button and selecting the *Print* command.

FILE Menu

FILE COMMAND: *Copy Profile*
This command allows profiles to be copied from one file to another. It is only functional when there are no profiles currently displayed on screen. When *Copy Profile* is selected, a dialog box appears. The left half contains Source information and the right side contains Destination information. To open the file where the profiles are currently located, the Source file, click on the *Open* command button in the Source field. Select a file name and click on *Open*. The names of the profiles in that file will appear in the source field. Click on the first profile that you wish to copy into another file. Note that the name appears in the Profile Name field. To open the destination file, click on the *Open* command button in the destination field. Select the file that you want the profile(s) to be copied into. That file name will appear in the top of the Destination field, with the profiles currently located in the file listed below. Click on the *Copy* command button. The highlighted profile in the Source field will now appear on the Destination field profile list. Repeat this procedure for as many profiles as desired. When you have finished, click on *Done*. The profile(s) will now be located in both the Source file and the Destination file. This feature is necessary for moving profiles to the same file as this is required for molecular weight analysis.

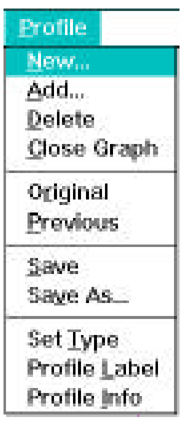
FILE COMMAND: *Copy to Clipboard*
This feature allows you to transfer data points (x-Distance and Y-O.D. or Counts) from the active profile to the Microsoft Windows Clipboard. From there the data points can be transferred to many Windows programs, such as Excel or Microsoft Word, for additional analysis or publication. If no profiles are currently displayed, selecting this function will display a dialog box prompting you for a FILE name. A second dialog box will then appear prompting for a profile name. Select one by clicking on it, and then clicking on *Add*. The data points will automatically be copied to the Windows Clipboard. If profile(s) are currently displayed on screen, the data points from the active profile will be copied to the Windows Clipboard. The Windows Clipboard accepts a maximum of 2000 data points, therefore; it may be necessary to clip the profile before copying.

FILE COMMAND: *Delete Profile . . .*
This command is only functional when there are no profiles currently displayed on screen. When this command is selected, the Windows entry box will appear which provides; a scrolling list box of the FILE names within the current drive and directory, and an entry box for entering the FILE name to open for deleting profiles. Notice that the entry box will have "***.SCN**" highlighted, indicating that .SCN must follow a FILE name. Move the mouse pointer to the FILE name in the list box containing the profile(s) that you wish to delete, and double click with the left mouse button.

<u>Locating FILEs in other drives and directories</u>
Changing directories within current drive:
To change directories within the current drive, using the scroll bars if necessary, double click on [**..**]. The list box will now list the directories available on the current drive. Double clicking on the desired directory will open that directory and list the existing FILEs within that directory which have the FILE extension (.SCN).

After opening a profile file for deletion of profile(s), a new dialog box will appear listing all the profiles within the FILE. To delete a single profile, point and click once on the profile name to highlight it, then click on the *DELETE* Command Button. To delete more than one profile at a time, press and hold <SHIFT> while pointing and clicking on all the profiles you wish to delete. When you have finished deleting profiles, point and click on the *CANCEL* or *EXIT* Command Buttons. It is important to note that at this point the profile names have been deleted from the FILE, however, disk space occupied by the deleted profiles will not be freed until the *Compact FILE* command is executed as outlined below.

FILE COMMAND: *Compact FILE*
This command is also only functional when there are no profiles currently displayed on screen. Disk space occupied by profiles deleted using the *Delete Profile . . .* command is not freed up until either the Profile Analyst program is exited, or the *Compact FILE* command is used. By using the *Compact FILE* command, it is possible to free up the disk space occupied by the deleted profiles without exiting the Profile Analyst software program.



PROFILE Menu

PROFILE COMMAND: *New . . .*
The *New . . .* command will display a new graph containing a selected profile. A maximum of four different graphs can be displayed at once, with each graph containing only one profile. To display a new graph containing a profile from within the profile list box, simply double click on the desired profile name, or enter the profile name complete with extensions and appendixes and then click on the *ADD* Command Button.

The profile-information dialog box contains a list box of all of the profiles contained within the file which has been selected. After you point and click on one of the profile names within the list box, the Type and Operations boxes will now show all the identifying characteristics of the profile. The Type box provides the following information:

Type Box
Raw - This identifies the highlighted profile as being an unprocessed profile. This type of profile is always the original profile. Raw profiles will never have a character appendixed to their name.

Processed - This identifies the highlighted profile as being processed in some manner. Processed profiles will always have at least one of the operations in the operations box highlighted, identifying the manner in which it has been processed. Processed profile names will always be followed by a character appendix, identifying it as being a processed profile.

Standard - This identifies the highlighted profile as having been tagged as a standard profile. The processed standard profile is used in the generation of a standard curve for molecular weight determination of unknown samples.

Background - This identifies the highlighted profile as having been tagged as a background profile. The background profile is data associated with the background of the electrophoretic sample, but not the sample itself. A background profile can be subtracted from a standard or unknown profile to remove all area which is background data. A background profile can be displayed, but, the Profile Analyst II software will not allow it to be processed in any manner.

Unknowns- This identifies the highlighted profile as having been tagged as an unknown profile. An Unknown profile is any one-dimensional data which contains sample data, and is not a standard profile.

PROFILE COMMAND: *Add . . .*
The *Add . . .* command will add a profile to an active graph. This command can be used until a maximum of four profiles are displayed, in one or more graphs.

PROFILE COMMAND: *Delete*
The *Delete* command will delete only the active profile (denoted by its color).

PROFILE COMMAND: *Close Graph*
This command closes the active graph with all its contents. To close a graph which does not contain an active profile, double click on one of the profiles within that graph making it active and then select the *Close Graph* command.

PROFILE COMMAND: *Original*
This command removes the active profile and replaces it with the original version of the profile. The *Original* command is necessary to restore a profile to original form once any "data modification" has been performed upon it.

PROFILE COMMAND: *Previous*
When this command is selected, the program will toggle back and forth between the current and the previous profiles which has had a modification performed upon it.

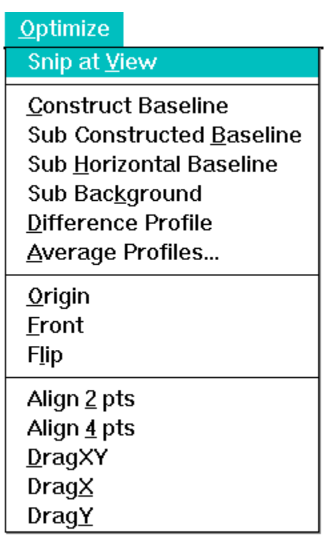
PROFILE COMMAND: *Save*
When this command is selected, the active profile will be saved with any modifications which have been performed. When a raw profile is received, the data is stored in the following format: FILE.SCN. As individual raw unprocessed profiles are created and stored into the files, they are sequentially given an extension to their root name. Using the *Save* command will not write over the original profile data, instead it will save processed profiles (background subtracted, peaks identified, etc..) using the same profile name along with a character appendix, i.e., Profile1-a. Molecular weight standards data table files are stored in the following format: standardfile.MW.

PROFILE COMMAND: *Save As . . .*
When this command is selected, the active profile can be saved using any root name desired up to fifteen characters in length. The program will automatically add the appendix "a" identifying the profile as a processed profile. Note: no extension is necessary when enter the profile name, as the program will automatically add the following extension and appendix ".000-a".

PROFILE COMMAND: *Set Type*
This command allows you to change the profile type: Standard, Background, and Unknown. The name of the active profile will appear at the top of the box, and its currently set type will be selected. Click on the radio button corresponding to the type you wish to change the active profile to. After making the designation, click on *Set*. Select *Save* under the *PROFILE* menu to implement this change. Once changed, the profile will be saved both with it's original type and the new type designation. This command is useful for identifying a profile as a background profile to be subtracted from other profiles; or as a standard profile for calculating a molecular weight standard curve. The default profile type is unknown.

PROFILE COMMAND: *Label Profile*
This command adds one label to your profile. Type in the text (up to 80 characters) and click on the *Set* button. You will now see the cursor in the shape of a pointing hand. Move the cursor to the location within the graph where you want the left edge of the label to appear and click the mouse button. To relocate or change the profile label, simply repeat the above procedure. To add a profile label to any other profile which is currently displayed on screen, simply double click on that profile with the pointer to make it active, then repeat the above procedure.

PROFILE COMMAND: *Profile Info*
When this command is selected, a "hand pointer" icon will appear for pointing and clicking on any profile within the active graph. By placing the "finger" of the icon on one of the profiles in the graph, and clicking with the left mouse button, information regarding that profile will appear on screen. Pointing and clicking with the icon on another profile within the active graph will update the profile information table to reflect that profile. When finished, click on the *OPERATION* menu, then click on either *Complete* or *Cancel* to remove the profile information box.



OPTIMIZE Menu

OPTIMIZE COMMAND: *Snip at View*
When this command is selected, view modifications performed with the *Zoom* function (accessed by pressing the right mouse button when the cursor is placed in a profile graph) will be transformed into a data modification. The user can therefore zoom in on any section of the displayed profile and analyze that section of the profile exclusively by pointing and clicking on the *Snip at View* command. Once the *Snip at View* command has been used, either the *Original* or *Previous* command must be used to restore the original or previous data.

OPTIMIZE COMMAND: *Construct Baseline*
The command lets you draw a valley to valley baseline which is useful for profiles containing a drifting baseline. To use this command, point and click on the *OPTIMIZE* menu and select *Construct Baseline*. Move the pointer to a spot to the left end and off of the profile and click. Then follow the contour of the profile, moving from valley to valley pointing and clicking. Finally point and click on a point to the right end and off of the profile. Be sure that you always extend the constructed baseline beyond the left and right ends of the profile. Once the baseline is constructed, point and click on the *OPERATION* menu. Select the *Complete* command to draw the baseline. Next, select *Sub Constructed Baseline* to subtract the baseline.

OPTIMIZE COMMAND: *Sub Construct Baseline*

The *Sub Construct Baseline* command is only operational if the *Subtract Baseline* Check Box is deactivated under the *SET UP* Menu, and if a baseline has been constructed using the *Construct Baseline* command. The constructed baseline will then be filled in and remain visible after the *Complete* command, under the *OPERATION* Menu has been selected. This allows the constructed baseline to be displayed in the same manner as a profile for comparison. To perform the subtraction of the constructed baseline from the active profile, simply point and click on the *Sub Construct Baseline* command.

OPTIMIZE COMMAND: *Sub Horizontal Baseline*

When this command is selected, a horizontal baseline can be selected and automatically subtracted from the active profile. This command is useful for subtracting background from profiles containing a uniform background level. To subtract a horizontal baseline from the active profile, simply point and click on *Sub Horizontal Baseline*, the mouse pointer will change to a horizontal line which is adjustable in the vertical direction. Move the horizontal line cursor up or down until the desired baseline position is located, then click on the left mouse button to automatically subtract the horizontal baseline.

OPTIMIZE COMMAND: *Subtract Background*

When this command is selected, actual background profiles from the image can be subtracted from the active profile. This method of background subtraction is usually the most accurate and should be the method of choice when available. To perform the subtraction of a background profile from an active profile, click on the OPTIMIZE Menu, then click on the *Subtract Background* command. A list box will display all background profiles in the active profile file. Only profiles designated as background profiles (using the *Set Type* command from the PROFILE menu and then subsequently saved) will be available for subtraction. Once a background profile is selected it is immediately subtracted from the active profile, and the resulting profile is then displayed.

OPTIMIZE COMMAND: *Difference Profile*

The *Difference Profile* command allows the subtraction of any profile (except a background profile) from the active profile, creating a difference profile. This difference profile can be subsequently analyzed in the same manner as an original profile, however, it cannot be saved. The *Difference Profile* command is useful for the determination of the difference between two profiles such as a control and an experimental sample. To subtract one profile from another both profiles must be displayed in the same graph. When two profiles are displayed in the same graph, the program will automatically subtract the non-active profile from the active profile. If more than two profiles are displayed in the same graph, when the *Difference Profile* command is selected a hand will appear. Point the hand to the profile which you wish to subtract the active scan from, and click the left mouse button. Once the difference profile has been executed, the resulting difference profile will be displayed as the active profile.

OPTIMIZE COMMAND: *Average Profiles*

The *Average Profiles* command allows you to average any number of different profiles. Selecting this command when there are no profiles currently displayed on screen will display an entry box prompting for a filename. Click on the filename containing the profiles to be averaged. Note: If the profiles are not within the same file, the *Copy Profile* command must be used to move profiles to a common file for averaging. Click on all the profile names which you wish to be averaged and then click on the *Avg* button. Only the highlighted profiles will be averaged. The resulting average profile will then be displayed on screen.

If two to three profiles are currently displayed on screen in the same graph, or in separate graphs, selecting this command will average all of the profiles displayed on screen. The resulting average profile will be displayed as the active profile in the active graph, along with all of the original profiles in their respective graphs. Note: To average more than three profiles, select the *Average Profiles* command without any graph displayed in the Display Window, as described above.

OPTIMIZE COMMAND: *Origin*

The *Origin* command allows the identification of the gel lane origin within the profile. The *Origin* command is used in conjunction with the *Front* command for the determination of peak relative mobility, and subsequent molecular weight analysis. To specify the origin of the displayed active profile, click on the *OPTIMIZE* Menu, then click on *Origin.* The mouse pointer will change to a scissors icon which should be placed at the origin resolving interface, near the left edge of the profile (X minimum). Press the left mouse button and the portion of the profile left of the origin will be deleted with the point of “cut” now being the new origin.

OPTIMIZE COMMAND: *Front*

The *Front* command allows the identification of the sample lane front within the profile. The *Front* command is used in conjunction with the *Origin* command for the determination of peak relative mobility, and subsequent molecular weight analysis. To specify the front of the displayed active profile, click on the *OPTIMIZE* Menu, then click on *Front* . The mouse pointer will change to a scissors icon which should be placed at the dye front, near the right edge of the profile (X maximum) on or above the front location. Press the left mouse button and the portion of the profile right of the front will be deleted with the point of “cut” now being the profile front.

OPTIMIZE COMMAND: *Flip*

The *Flip* command is used to flip a profile which was drawn in the incorrect orientation, with the dye front near the left edge of the profile. To flip a profile simply click on the *OPTIMIZE* Menu, then click on *Flip.* The data point sequence of the displayed active profile will then be reversed, with the origin of the profile now located near the X minimum location.

OPTIMIZE COMMAND: *Align 2 pts*

The *Align 2 pts* command allows two overlaid profiles to be aligned along the X-axis in relation to two corresponding points on each profile. By aligning the overlaid profiles, the differences in profile object drawing position or sample shape can be compensated. Thus comparison of peak mobilities among different profiles is facilitated with the use *Align 2 pts* command. To align any two profiles which are overlaid in the same graph, click on the *OPTIMIZE* Menu, then click on *Align 2 pts.* The mouse pointer will be replaced by a downward pointing hand. Move the "finger" of the hand to a point on one of the profiles, such as the top of a peak, which you wish to align with a point on the other profile, and click the left mouse button. A small circle will remain identifying the first point location. Now move the hand to a desired or corresponding point along the second profile, and click or press and release the left mouse button. Once a point on the second profile has been identified, the program will automatically align the two profile along the X-axis.

OPTIMIZE COMMAND: *Align 4 pts*

The *Align 4 pts* command allows two profiles to be aligned not only along the X-axis, but along the Y-axis so that the profiles can be precisely overlaid on top of each other. By providing two "landmark" points along each profile, the *Align 4 pts* command uses a stretching routine, allowing for slight mobility differences between different gel lanes. To align and stretch any two profiles which are overlaid in the same graph, click on the *OPTIMIZE* Menu, then click on *Align 4 pts.* The mouse pointer will be replaced by a downward pointing hand. Move the "finger" to a point on one of the profiles, such as the top of a peak, which you wish to align with a point on the other profile, and click the left mouse button. A small circle will remain identifying the first point location. Now move the hand to a corresponding point along the second profile, and click or press and release the left mouse button. Repeat this procedure with the remaining two points. Once all four points have been identified, the program will automatically align the two profiles stretching the second profile as necessary.

OPTIMIZE COMMAND: *DragXY*

The *DragXY* command allows an active overlaid profile to be dragged in any direction around the graph. To drag an active profile in a XY direction, click on the *OPTIMIZE* Menu, then click on *DragXY.* The mouse cross hair will be replace by a vertical fish hook icon. Move the fish hook icon to a location within the graph where you wish to move the active profile, and click the left mouse button. The active profile will now appear at the new location. By pressing the left mouse button instead of clicking, a connecting line will be drawn from the profile to the new location and the profile will not move until the mouse button is released.

OPTIMIZE COMMAND: *DragX*

The *DragX* command allows an active overlaid profile to be dragged only in the X direction along the graph. This is useful for quickly aligning many profiles within the graph to facilitate interpretation while maintaining the profiles original Y position. To drag an active profile in the X direction, click on the *OPTIMIZE* Menu, then click on *DragX.* The mouse cross hair will be replace by a horizontal fish hook icon. Move the fish hook icon to any location on the active profile, and click the mouse button. Next, move the fish hook to the new X location, and hold down the mouse button. Notice that a connecting line will form between the two points. The profile will move when the mouse button is released.

OPTIMIZE COMMAND: *DragY*

The *DragY* command allows an active overlaid profile to be dragged only in the Y direction along the graph. This can be useful for quickly aligning or organizing many profiles within the graph to facilitate interpretation while maintaining the profiles original X (distance) position. To drag an active profile in the Y direction, click on the *OPTIMIZE* Menu, then click on *DragY*. The mouse cross hair will be replace by a vertical fish hook icon. Move the fish hook icon to any location on the active profile, and click the mouse button. Next, move the fish hook to the new Y location, and hold down the mouse button. Notice that a connecting line will form between the two points. The profile will move when the mouse button is released.

ENHANCE Menu

ENHANCE COMMAND: *Criteria*

The *Criteria* command allows you to specify the level of smoothing or enhancement for your particular sample data. When finished setting the desired parameters, click on the *Set* command button.

Window Size- This entry field allows you to enter the filtering window size value used by the *SMOOTH* command. Enter a value from 3 (minimum smoothing), up to 99 (maximum smoothing). This value determines the number of data points which the moving weighted algorithm will use to smooth the data.

Iterations- This entry field allows you to enter the number of iterations (passes) you wish to have the *SMOOTH* command perform using the specified window size. Enter a value from 1 to 99 smoothing passes.

En Boost- The enhancement boost factor is a constant which is multiplied by high spatial frequencies extracted from the data. The result is then added back into the original data. Enter a boost value between 1 and 10 as desired.

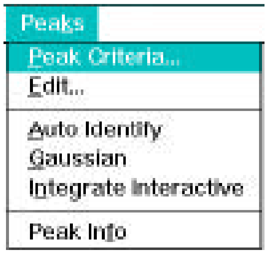
Display Each- When this check box is activated, the program will display each smoothed result profile when an iteration is specified.

ENHANCE COMMAND: *Smooth*

When this command is selected, the program will smooth or suppress high frequency noise within the active profile according to the window size and number of iterations assigned under *Criteria*. The algorithm used by the *Smooth* command is simply a moving weighted average of nearby data points. The larger the window size the more the data will be suppressed down to a lower frequency. The application of a moving weighted average filter to smooth the original data will change the shape of the curve, however, it will not change the area under the curve. That is, for every peak reduced an equivalent amount of area is distributed to the surrounding valleys.

ENHANCE COMMAND: *Enhance*

When this command is selected, the program will enhance the active profile according to the enhancement boost factor (En Boost) assigned under *Criteria*. The enhancement feature is designed to boost high spatial frequencies of partially resolved peaks. This is done by extracting the high spatial frequencies, multiplying them by a constant (the boost factor) and adding them back into the original data. In the enhancement process, we have extracted rather than suppressed components above the enhancement frequency. Note: Do not use boost factors so high that the results go off scale. If this rule is followed, the measured areas of peaks will be accurate.



PEAKS Menu

PEAKS COMMAND: *Peak Criteria . . .*
This command allows you to enter specific peak criteria parameters for use with the automatic peak identification routine. The criteria which you set are exclusion values which will determine which peaks within the profile will be identified and integrated. Any or all of the peak identification parameters can used to eliminate unwanted peaks when using the automatic peak identification routine.

Amplitude- This entry field allows you to select a minimum peak amplitude, whereby all peaks with a maximum peak height **below** this value will not be identified.

Width- This entry field allows you to select a minimum peak width (mm) "cut-off" value, whereby all peaks with a width less than this minimum value will not be identified.

Area- This entry field allows you to select a minimum peak area "cut-off" value, whereby all peaks with an area less than this minimum value will not be identified.

Sm Win- The Smoothing Window entry field allows you to select a smoothing window value for use in the programs automatic peak identification routine. Using a smoothing window has the same effect as the *Enhance* Menu's *Smooth* command. Entering increasing smoothing window values along with at least one iteration, identifies fewer unwanted peaks. **Note**: for the smoothing window to be operational a smoothing iteration value of at least one must be entered.

Sm Iter- The smoothing iterations entry field allows you to select the number of passes which the smoothing window will make over the profile. The more iterations that are performed, the more accurate the smoothing and subsequent peak identification will be. If the automatic peak identification routine is still identifying unwanted noise, either increase the smoothing window size, enter peak amplitude, width, or area criteria's, or perform an actual smoothing of the data points using the *Smooth* command.

Calc'd Win- This is only a message box displaying the smoothing window value calculated by the program during the last automatic peak identification. This value will be zero when the program is first started. The calculated smoothing window value serves a guide to the user when setting the smoothing window value under *Sm Win.* Increase the smoothing window to identify fewer peaks and decrease the smoothing window to identify more peaks.

PEAKS COMMAND: *Edit . . .*

The *Edit . . .* command allows you to manually create, adjust, or delete any peak within the active profile. When the *Edit . . .* command is selected, the PEAK EDIT window will replace the DISPLAY window, providing a new menu for peak editing. The following menu items are available within the PEAK EDIT window:

EDIT- The *EDIT* Menu contains the *Cancel* and *Exit* commands. Both commands will exit the PEAK EDIT window and return to the DISPLAY window. When the *Exit* command is used, all changes made in the PEAK EDIT window will be reflected in the DISPLAY window. To cancel editing done in PEAK EDIT window and return to the DISPLAY window press the *Cancel* command.

CREATE- The *CREATE* command allows you to quickly manually identify and number peaks within the active profile. When selected, the pointer will change to a "magic wand" icon. Simply move the point of the "magic wand" to the origin of the peak you wish to identify, and press the left mouse button. Continue pressing the left mouse button while dragging the "magic wand" to the right until the end (front) of the peak is reached. The area under the curve will fill in as the "magic wand" is dragged across the profile. If the boundaries of the peak you are creating overlap a previously created peak, the area of overlap will be not be filled in. When finished creating peaks, either click on the *CANCEL* or the *Exit* command under the *EDIT* menu to return to the DISPLAY window.

ADJUST- The *ADJUST* command allows you to quickly adjust the boundaries of a previously identified or created peak. When this menu/command is selected, the pointer will change to a "tuning fork" icon. Simply place the arrow head of the icon over the peak which you wish to adjust and press the left mouse button. The "tuning fork" icon will now change to a "magic wand" icon allowing you to redraw the area for that peak as outlined above under *CREATE*. When the mouse button is released after the adjustment, the icon will change back to the "tuning fork" allowing you to select another peak for adjustment. Click on either the *CANCEL* command or on the *Exit* command under the *EDIT* menu to return to the DISPLAY window.

DELETE- The *DELETE* command allows you to rapidly delete any peak within the active profile. When this command is selected, the pointer will change to a "hand grenade" icon. Simply place the arrow head of the icon over the peak which you wish to delete and press the left mouse button. The peak will immediately disappear as the graph is updated. Click on either the *CANCEL* command or the *Exit* command under the *EDIT* menu to return to the DISPLAY window.

CANCEL- This command allows you to cancel the current edit operation.

PEAKS COMMAND: *Auto Identify*

The *Auto Identify* command performs an automatic identification of all peaks within the active profile, which meet the criteria set under *Peak Criteria . . .* The integration algorithm used in the automatic peak identification program is a trapezoidal integration algorithm. To use the *Auto Identify* command, simply point and click on PEAKS and select *Auto Identify*. The area values for each identified peak can be found using the *Peak Info* command or by saving the profile and printing a report.

PEAKS COMMAND: *Gaussian*

The *Gaussian* curve fitting command is used for the integration of overlapping peaks. Because overlapping peaks are not "baseline" resolved, using the integration values determined with *Auto Identify* command would not be precisely accurate. The *Gaussian* curve fitting command calculates what the normal progression of the peak would be if it were not smeared (completely resolved), providing for more accurate quantitation values. The *Gaussian* curve fitting command can only be used once the peaks of interest have been identified using the *Auto Identify*, or *Edit . . .* commands. Once all the desired peaks within the profile have been identified, simply point and click on *PEAKS*, and select *Gaussian*. Instead of filling in the area under the curve, the *Gaussian* curve fitting command will overlay a different colored trace for the area boundary which was integrated for each peak. The area values for each identified peak can be found using the *Peak Info* command or by saving the profile and printing a report. All identified peaks within a particular profile may not be identified by the Gaussian function if the peak and /or the profile do not meet certain criteria required by the function.

PEAKS COMMAND: *Integrate Interactive*

The *Integrate Interactive* command allows you to quickly get area values for any peak or segment of the active profile. These values are not retained anywhere within the program and must be written down if they are to be used further. To use the *Integrate Interactive* command, simply click on PEAKS, and select *Integrate Interactive.* The pointer will change to an "integral" icon. Place the icon at the beginning of the area you wish to integrate, and press the left mouse button. The area box in the lower right corner will now display an area value of zero. Drag the icon to the end of the area you wish to integrate and release the mouse button. The area box will now display the integral value for the area just identified. Every time the left mouse button is pressed the area value will zero, and a new area can be integrated by dragging the icon. Please note that the icon must be dragged from left to right for proper integration. To exit the *Integrate Interactive* command, click on *OPERATION* and select *Cancel.*

PEAKS COMMAND: *Peak Info*

The *Peak Info* command provides information regarding any peak which has been identified within the active profile. To use the *Peak Info* command, point and click on PEAKS and select *Peak Info.* The pointer will change to a "hand" icon. Using the icon, point to any peak area, and click in the left mouse button.

The following is the information provided in the Peak Information dialog box:

Number- This identifies which peak the peak information box is currently listing information for.

Name- If the peak is a standard from a standard profile which has had a molecular weight standard curve calculated, then name belonging to the peak would be displayed here.

Start- This identifies the x location (mm) where the peak starts (origin).

End- This identifies the x location (mm) where the peak ends (front).

Amplitude- This identifies the maximum peak height (O.D. or Counts.)

Amp X Loc- This identifies the x location (mm) of the peak amplitude.

Width- This identifies the width of the peak (End minus Start).

Area- This identifies the area (O.D. or Counts/ mm) associated with the peak. In terms of absolute quantitation, this value would be equivalent to concentration.

Percent Total- This identifies the percentage of the total area under the profile curve which is associated with that peak.

Percent Selected- This identifies the percentage of the total area selected which is associated with that peak. The total selected area is the area comprised of only the peaks which have been identified.

Distance- This identifies the distance (mm) the peak amplitude is from profile origin.

RF- This is the relative mobility value calculated for the peak. This is the ratio of the distances traveled by the spot and the dye front. This value is always less than one.

MW- The is the molecular weight value for the peak which was determined using the molecular weight function. A value will only be displayed if the molecular weight of the peak has been previously calculated.

Amount- The amount in peak will only be calculated and displayed here if a profile track loading amount was entered during the capture process. This amount in peak value is calculated from the percent total profile area attributed to this peak.

Mean- The mean value will only be displayed if Gaussian curve fitting was performed. This is the mean x value (mm) for the curve.

Std Dev- The standard deviation value is one standard deviation from the calculated mean. This value will only be displayed if Gaussian curve fitting was performed.

Amp- The Gaussian curve amplitude (max. curve height) will only be displayed here if Gaussian curve fitting was performed.

SCALE Menu

SCALE COMMAND: *Absorbance (OD)*
If the data units of the currently active scan are % transmission/reflectance, or %absorbtance, selecting this command will convert the data units to absorbance (OD), and the Y axis of the graph will reflect this change. The OD scale is used for concentration quantities and represents $-\log_{10}$ (data/calibration).

SCALE COMMAND: *% Transmission*
If the data units of the currently active scan are absorbance, or % absorbtance, selecting this command will convert the data units to % transmission, and the Y axis of the graph will reflect this change. This scale is useful for analyzing data which is in the % transmission/reflectance mode. Such data can be from photographs of ethidium bromide gels, whereby the bands in the photograph are white, and the background is black.

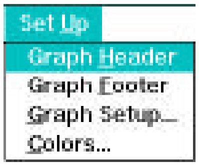
SCALE COMMAND: *% Absorbtance*
If the data units of the currently active scan are absorbance, or % transmission / reflectance, selecting this command will convert the data units to % absorbtance, and the Y axis of the graph will reflect this change.

OPERATION Menu

OPERATION COMMAND: *Complete*
The *Complete* command will finish in any operation which is currently in progress. This command is used after manually constructing a baseline. *Complete* instructs the program to fill in the constructed baseline or to go ahead and subtract the baseline from the active profile, depending upon whether the *Baseline Subtraction* box is checked under the *Graph Set Up* menu. Additionally, the *Complete* command will terminate the *Peak Info* and *Graph Info* commands.

OPERATION COMMAND: *Cancel*
The *Cancel* command will cancel any operation currently in progress. The *Cancel* command will be highlighted when available for canceling a operation in progress. Simply click on the OPERATIONS Menu and select *Cancel* and the operation will be canceled, bringing the program back to its previous status.

OPERATION COMMAND: *Redraw*
The *Redraw* command will redraw all profiles displayed and removes any profile or window artifact which may have been left behind from a previous function. Simply click on the OPERATIONS Menu and select *Redraw* and all profiles will be re drawn.



SETUP Menu

SETUP COMMAND: *Graph Header*
The *Graph Header* command allows you display text at the top of the graph containing the active profile. When selected, a dialog box appears where you can enter text and set horizontal justification. Click on the *Set* button when finished. This is a convenient way to identify a graph for printing, and while it is on the screen. The header however is not saved once the graph has been closed. This command is only functional when a profile graph is displayed.

SETUP COMMAND: *Graph Footer*
This command is identical to Graph Header, but the text appears at the bottom of the graph. This command is only functional when a profile graph is displayed.

SETUP COMMAND: *Graph Setup . . .*
This command allows you to have complete control of the graph display parameters. When selected the following dialog box will appear:

Axis Calculations Field-
Range of Active Profile- Graph axis scaling will adjust according to only the active profile, and will not take into account the range of other profiles within the same graph.

Range of All Profiles- Graph axis scaling will adjust according to the range of all profiles within the graph.

Axis Padding (%)- This entry field allows you to enter the percentage of the graph which you want to have as padding around the profiles. If no padding is selected, the zero values for the profile(s) will lay directly on the X-axis. The default value is 10% padding.

Axis Labels Field-
Labels On All Graphs- Axis labels will be drawn on all graphs when multiple graphs are displayed. This is the default setting.

Labels On Single Graph- Axis labels are drawn when a single graph is displayed, however, when a new graph is added no axes will be displayed on any of the graphs.

Labels On No Graph- No axis labels will be drawn on any of the graphs.

Graph Operations-
Fill Peaks On Identify- Each peak on the graph will be filled in once it is identified using either the automatic or manual peak identification methods. If this box is not checked, peaks will not be filled in, and the only way to identify the peaks in the graph will be if the labels and perpendiculars are displayed.

Display Peak Labels- Peak labels, numbers, and molecular weight standard names, will be displayed above the corresponding peak.

Display Perpendiculars- When this box is checked, perpendiculars will be displayed identifying the X (mm) boundaries for each peak.

Other Options-

Subtract Baseline- When this box is checked, the subtract constructed baseline function is changed from a three step operation to a two step operation. The default, two step method is as follows: 1) A baseline is constructed using the *Construct Baseline* command from the *Optimize* Menu. 2) The *Complete* command is selected from the *Operation* Menu, and the constructed baseline is automatically subtracted from the active profile.

When this check box is not checked, the subtract constructed baseline function becomes a three step operation: 1) A baseline is constructed using the *Construct Baseline* command from the *Optimize* Menu. 2) The *Complete* command is selected from the *Operation* Menu, the constructed baseline becomes filled in for viewing and is not subtracted from the active profile until; 3) the *Sub Constructed Baseline* command from the *Optimize* Menu is selected.

Cross Hair Cursor- When this box is checked, the mouse cursor is changed from an arrow to a cross hair cursor. This is the default option.

SETUP COMMAND: *Colors*

The *Colors* command allows you to adjust the colors of all profiles and the graph background. This enables you to easily differentiate the profiles. The default colors for the profiles and graph background are: Profile 0= white, Profile 1= red, Profile 2= yellow, Profile 3= blue, Active profile= Fuschia, Graph background=black.

To change the color from its default value perform the following steps:

• Select the *Colors* command from the *SET UP* Menu to display the Colors dialog box.

• Click on the background or appropriate profile command button. The color window will now display the color currently assigned to that profile or background.

• Adjust any or all of the 3 color wheels by pressing either the up or down arrow until the desired color shade is displayed in shade window.

• Set the color by clicking on the *Fix* command button. Continue this procedure for any or all of the graph options as desired. When finished simply click on the *Exit* command button.

To reset the color or gray shades to their default values, simply click on the *Default* command button, then the *Exit* command button.

Right Mouse Button Commands

When a profile is displayed in the DISPLAY function, the **right** mouse button can be pressed to activate an additional command menu. This small menu box contains commands which act upon the active profile, and will only be displayed when the cursor is in the active graph. If more than graph one graph is displayed, the menu box will only be active in the graph containing the active profile. The following are the commands available in this menu box:

Zoom-
When the *Zoom* command is activated, the cursor will change to a magnifying glass. To zoom a specific section of the active profile, place the magnifying glass above and to the left of the area you want to box for zooming. Press the left mouse button and drag the mouse down and to the
right, until the desired area is boxed. Release the left mouse button. To work only with the zoomed section, click on the *Optimize* Menu and select the *Snip at View* command. This performs a data modification, eliminating all sections of the profile not currently in view.

Scroll-
When the *Scroll* command is selected, the cursor will change to a scroll icon, allowing you to drag the profile(s) about the graph. Using Scroll command will drag all profiles within the graph, and maintain their exact coordinates. This command differs from the drag commands in that the coordinates of the profiles are always maintained. When finished scrolling, press the right mouse button and select the *End* command.

End-
The *End* command will terminate either the *Zoom* or Scroll commands.

Previous-
The *Previous* command will toggle between the current display and the previous view modification which was performed. If you wish to go back to the zoomed section simply select *Previous* again. This command will not be operational if a data modification has been performed using the *Snip at View* command.

Restore- If a series of view modifications have been performed, e.g., multiple levels of zoom, and you wish to restore the original profile or the last data modification, simply select the *Restore* command.

Print- The *Print* command allows you to print the active graph and its contents. This is the only way to print the actual profiles. When this command is selected, a dialog box will appear stating that the graph is being sent to the printer. To cancel the *Print* command, click on the *Cancel* command button in the dialog box.

Molecular Weight Function
These commands allow for the entering of standard molecular weight values into a table, calculation of a relative mobility vs. log molecular weight standard curve, and calculation of an unknown molecular weight.

MODE Menu
The mode menu is continuous throughout the Profile Analyst software program, allowing the user to move rapidly from one function (Display, Weight, and Report) to another without having to go through multiple menu levels. Also available in the *MODE* Menu is the *Exit* command, which allows the user to exit to the main menu or to another function.

FUNCTIONS Menu

FUNCTIONS COMMAND: *Size Standard Editor*
When this command is selected, the Standard Editor dialog box will appear allowing you to either create a new standard molecular weight file (table) or edit an existing one. Each standard name and corresponding molecular weight in a standard molecular weight file is associated with a peak in a standard profile, using the *Calculate Standard* function.

Size Standard Editor dialog box.

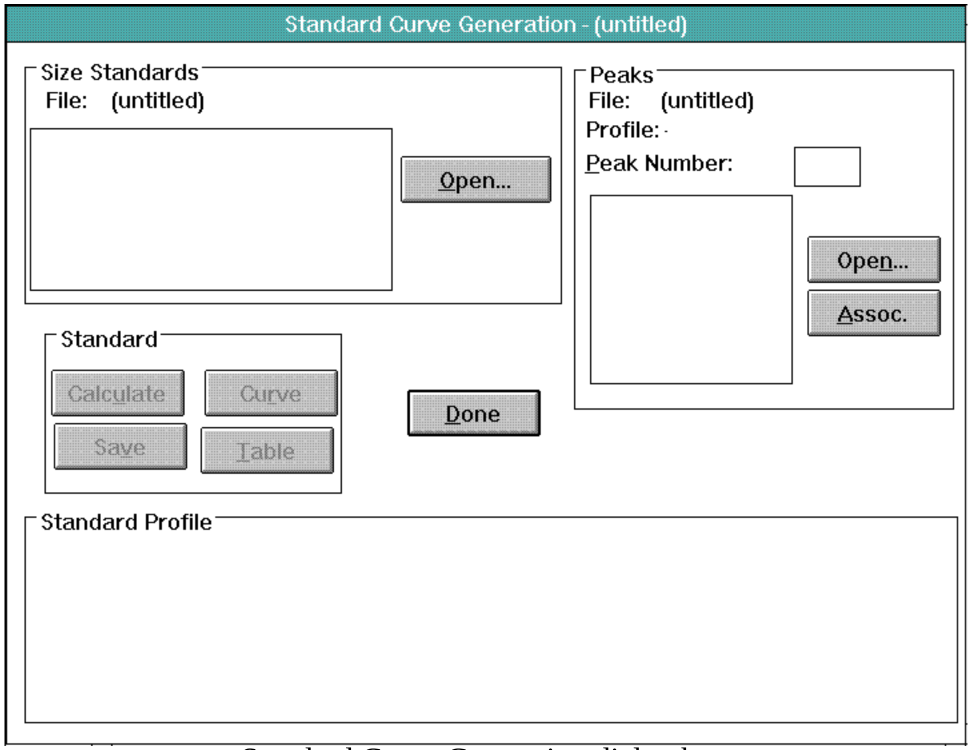
To create a new standard molecular weight file:
Click on the *FUNCTIONS* menu and select the Size Standard *Editor* command. A dialog box appears. Enter the names and molecular weight values for your molecular weight standards by simply typing in the standard name, pressing <Tab>, and entering the corresponding weight. **Note: The DNA size standards must be entered in terms of bases and not kilobases.** Finally click on the <Enter> command button or press <Return>. Continue this procedure until all the standard names and molecular weights have been entered. Note that the order of entry is not important as they will automatically by placed in order of decreasing molecular weight. When finished, click on the *Save* command button. A dialog box appears prompting for a filename. Enter a filename and click on the *OK* command button. The filename will automatically be appended with a .MW extension.

To edit an existing standard molecular weight file:
Click on the *FUNCTIONS* menu and select the Size *Standard Editor* command. A dialog box appears. Click on the *Open* command button. A standard Windows entry box appears listing only the existing standard molecular weight files within the current drive and directory. Double click on the standard file that you wish to open. Once opened, the name of the file will be in the title of the dialog box, and the first standard and corresponding molecular weight of the file will be displayed in the entry boxes.

> To add a standard to the weight table, type over the currently highlighted name and molecular weight and then click on *Enter*. Note: the standard and molecular weight you've typed over is not deleted from the table.
>
> To delete, or change any of the values within the table use the *Next* or *Previous* commands to step through the table until the desired standard and molecular weight are displayed. To delete that table entry press Delete and both the standard and corresponding molecular weight will be deleted from the table. To change any standard or molecular weight value for an entry, highlight that value and then enter the new value. Click on *Enter* and the table will be updated to reflect the changes.

To start a new standard molecular weight file:
Click on the *New* command button and a dialog box will appear asking if you want to save changes to the current file. Answer as appropriate. The edit fields will now clear allowing entry of data for a new standard molecular weight file.

FUNCTIONS COMMAND: *Calculate Standard Command*
The *Calculate Standard* command is used to calculate a relative mobility vs. log molecular weight standard curve from standard profile data. When this command is selected, the following full screen dialog box appears with interior fields: the size standards field, the peak field, and the standard field.



Standard Curve Generation dialog box.

Size Standards Field

The size standard field contains a scrolling list of standards and their corresponding molecular weights which can be associated with peaks within a profile file. The *Open* command button in the standard field opens an entry box which prompts for a standard molecular weight filename. This entry box will only list filenames which have the .MW extension. Double click on a standard molecular weight file to open it. To view the complete standard and molecular weight table it may be necessary to use the scroll bar.

Peak Field

The peak field displays a scrolling list of the peaks contained within a standard FILE which is opened. Clicking on the *Open* command button in the peaks field will display a dialog box which prompts for the name of the file in which the molecular weight standards profile is stored. Select a standard profile file by double clicking on it. A second dialog box will then appear prompting for the standard profile name. Double click on the desired standard profile name. The table will now list all the peaks which have been previously identified in the profile. The profile itself will also be displayed in the graph in the lower part of the screen. The *Assoc.* command button is used to associate a highlighted standard listed in the Size Standards field, to a appropriate highlighted Peak number in the Peak field. To continue associating size standards and peaks down the tables, simply continue to press the *Associate* command button. Note that the size standards are listed in descending order of molecular weight and that the peaks are listed in descending order of relative mobility.

Standard Field

The standard field contains the command buttons associated with calculation of the standard curve. Once the peaks have been associated with the correct size standard name and weight, the *Calculate* command button will become highlighted. Click on the *Calculate* command button to specify any or all of the Regression types which you would like to have calculated. After clicking on the desired Regression type click on the *OK* command button. The regressions will then be quickly calculated, the Regression type dialog box will disappear, and the Calculate button will be grayed. To see the relative mobility vs. log (MW) scatter plot and the different regression curves calculated, click on the *Curve* command button. A scatter plot graph of relative mobility vs. log (MW) will then appear. To display the different regression curves on the graph, simply click on any or all of the check boxes along the bottom. To remove any of the curves, simply click again on the appropriate check box. The graph will update showing the selected regression curves in different colors. This feature is useful for determining which regression type provides the best fit of the data points. Due to the sigmoidal shape of many standard curves, the third or fourth order regression will often give the best fit of the curve. The user is however cautioned when using higher orders of regression (Appendix C). To close the scatter plot double click on the box in the upper left corner. Once the standard curves have been calculated, the *Save* command button must be selected in order to use this data for calculation of an unknown. The *Table* command button will display a list of the associations. To quickly close the table double click in the upper left hand corner on the menu bar.

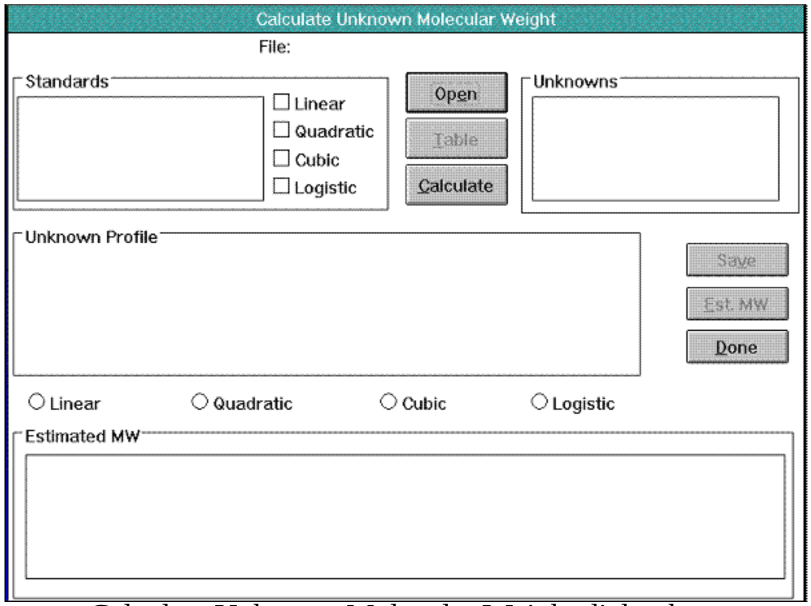
Standard Profile Graph

The standard profile which is selected for generation of the standard curve is displayed in this field. The peaks which have been identified in this profile are shaded and have a peak number located above it. If the number of peaks which have been identified in the profile is greater than the number of standards listed in the molecular weight file, then either the molecular weight file or the profile will have to be edited.

When the cursor is moved into the profile graph the cursor becomes a vertical line which when moved left and right in the graph will correspond to the X (mm) readout value on the bottom of the graph.

FUNCTIONS COMMAND: *Calculate Unknown Command*

The *Calculate Unknown* command allows for the calculation of an unknown sample's molecular weights using a standard curve generated with the *Calculate Standard* command. Both the standard profile(s) and unknown profile(s) must be contained within the same file to be available for use in the *Calculate Unknown* command. When this command is selected, the following full screen dialog box appears with interior fields: standards, unknowns, unknown profile, and estimated molecular weight.



Calculate Unknown Molecular Weight dialog box.

Standards Field

The standards field will display all the standard profile names within the opened file, which are available for use in calculation of the unknown molecular weights. Only standard profiles which have had peaks identified and a standard curve calculated will be listed in this box. When a standard profile is highlighted in the list box, the regression radio button(s) on the right will be highlighted identifying the types of regressions which have been calculated using this profile. Use the scroll bar if necessary to highlight a desired standard profile.

Unknowns Field

The unknowns field will display all the unknown profile names within the opened file, which are available for use in calculation of unknown molecular weights. All profile names within this list box will have an appendix, e.g. sample.000-a, as only profiles which have had peaks identified will be available for molecular weight calculation. Use the scroll bar if necessary to highlight a desired unknown profile. Note that the unknown profile(s) must be from the same file as the standard profile(s).

Unknown Profile Field

The unknown profile field displays a graph of the unknown profile and provides an on-screen readout of the X (mm) location, relative mobility (Rf), and the calculated molecular weights, at the position of the cursor. When the cursor is moved into the graph it changes to a vertical line, enabling you to precisely position the mouse. The default on-screen readout values are for the cubic regression curve type. To change the on-screen readout values to reflect the specific regression curves, simply click on the appropriate radio button below the graph.

Estimated MW Field

The estimated MW field displays a table of the relative mobility (Rf), calculated molecular weights (linear, quadratic, cubic, and logistic), for each peak identified within the unknown profile. When the *Table* command button is pressed, Rf, MW, and standard names for each peak of the standard profile are displayed in this field. To view the estimated molecular weights again, click on the *Est. MW* command button to the right of the field. Use the scroll bar if necessary to view the full list.

Calculating Unknown Molecular Weight-

To calculate the molecular weight of an unknown, click on the *Open* command button. The Open file entry box will be displayed, listing all files within the current drive and directory. Double click on the file containing the standard and unknown profile(s) you wish to use in the calculation. Note: the standard and unknown profile(s) must be contained within the same file. Once opened, the file name will be displayed at the top of the dialog box, and the standard and unknown profiles available for calculation will be displayed in their respective fields. Scrolling if necessary, highlight both the standard profile and the unknown profile, which you wish to use for the molecular weight calculation. Click on the *Calculate* command button and the unknown profile graph will be displayed, along with the estimated molecular weights.

Move the cursor into the graph, and read the relative mobility and molecular weight values for any position along the profile. Click on the regression radio buttons so that the readout values reflect those regression curves. To save the calculated molecular weights for the unknown profile click on the *Save* command button before exiting using the *Done* command button. To calculate the molecular weight of another unknown profile in the opened file, using the same standard profile, highlight the new unknown profile name then press *Calculate*. The new unknown profile graph will be displayed and the estimated molecular weight table will be updated to reflect the new profile. If calculation of unknown using another standard profile is desired, simply highlight the unknown profile and the new standard profile then press *Calculate.*

Reports Function

The Reports commands allow you to print, view, or export a report on a profile contained within any file. The reports include: capture settings, information about the profile itself, area report if peaks have been identified, and molecular weight report if molecular weights have been calculated. Note: to print the profile in a graph, the profile must be displayed using the DISPLAY function and printed by pressing the right mouse button and selecting the *Print* command.

FUNCTIONS Menu

FUNCTIONS COMMAND: *Contents*

When this command is selected, a dialog box appears containing check boxes for each section of a report which can be viewed or printed. Each report contains profile name, profile type, the file name where it exists, and the date when it was printed. The following is a description of the report sections which can be printed:

Capture Settings- A list of the capture settings will be printed.

Profile Data - A section of general profile information will be printed. This section includes origin and front X (mm) location, number of peaks identified in the profile, and total area under the profile curve.

Peak report- Information about each identified peak will be printed. This information includes: identification of whether Gaussian modeling has been performed, peak number, position within the profile, distance from the origin, peak relative mobility, maximum peak height, peak width, percentage of total area peak comprises, and amount of material in peak.

Molecular Weight Report- The molecular weight report section will be printed. If the profile being printed is a standard profile then the report will be a standard report consisting of: the peak number, peak name, relative mobility (Rf), and known molecular weight. Additionally, information regarding the standard curves will be provided. This includes: the equation, the coefficient values, R squared, mean Rf, mean Rf standard. error, for each order of regression. Normally, the regression order providing the highest R squared value, and the lowest standard error would provide the most accurate molecular weight values. The user is cautioned when using higher orders of regression (see Appendix C).

If the profile is an unknown profile then the report will consist of: the unknown peak number, its relative mobility, and the molecular weights as calculated using the first, second, and third order regressions.

Top Printer Margin- This command adjusts the left printer margin. Simply enter a value between 0 and 8.

After selecting the desired options to be printed, click on the *Set* command button.

FUNCTIONS COMMAND: *View*

When this command is selected, a dialog box appears, prompting for a filename. Select the file name and click on the *Print* command button. A second dialog box appears prompting for profile name. Select the profile name and click on the *Add* command button. The report sections selected with the *Contents* command will be displayed. Two full reports can be viewed at one time to allow profile report comparison. To print active reports at this point, select the *Print* command from the FUNCTIONS menu. To make either of the viewed reports active, click on the title bar of the report you wish to print.

FUNCTIONS COMMAND: *Print*

When this command is selected, the open file entry box will appear. Double click on the file which contains the profile report you wish to print. A new dialog box will appear containing a list box of the profiles contained within the file you have opened. Click on a profile name to highlight it. The two fields to the right contain radio buttons which will highlight, identifying the type of profile and any operations which have been performed on it. Click on the *Add* command button to send this profile report to the printer.

FUNCTIONS COMMAND: *Clipboard*

Selecting this command will copy only selected sections from a viewed profile report to Windows Clipboard which can then be transferred to other applications, such as Microsoft Excel or Word. Report data will be transferred in a ASCII tab delimited format.

FUNCTIONS COMMAND: *Save as*

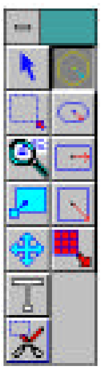
Selecting this command allows you to save the reports. Enter in the appropriate report file name followed by the extension .rpt and click on OK. Reports are saved in a ASCII tab delimited format.

5.2 Volume Analyst - Volume Integration

Volume Tools

Choosing *Volume Analysis* from the ANALYSIS menu displays the Volume Tools on the right side of the Tool Palette. When displayed, the Volume tools consists of five boxes, each of which represents a specific object used to identify spots on an image and generate a volume integration report. Double clicking on the tool icon displays the preferences for that specific tool.

When a tool is selected and used, the specified object will be displayed on the image. All of the objects (circle, ellipse, rectangle, square, and grid) can be modified by using the object tool which is activated when the cursor is placed on an active object. A maximum of 200 volume objects can be created and displayed on an image. Conversely, volume objects can be deleted from an image using the EDIT menu command *Delete* or *Clear all*. Selecting *Analyze Volume Profile(s)* from the ANALYSIS menu displays the volume integration report for all objects within the image.



Circle Tool

• Click on the circle tool icon.

• Using the mouse, move the cross-hair cursor to a point above and to the left of the spot in the image you want to quantitate.

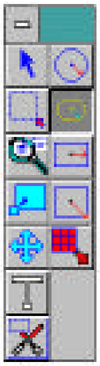
• Click the left mouse button, and drag the mouse downward and to the right. A square will appear on the graphics screen. Moving the mouse adjusts the size of the square so that it completely encloses the spot. A circle will appear surrounding the spot when the mouse button is released.

• *Object Altering tools*- When placed on the circle, these tools allow the object to be modified to better fit the data.

Movement arrow- Allows the circle to be moved to another location

<Ctrl> +Movement arrow- Clicking the movement arrow while holding the <Ctrl> key down, duplicates the circle

Sizing square- When placed on the square surrounding the circle, this option allows the circle to be reduced or enlarged



Ellipse Tool

• Click on the ellipse tool icon.

• Using the mouse, move the cross-hair cursor to a point above and to the left of the spot in the image you want to quantitate.

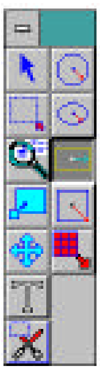
• Click the left mouse button, and drag the mouse downward and to the right. A square will appear on the graphics screen. Moving the mouse adjusts the size of the square so that it completely encloses the spot. An ellipse will appear surrounding the spot when the mouse button is released.

• *Object Altering tools*- When placed on the ellipse, these tools allow the object to be modified to better fit the data.

Movement arrow- Allows the ellipse to be moved to another location

<Ctrl> +Movement arrow- Clicking the movement arrow while holding the <Ctrl> key down, duplicates the ellipse

Sizing square- When placed on the square surrounding the ellipse, this option allows the ellipse to be reduced or enlarged



Rectangle Tool

• Click on the rectangle tool icon.

• Using the mouse, move the cross-hair cursor to a point above and to the left of the spot in the image you want to quantitate.

• Click the left mouse button, and drag the mouse downward and to the right. A square will appear on the graphics screen. Moving the mouse adjusts the size of the square so that it completely encloses the spot. A rectangle will appear surrounding the spot when the mouse button is released.

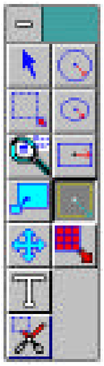
• *Object Altering tools*- When placed on the rectangle, these tools allow the object to be modified to better fit the data.

Movement arrow- Allows the rectangle to be moved to another location

<Ctrl> +Movement arrow- Clicking the movement arrow while holding the <Ctrl> key down, duplicates the rectangle

Sizing square- When placed on the rectangle, this option allows the rectangle to be reduced or enlarged

Rotating arrow- Clicking the mouse on the center dot and moving the mouse vertically rotates the rectangle. Note: For fine adjustments of the angle of rotation, click on the center dot, hold the mouse button down, and move the cursor farther away from the rotating point.



Square Tool

• Click on the square tool icon.

• Using the mouse, move the cross-hair cursor to a point above and to the left of the spot in the image you want to quantitate.

• Click the left mouse button, and drag the mouse downward and to the right. A square will appear on the graphics screen. Moving the mouse adjusts the size of the square so that it completely encloses the spot. A square will appear surrounding the spot when the mouse button is released.

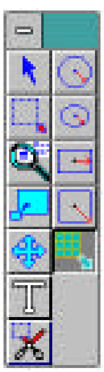
• *Object Altering tools*- When placed on the square, these tools allow the object to be modified to better fit the data.

Movement arrow- Allows the square to be moved to another location

<Ctrl> +Movement arrow- Clicking the movement arrow while holding the <Ctrl> key down, duplicates the square

Sizing square- When placed on the rectangle, this option allows the square to be reduced or enlarged

Rotating arrow- Clicking the mouse on the center dot and moving the mouse vertically rotates the square. Note: For fine adjustments of the angle of rotation, click on the center dot, hold the mouse button down, and move the cursor farther away from the rotating point.



Grid Tool

• Double click on the Grid icon to display *Grid Preferences.*

<u>Options Field-</u>
• Select either a 8 x 12 (96 well Bio-Dot apparatus format), 8 x 6 (48 well Bio-Dot SF Slot Blot format), or custom grid configuration. For custom configuration, designate the number of columns and rows.

<u>Grid Field-</u>
• Custom configuration allows you to choose the shape of the cells in a grid. If using a custom configuration, select the shape (circle, ellipse, square, or rectangle) of the grid cell. Click on the radio button. The 96 well Bio-Dot configuration always use circle cells, and the 48 well Bio-Dot SF configuration always use rectangle cells.

• Using the pull down menus, select the color of the grid, and the color of the sizing cell.

• Custom configuration also allows you to group or ungroup the cells in a grid. Check the *Grouped* box if you wish to move the grid components as a group. Leave the box unchecked if you want to move each cell independently. This may be preferred if spots vary significantly in location. Ungrouped cells can be later grouped and moved as one using the *Block Select* tool from the Tool Palette.

• Check the *Save Settings* box and Click OK.

Making the Grid-

• Using the mouse, move the cross-hair cursor to a point above and to the left of the rows and columns of spots in the image.

• Click the left mouse button and drag the mouse downward and to the right. A square will appear on the graphics screen. Moving the mouse adjusts the size of the square so that it completely encloses all of the spots. A grid with the specified cells will appear when the mouse button is released.

Moving the Grid-

• Ungrouped - Each individual cell can be moved by first activating the grid cell of interest. Once the cell is active, the cursor becomes the moving arrow tool. Hold down the left mouse button and move the cell to another location.

• Grouped- A grouped grid can only be moved if the entire grid is active. This is accomplished by placing the cursor on a point of the grid other than the sizing cell and clicking the mouse. Once the grid is active, the cursor becomes the moving arrow tool. Hold down the left mouse button and move the cell to another location. More than one grid cell can be moved by using the block select tool to activate multiple cells.

• Cut and Paste - Activate the grid by clicking on it with the mouse and select *Cut* from the EDIT menu. The grid will disappear from the screen. Next, select Paste from the EDIT menu. Place the cursor, now a Capital P, to the new grid location, and click the left mouse button.

Sizing the Grid-

• Ungrouped - Ungrouped cells **cannot** be sized as a complete grid.

• Grouped - With an active grid, place the cursor at the edge of the grid. Notice that the cursor has changed to the sizing square. Click the mouse button and drag the sizing square until the grid is the proper length and width. **Note:** The grid can only be reduced to the size in which all of the cells are contained in the grid. To further reduce the size of the grid, reduce the size of the cells.

Sizing Cells in a Grid-

• Ungrouped - Click the hand pointer on any cell of the ungrouped grid to activate the cell. Place the cursor on edge of the cell (the dotted square if the cell is a circle or ellipse). Notice that the cursor has changed to the sizing square. Click the mouse button and drag the sizing square until the cell is the appropriate size, and release the mouse button. All of the cells within the ungrouped grid will change to the same size as the modified cell.

• Grouped - Click the hand pointer on the sizing cell (upper left cell) of the grouped grid. Place the cursor on edge of the sizing cell (the dotted square if the cell is a circle or ellipse). Notice that the cursor has changed to the sizing square. Click the mouse button and drag the sizing square until the cell is the appropriate size, and release the mouse button. All of the cells within the grid will change to the same size as the modified cell.

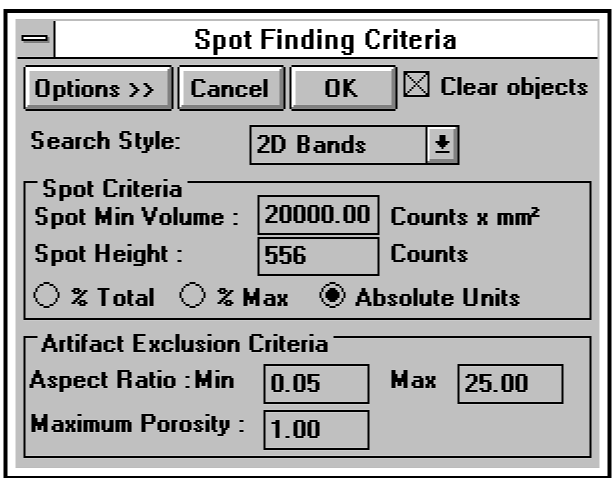
Rotating a Grid-
• Grouped - Place the mouse on the center dot of the grid. Notice that the icon has changed to a rotating arrow. Click and hold the mouse button on the center dot. Moving the mouse vertically while holding down the left mouse button, rotates the column.

Duplicating a Grid
• Copy and Paste - Activate the grid by clicking on it with the mouse and select *Copy from* the EDIT menu. Next, select Paste from the EDIT menu. Place the cursor, now a Capital P, to the new grid location, and click the left mouse button. Grids can be copied and pasted within the same image, or between two different images.

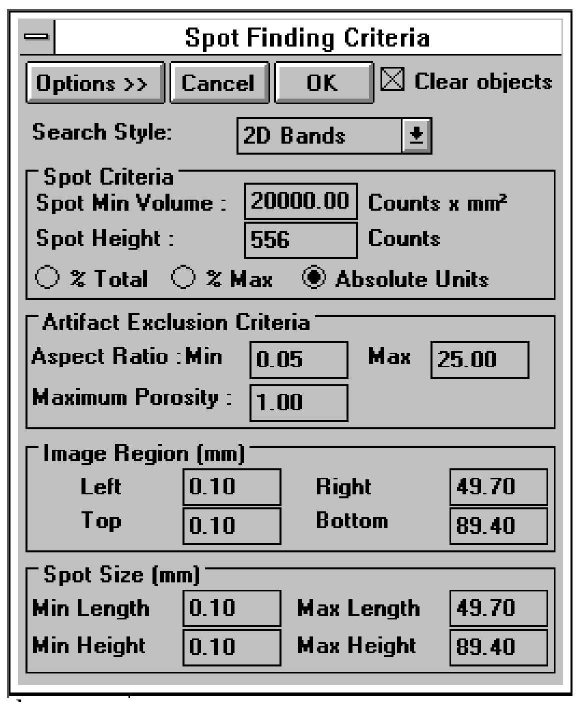
• *<Ctrl> +Movement arrow*- Clicking the movement arrow while holding the <Ctrl> key down.

Automatic Spot Detection

Selecting *Volume Analysis* from the ANALYSIS menu also activates the *Automatic Spot Detection* command. Selecting this command displays a spot finding criteria dialog box. Within this box, image region, spot size, spot criteria, and artifact exclusion criteria can be set.



a.



b.

Spot finding criteria dialog box, (a) unexpanded, (b) expanded with *Options>>* button.

Search Style - This pull-down menu allows rapid selection of 2D bands or spots.

Spot criteria - Minimum peak density and peak height of the spots to be found. Spot minimum volume is the smallest peak density in O.D or Counts.x mm^2. Spot height is the smallest peak height or amplitude in the z direction in O.D or Counts. Spot criteria can be defined as % total, % maximum, or absolute units. Spot criteria (spot minimum volume and spot height) is critical in detecting variations in the number of spots detected.

Artifact exclusion criteria - Consists of aspect ratio and porosity criteria and controls the identification of narrow, elongated spots which may be smears or artifacts.

Aspect Ratio:
Aspect ratio limits peak identification of smears in the x or y direction. Aspect ratio is calculated from the dimensions of the object by the formula (Y size)/(X size). Tall thin objects will have high aspect ratios and low flat objects will have small ratios. A round peak will have a ratio of 1.0. The aspect ratio is useful in rejecting artifacts since most artifacts have either very small or very large aspect ratios, whereas most peaks will have an aspect ratio close to 1.0. A range from 0.25 to 4.0 would allow a 4:1 elongation in either direction and is a reasonable starting range.

Porosity:
Porosity limits peak identification of smears in the diagonal direction. Porosity is useful in distinguishing spots from smears, streaks, and scratches. It is determined by drawing a rectangle around each object. If the object is a spot, most of the rectangle is filled. If the object is a smear, little of the rectangle is filled. The fraction of the rectangle not filled is the porosity.

Porosity = 1-[(area of the rectangle) / (X size) x (Y size)]

A spot has a porosity close to 0. A smear has a porosity close to 1.0 (Most of the area of the rectangle is not filled by the object. As a rule of thumb, an object with a porosity over 0.7 is most likely an artifact. However, each sample should be tested experimentally to determine the porosity limits.

Image region criteria - Visible when the *Options>>* key is selected, this criteria limits the specific area of an image for spot detection. Using the image rulers, left, right, top, and bottom millimeter dimensions may be entered. This criteria will default to the entire image area.

Spot size criteria - Visible when the *Options>>* key is selected, this criteria limits the minimum size of the spots to be found.

Note: All volume objects displayed through the use of the automatic spot detection feature cannot be moved, copied, or duplicated. Only volume objects drawn manually can be moved, copied, or duplicated.

***The number of spots found is dependent upon the set criteria. All spots on an image may not conform to a single criterion, and therefore multiple analyses may be required to find all spots on some samples.**

Volume report

The volume report provides quantitative information about each object on the image. Within the report, values can be sorted; column values can be hidden; standard values can be entered; and a standard curve can be generated. The volume report results include:

***Label* -** Identifies the object's Label as it is displayed on the screen. Clicking on the active cell and using the <Enter> key of selecting Modify from the EDIT menu opens the Edit Volume Report dialog box and allows you to change the label of the object.

***Name* -** Identifies the object's name as it is displayed in the edit volume report window. Clicking on the active cell and using the <Enter> key of selecting Modify from the EDIT menu opens the edit volume report window and allows you to change the name of the object.

***Type* -** Sample types include *Unknown* , *Standard*, or *Background*. Clicking on the active cell and using the <Enter> key of selecting Modify from the EDIT menu opens the edit volume report window and allows you to change sample type.

***% Vol* -** The percent volume of the object in relation to the total volume of all objects identified

***Area* -** Area of the identified object is in units of mm^2.

Pixels- Total number of pixels within the object

Volume - The integrated volume of the object is (O.D. * Area) or (Counts * Area).

Mean - Average O.D. or Counts of the pixels within the object, where O.D. or Counts = Volume/Area.

Density- Volume / mm^2

Maximum- Maximum O.D. or Counts value within the object.

Minimum- Minimum O.D. or Counts value within the object.

***Standard Deviation* -** Standard deviation from the mean.

X Position - X coordinate for the maximum value on the object.

Y Position - Y coordinate for the maximum value on the object.

Adj Volume - The integrated volume of the object adjusted for background removal.

Concentration - Concentration of unknowns is calculated from the standard curve, created by specifying standard concentrations and regression analysis.

Molecular Weight - Molecular weight of unknowns is calculated from the standard curve, created by specifying standard Y position and regression analysis

pI - The iso-electric point of unknowns is calculated from the standard curve, created by specifying standard X position and regression analysis.

Note that the column display default settings within the volume report do not include *Position, Molecular Weight,* or *pI* columns. These columns must be displayed before calculating molecular weight or pI. To display hidden columns, select the *Format Columns* command under the

OPTIONS menu. Next, click on the plus sign in the *Column Selector* dialog box until the hidden column parameter is displayed. Uncheck the *Hide* box and click OK. The volume report may need to be expanded to display the new columns; or alternatively, the columns can be viewed using the scroll bar on the volume report.

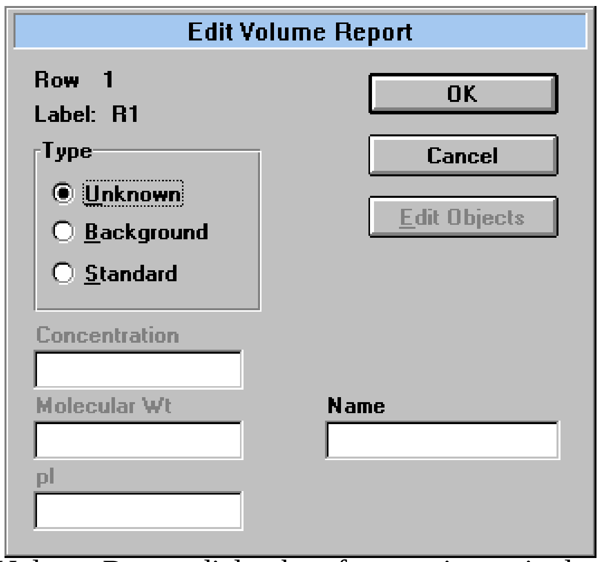
Editing the Volume Report

Double clicking on an active object(s) within the image or single clicking on a cell (highlight) of the volume integration report and selecting <ENTER> will open up an *Edit Volume Report* dialog box. The *Edit Volume Report* dialog box displays the object label and type parameters, standard value entry boxes, and spot width and height.

Object Type - Using the radio buttons, specify whether the object is either an Unknown, Background or Standard. If the object is a standard, enter the standard concentration, molecular weight, or pI in the appropriate box. Note that standard values can be entered only if the standard column is displayed in the volume report.

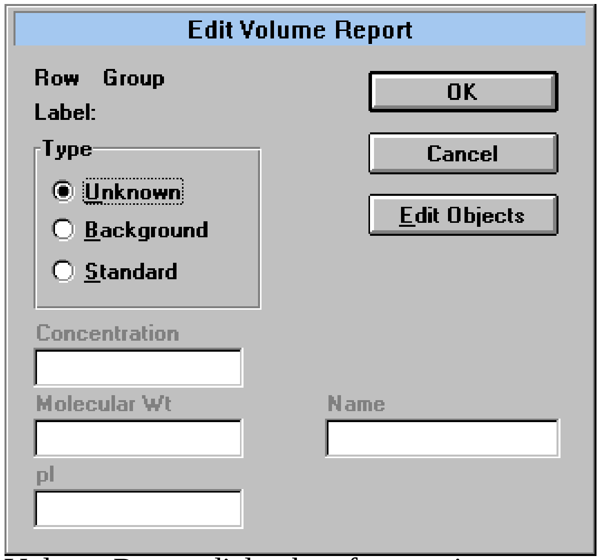
Object label - This identifies the object used, R=Rectangle, G=Grid, etc..

Name- This is the object's name.



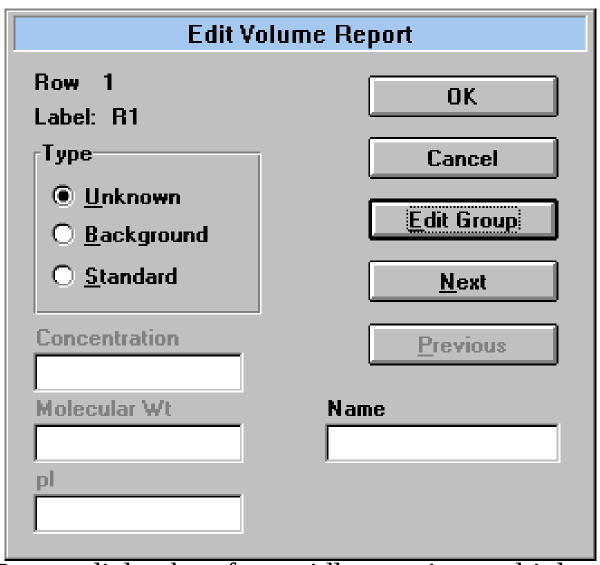
Edit Volume Report dialog box for entering a single value.

The *Edit Volume Report* dialog box also allows an active group of objects to be associated with one object type, background, or standard value. This feature is useful to average the background values of many background objects or to give many standard objects the same standard value. When the *Edit Volume Report* dialog box is opened when many objects are active, *Group* will be displayed in the Row # heading. This indicates that any object type or standard value change will affect all of the active objects.



Edit Volume Report dialog box for entering group values

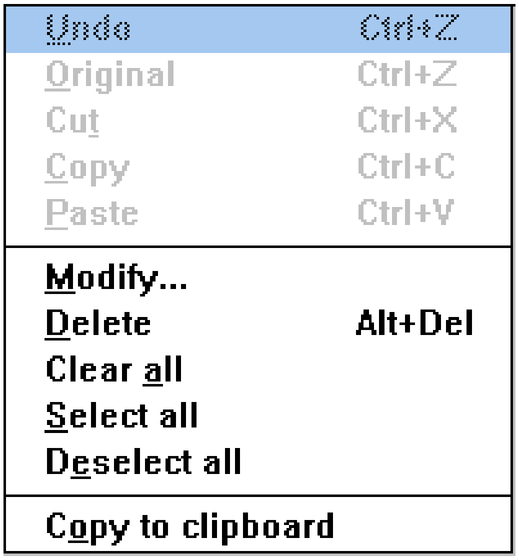
Additionally, the *Edit Volume Report* dialog box allows rapid entry of different standard values using the Edit *Objects* command button. Clicking on the *Edit Objects* command button, entering a standard value, and then clicking on the *Next* control button, allows each of the objects to rapidly be given a different standard value. This function is only active when more than one standard objects are active.



Edit Volume Report dialog box for rapidly entering multiple standard values

6.5 Volume Menus

EDIT Menu



EDIT COMMAND: *Undo*
This function reverses the last image operation (filter, rotate, or clip) performed.

EDIT COMMAND: *Original*
This function displays the original, unmodified image.

EDIT COMMAND: *Cut*
Selecting this command will cut any selected object and place it into the clipboard. These objects can be pasted into most clipboard compatible programs.

EDIT COMMAND: *Copy*
Copy will duplicate any selected object(s) into the clipboard and allow it to be pasted within an active image.

EDIT COMMAND: *Paste*
This function will paste any object(s) cut or copied into an active image.

EDIT COMMAND: *Modify*
This function will open the Edit Volume Dialog box when a volume report cell is highlighted.

EDIT COMMAND: *Delete*
This function will delete any active cells within a volume report.

EDIT COMMAND: *Clear all*
Selecting this command will delete all active cells within a volume report.

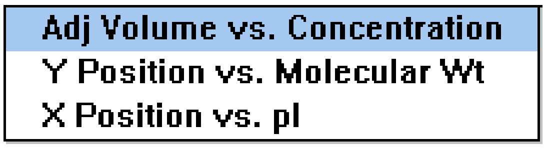
EDIT COMMAND: *Select All*
This command selects all cells within a volume report and makes them active.

EDIT COMMAND: *Deselect All*

This command deselects all cells within a volume report and makes them inactive.

EDIT COMMAND: *Copy to clipboard*
Selecting this command will copy selected rows from the results table to Windows Clipboard which can then be transferred to other applications, such as Microsoft®Excel® or Microsoft® Word®. This command is active only when result tables are displayed.

VIEW Menu

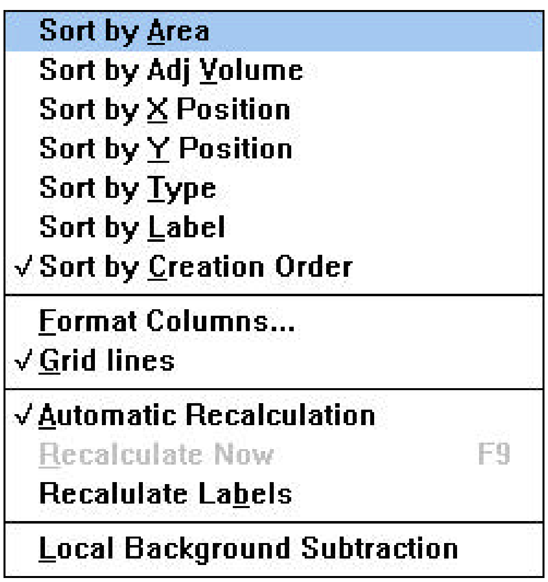


VIEW COMMAND: *Adj. Volume vs. Concentration*
This option creates the concentration standard curve based on adjusted volume.

VIEW COMMAND: *Y Position vs. Molecular Weight*
This option creates the molecular weight standard curve based on the object's Y position.

VIEW COMMAND: *X Position vs. pI*
This option creates the pI standard curve based on the object's X position.

OPTIONS Menu



OPTIONS COMMAND: *Sort by Area*
When this command is selected, objects in the Volume report will be sorted and displayed in order of increasing Area

OPTIONS COMMAND: *Sort by Adj Volume*

When this command is selected, objects in the Adjusted Volume report will be sorted and displayed in order of increasing Volume.

OPTIONS COMMAND: *Sort by X Position*
When this command is selected, objects in the Volume report will be sorted and displayed in order of increasing X Position.

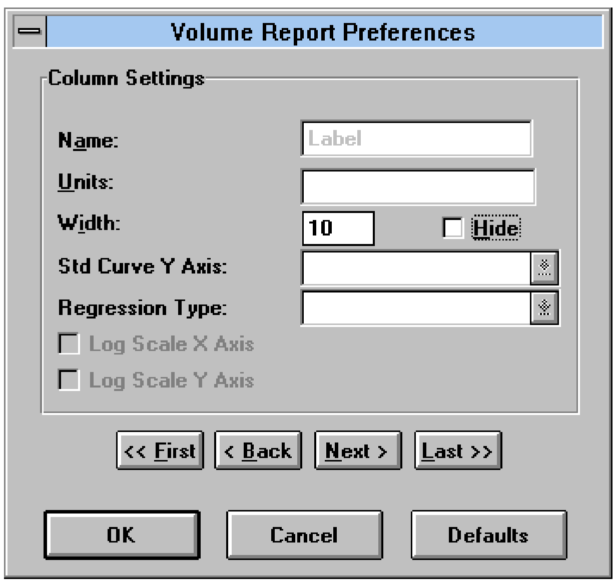
OPTIONS COMMAND: *Sort by Y Position*
This command allows objects in the Volume report to be sorted and displayed in order of increasing Y Position.

OPTIONS COMMAND: *Sort by Type*
This command allows objects in the Volume report to be sorted by type.

OPTIONS COMMAND: *Sort by Label*
This command allows objects in the Volume report to be sorted by label.

OPTIONS COMMAND: *Sort by Creation Order*
This command allows objects in the Volume report to be sorted and displayed in creation order.

OPTIONS COMMAND: *Format Columns...*
This option allows specific columns to be hidden from view and column widths to be modified to provide a more concise volume report. Selecting this command also allows the standard curve Y axis and regression type to be set for Concentration, Molecular Weight, and pI.



OPTIONS COMMAND: *Grid Lines*
Selecting this option displays or hides the Volume Report grid lines

OPTIONS COMMAND: *Automatic Recalculation*
Selecting this option allows the values in the Volume Report to be automatically recalculated. This option is convenient when values are modified.

OPTIONS COMMAND: *Recalculate Now*
This option allows you to recalculate values at any time.

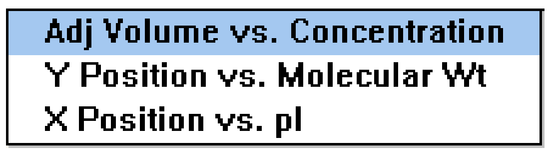
OPTIONS COMMAND: *Recalculate Labels*
This option allows you to renumber volume object labels.

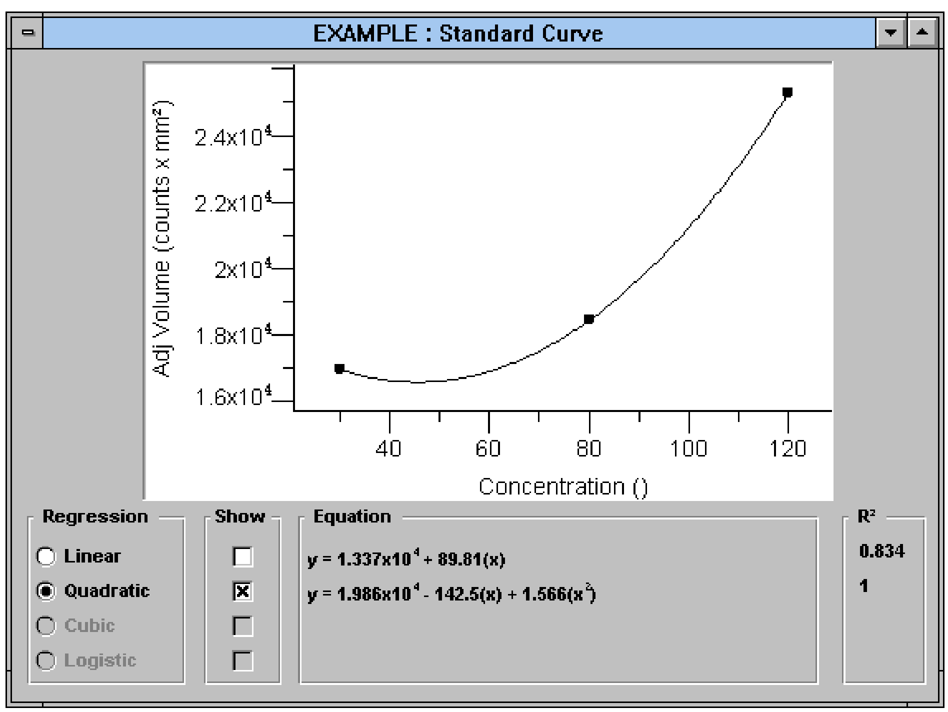
OPTIONS COMMAND: *Local Background Subtraction*
This option subtracts the mean pixel value concentrically outside each individual object perimeter.

Description of the Standard Curve

The volume integration module of Molecular Analyst/PC permits sophisticated graphing of the standard curves using a variety of graph types, regression analyses, and graph axes. From the standard curve graphs, concentration, molecular weight, and isoelectric point can be calculated.

With the volume analysis report displayed and objects identified as either background, standards, or unknowns, selecting the command *Adj Volume vs. Concentration* from the VIEW menu displays the plot of the standard curve(s)

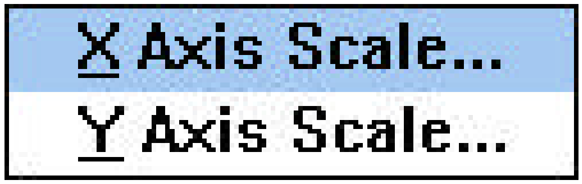




Notice within the standard curve is a legend with linear, quadratic, cubic, and logistic curve check boxes. The possible regression analyses include: Linear Regression, Quadratic Regression, Cubic Regression, and 4-Parameter Logistic Regression. Linear Regression requires a minimum of two (2) standard data points, Quadratic Regression requires a minimum of three (3) standard data points, and Cubic Regression requires a minimum of four (4) standard data points. 4-Parameter

Logistic Regression requires a minimum of five (5) standard data points. The type of regression curve can also be displayed or hidden on the graph by using the regression curve boxes located within the standard curve. Moreover, multiple regression analysis curves will be displayed given that the minimum number of standard data points for that curve are available (i.e.. given 3 standard data points, both linear and quadratic curves will be displayed).

Lastly, two different axes are available to produce the standard curve: *Linear,* and *Log.* To view each type of graph axis, simply click the OPTIONS menu when the standard curve is displayed.



Start with the simplest regression method, Linear, and examine the fit of the data in each set of axes: *Linear-Linear, Linear-Log, Log-Linear, and Log-Log.* The goal is to find the set of curve fitting conditions that best matches the data. Move to the next order of regression, Quadratic, Cubic, and finally Logistic. To calculate concentration of the unknowns for a specific regression analysis, simply click on the standard curve radio button. This will display the regression formula and standard error for that curve, and calculate the unknown concentrations in the volume report. To view a different curve, simply select the appropriate radio button on the standard curve display window. The unknown values will automatically recalculate using the selected regression analysis.

Appendices

APPENDIX A: Disk Space Management

Managing the Data Base:

Periodically it will be necessary to delete unwanted files, created by the Profile Analyst module, from your hard disk in order to free-up hard disk space. The files can be archived on floppy disks for future use. Deletion of files is accomplished using Windows File Manager, while deletion of the profiles contained within the files is accomplished using the *Delete Profile ...* function from the DISPLAY window.

To delete files from a hard disk use the following procedure:
(1) Bring up File Manager by double clicking on the icon. (2) Specify which disk drive you wish to delete files from by clicking on the appropriate icon. (3) Point and click on the file which you wish to delete. (4) Click on FILE in the menu bar and then click on the DELETE command. The file will then be deleted from the disk drive. Continue this process until all of the unwanted files are deleted.

To delete unwanted profiles from within files, use the following procedure:
(1) After initiating the Profile Analyst II program, click on the *MODE* menu and select *Display* from the menu. (2) Click on the *FILE* menu and select *Delete Profile ...* from the menu. (3) Double click on the file name which contains the profile(s) which you wish to delete. A dialog box showing the profiles contained within the select file will appear. (4) To delete one profile, double click on the single profile to be deleted. To delete many profiles from the selected FILE, press and hold the <Shift> key and click the left mouse button on all profiles which you wish to delete. Click on the *Delete* command button to delete the selected profiles. The "Open File Name:" dialog box will then reappear enabling you to delete profiles from another FILE. If you are finished deleting profiles, click on the *Exit* command button to go back to the *Display* window.

APPENDIX B: Use of Estimated Molecular Weights

1. The user is cautioned when using the estimated molecular weights calculated with higher order regressions. Higher orders of regression often result in a "better fit" because they make more concessions to particular measurement errors in the data used to derive the standard curve. While this can increase the amount of useful information, it can also reduce the likelihood that the standard curve represents the sample data. This is because the precision with which the standards data are modeled is not always matched by the sample data.

2. The user is also cautioned when using sample data where the x and y values do not fall within the x and y range used to calculate the standard curve. Using these values outside of the standard curve x and y range may result in undefined or absurd values (the Profile Analyst will display 0.000 for values outside of the range). If higher orders of regression of standard data produce several undefined sample values, then it may not be the best model to choose for the standard curve.

APPENDIX C: Regression Analyses

Linear Regression:

When analyzing data, always start from the simplest set of curve fitting conditions. Begin with the linear regression. Examine the fit of the data in each of the four sets of axes: Linear-Linear, Linear-Log, Log-Linear, and Log-Log. The goal is to find the set of curve fitting conditions that best matches the data. The data appears in the graphs as circles, the regression appears as a bold, smooth line or curve. The correlation coefficient (r) gives an indication of how well the regression model fits the data. In effect, the correlation coefficient is the square root of the proportion of total variation that can be explained by the regression.

r = (explained variation / total variation)1/2

The amount of explained variation increases as the goodness of fit increases. In a perfect fit, all of the variation is explained, the explained variation equals the total variation, and the ratio of the explained variation to total variation becomes 1.000. Or, more simply, the closer the correlation coefficient is to 1.000, the better the fit of the regression. A perfect fit with a correlation coefficient of 1.000 would indicate 100% certainty of the fit. The correlation coefficient of 0.9995 in the graph above indicates 99.95% certainty of fit.

Quadratic Regression:

The next level of sophistication of curve fitting is the quadratic regression. Select *Quadratic* and examine the fit of the regression in all four combinations of axes scales. Again, the goal is to find the set of regression conditions in which the regression most closely matches the data. The goodness of fit in this case is measured by the standard error of the regression. The lower the standard error, the better the fit. Visual comparison alone indicates that the Log-Log fit is superior to the Log-Linear fit. The corresponding standard errors confirm this visual evaluation; the Log-Log fit has the lower standard error.

Removing outliers or tails from the standard curve may improve your fit significantly, and allow you to use a simpler level of curve fitting. Outliers are removed from the volume report.

Recall that a quadratic expression Y = a+bX+cX2 may be solved by setting the equation equal to zero and solving for X. The root is unsolvable if the expression under the square root sign is a negative value.

The quadratic formula describes a parabola, which appears as a bowl when plotted. The portion of the quadratic curve near the inflection point has a very shallow slope. A small difference in Y value in this portion of the curve translates into a large difference in the corresponding calculation of X. Reliability in this portion of the curve depends on the steepness of the parabola.

Cubic Regression:

Cubic regression is the next highest order of regression and is a second order quadratic regression analyses.

Logit-Log or Four Parameter Regression:

The four-parameter logistic model is the highest level of curve fitting available and is used to describe the smooth, symmetrical, sigmoidal relationship between electrophoretic mobility (distance migrated) and log of molecular size (kb): distance = a - d/1 + (size/c)b + d. The four parameters (a,b,c,d) are estimated by nonlinear least-squares curve fitting using the Marquardt-Levenberg algorithm, where a represents an upper plateau, b a slope factor, c the midpoint, and d the lower plateau. Estimates of size for unknown species are accompanied by estimates of the standard error (SE)

Mathematical Equations

For Linear Regression analysis:
Where x_i is the X value of "i", X is mean X value, y_i is the Y value of "i", and Y is the mean Y value, we define:

The slope, $ß = \sum(x_i-X)(y_i-Y)/\sum(x_i-X)$
The intercept, $a = y - ßx$
Then: $y = a + ßx$

For a Quadratic Regression analysis:
Where x_i is the X value of "i", X is the mean X value, y_i is the Y value of "i", Y is the mean Y value, a is the zero order coefficient of the regression, ß is the first order coefficient of the regression, and N is the number of sample pairs, we define:

$$S_{xx} = 1/N \sum (x_i-X)2$$
$$S_{xy} = 1/N \sum(x_i-X)(y_i-Y)$$
$$S_{yy} = 1/N \sum(y_i-Y)2$$
$$S_{xx} 2 = 1/N \sum(x_i-X)(x_i 2 -X 2)$$
$$S_x 2 {}_x 2 = 1/N \sum(x_i 2 -X 2) 2$$
$$S_{yx} 2 = 1/N \sum(y_i-Y)(x_i 2 -X 2)$$

Then: g is the second order coefficient of the quadratic equation.

$$g = (SxxSyx 2 - Sxx 2 Sxy)/SxxSx 2 x 2 -(Sxx 2) 2$$

ß is the first order coefficient of the quadratic equation.

$$ß = (SxySx 2 x 2 - Sx 2 ySxx 2)/(SxxSx 2 x 2 -(Sxx 2) 2)$$

a is the zero order coefficient of the quadratic equation.

$$a = y - ßx-gx 2$$

Standard error of the estimate:

$$S_D = \{\sum(y_i- a - ß x_i) 2 /N - 3\} 1/2$$

For a Cubic Regression analysis:
Cubic Regression has a third order coefficient of the of a quadratic equation.

For a Four Parameter Regression analysis:

$$Y=\{(a-d)/(1+(x/c)^b)\}+d$$

Where Y is the absorbance value, x is the size value, a is the upper plateau, d is the lower plateau, c is the midpoint, and b is the slope of the linear portion of the curve.

Oerter KE; Munson PJ; McBride WO; Rodbard D.
Computerized estimation of size of nucleic acid fragments using the four-parameter logistic model. Analytical Biochemistry, 1990 Sep, 189(2):235-43.

Appendix D: Edge Enhancement

The GS-670 Densitometer has a built-in image enhancement feature. This feature allows the edges of an image to be soften of sharpened. In normal mode, data is passed from the CCD array to the computer with no modification other than look up table scaling. In the soften mode, a pixel is averaged with the pixels in the next adjacent rows and columns. The average is weighted so that the overall intensity of the image does not change. Soften mode will minimize moire patterns. In sharp mode, a fraction of the value of pixels in the adjacent rows and columns is subtracted from each pixel. The coefficients are selected so that the overall intensity of the image does not change. This enhances any pixel which is different from its neighbors. The effect of this average is to sharpen edges. If a scan is noisy or the sample quality poor, it will also enhance defects. Extra sharp mode is the same as sharp except that a larger portion of the neighboring pixels are subtracted.

APPENDIX E: Troubleshooting

Image Display

Condition	Possible Cause/Solution
Windows not properly displayed in Gel Doc	-No video pass-through on computer or board -Incorrect video driver installed -VGADAC=ON line is not in the [MACH] directory in the WIN.INI file in WINDOWS
No scanning at all	-The proper interface cable was not used -The proper SCSI or GPIB card not used - GPIB or SCSI software not installed properly
The Sub Background command is not functional	-The background profile which was imported was not set as background using the PROFILE command Set Type -The background profile was not saved
The Point-to-Point baseline is not functional	-The baseline endpoints do not extend beyond the far left and right edges of the profile
No peaks were identified after using the Auto Identify command.	-The Peak Criteria is excluding all peaks within the profile -The profile was not selected as "Active"
The first and last peak of the profile is not identified	The profile was clipped too short using the OPTIMIZE commands Origin and *Front.*

APPENDIX F: Technical Support

Bio-Rad provides complete technical support on all features of the Profile Analyst II software. If you have any problems, or any questions on the use of the features, please call Bio-Rad Technical Service at: 1-800-4-BIO-RAD (1-800-424-6723)

Please report any reproducible software problems to Bio-Rad Technical Service so we can eliminate them from future versions. Information in this manual is subject to change without notice. Molecular Analyst software is a registered trademark of Bio-Rad Laboratories.

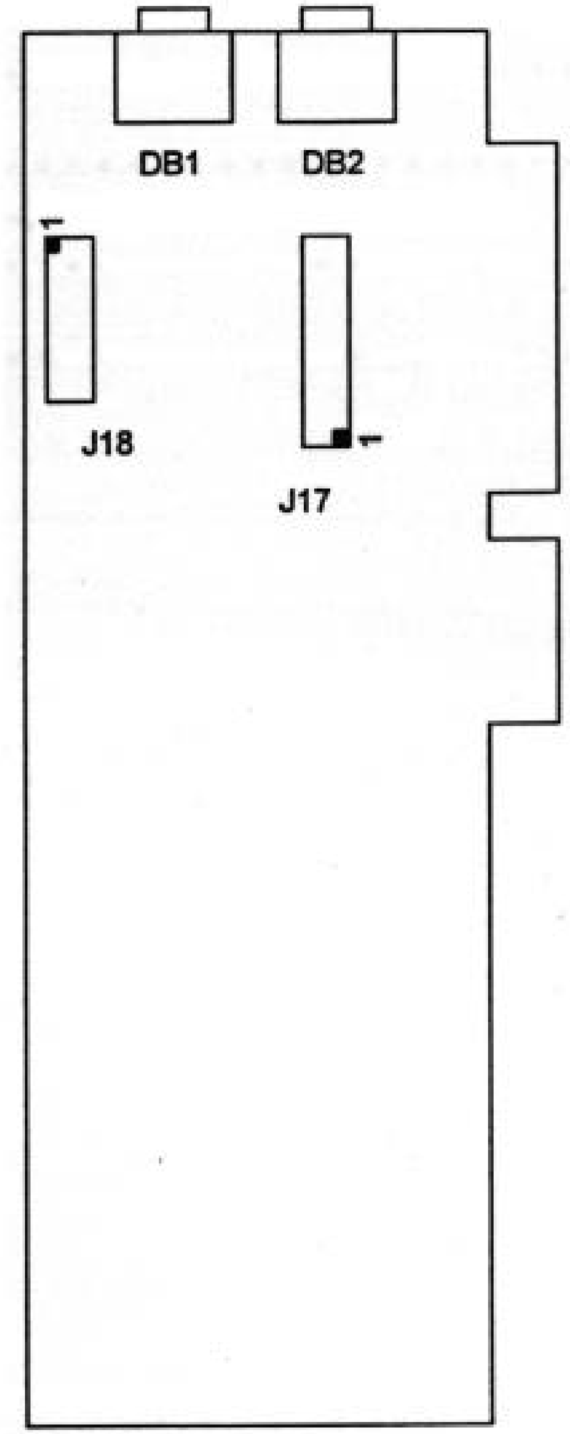
IBM is a registered trademark of International Business Machines. MS-DOS, Windows, Excel, and Word are registered trademarks of Microsoft Corporation.

APPENDIX G: Frame Grabber Board Setup and Base Address Configuration

DIP switch settings for the I/O Base Address

Frame Grabber Board VGA Feature Connector J18

Number 1 represents Pin 1 which should correspond to the red stripe on the ribbon cable provided with the frame grabber card.



APPENDIX H: XLI Laserpix Printer Card and Mitsubishi P68 Printer Settings

<u>Installing the XLI Laserpix Printer Card</u>

Step 1 - Verify That Your System Meets The Minimum Requirements

1. Printers

Printers	Computers
Laserjet 4/4m	80486 At 33 Mhz
Laserjet Ii,Iii	80386 At 25 Mhz

2. Memory (Ram)

	Recommended	Minimum
Laserjet 4/4m	24 - 32 Mb	16 Mb
Laserjet Ii,Iii	12 - 16 Mb	8 Mb

3. Disk Space (40 Mb Minimum)

	Recommended	Minimum
Disk where ***temp*** environment points to	20 - 50 MB	20 MB
Swap File	20 MB	-

4. Parallel Port

Step 2 - Install The Laserpix Hardware As Directed In Manual

Step 3 - Install The Laserpix Software As Directed In Manual

Step 4 - Review The XLI Installation Checklist

Step 5 - Installing Version 1.2 Special C

There were bugs found in the xli driver. It returned actual printable page dimensions that were inconsistent with other print drivers. This resulted in clipped images on the sides, depending on print mode, (left side clipped in portrait mode, bottom side clipped in landscaped mode.)   The driver also generated “Protection Faults” especially when printing big images at 1200 dpi. A new version of the xli driver is available, Version 1.2 Special C. This version corrects these problem.

1. Remove or rename any sd_xli.drv from your system.

Multiple copies of sd_xli.drv can be confusing.

Setup will automatically install the driver in the windows directory it finds in your system. Sometimes, there are multiple versions of windows on a system. This means there may be multiple copies of sd_xli.drv on a system. It then becomes confusing as to which version of the driver your system is accessing. Removing or renaming ALL sd_xli.drv from your system will eliminate this confusion.

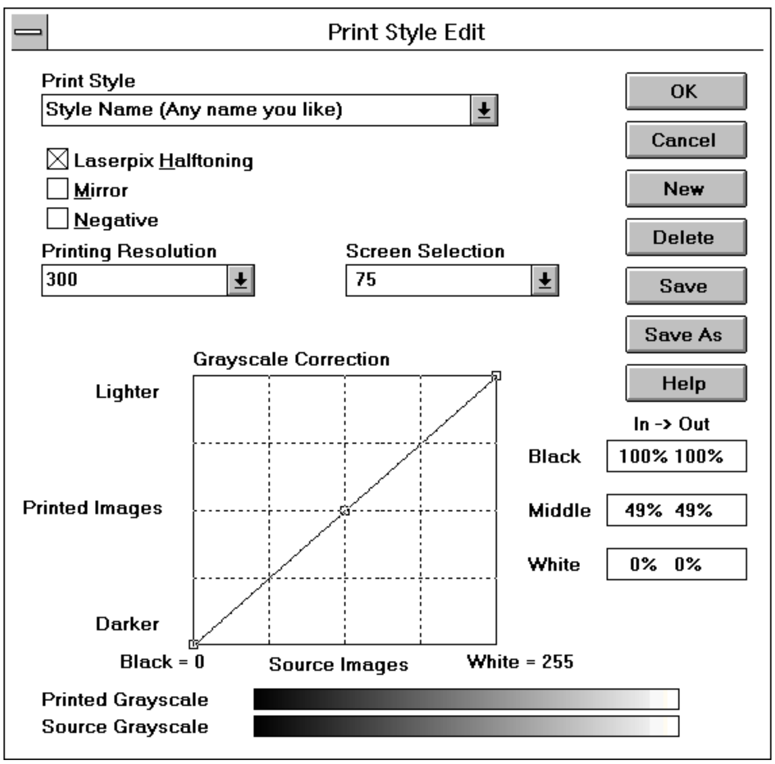
2. Run setup.exe

STEP 6 - PRINT STLYE RECOMMENDATIONS

XLI provides a set of pre-defined print styles. These print styles are defined by the page resolution and screen resolution**.** XLI allows you to create your own set of print styles. When printing images big or small use the following for best results.

Desired Print Result	Printing Resolution	Screen Resolution
Smooth uniformed background Less detail in image	300 dpi	75
Finer detail in image Less smooth background	300 dpi	150

You will find, increasing the page resolution will **NOT** necessarily improve print resolution in an image. This resolution will be noticeable when printing text documents. Therefore, (for quick results), it is recommended to use 300 dpi with one of the screen resolutions mentioned above, depending on desired result.



XLI INSTALLATION CHECKLIST

1. Parallel Cable

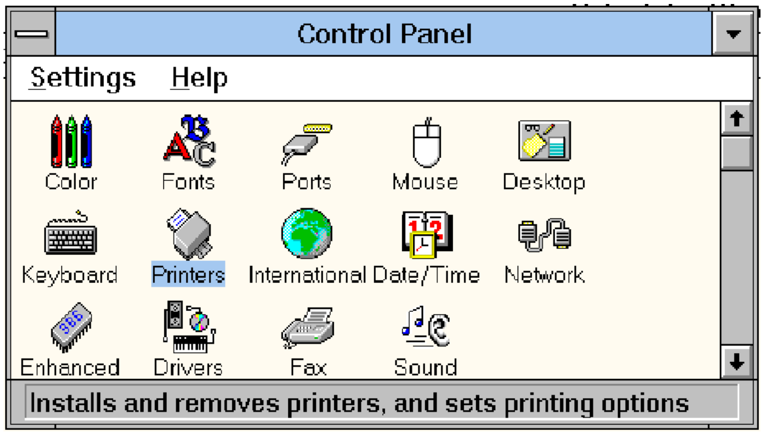
a.) Check the parallel cable that it is **NOT** faulty.
b.) Check the parallel cable that it is connected to the correct port and that it is connected securely.

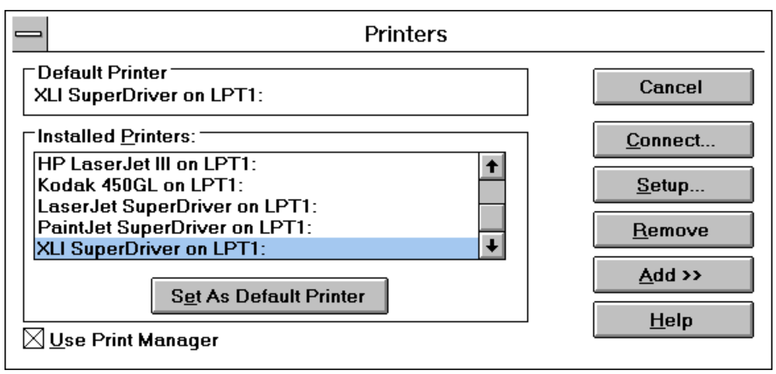
2. Video Cable

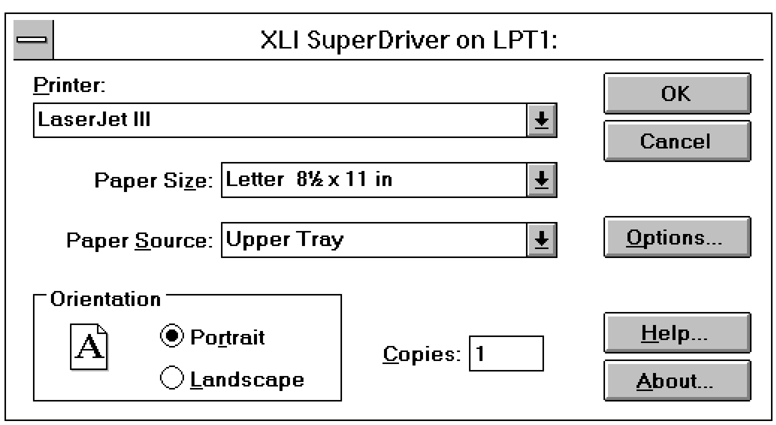
a.) Check the video cable, that it is **NOT** faulty.
b.) Check the video cable, that it is connected correctly and securely.

3. Select Correct Printer and Driver

a.) Select the correct printer and driver using **control panel - printers**

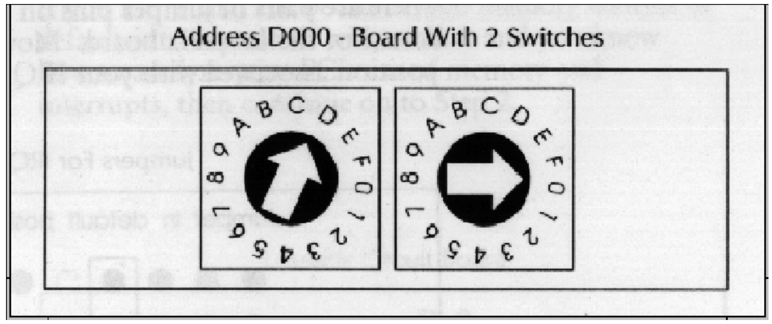


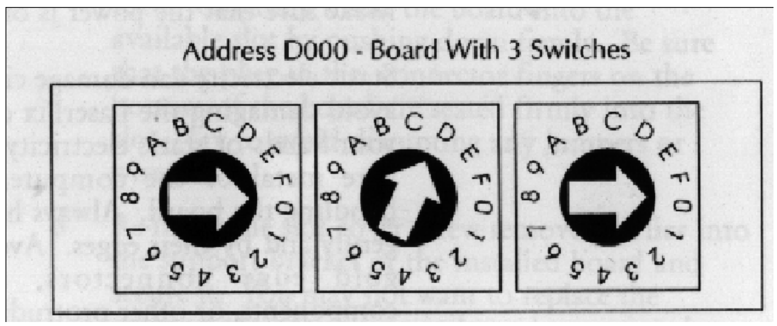




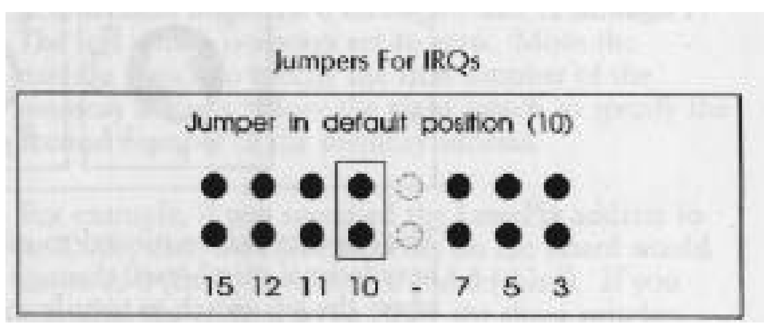
4. Verify That Memory and IRQ Settings are the same in hardware and software

a.) Turn machine off
b.) Remove XLI card from machine
c.) Write down the memory address





d.) Write down the IRQ setting



d.) Put XLI card back into machine securely
e.) Turn machine back on

f.) Verify settings match software settings using **control panel** - **printers - setup... - options... - HW Setting.**

g.) If software does **NOT** match hardware settings,

1. Type in the correct settings.
2. Restart windows

5. Verify XLI memory is excluded in config.sys

a.) Check that the xli memory is being excluded in the config.sys file.
ie.) XLI memory at D000 , X=D000-D1FF should be added to your emm386 line in your config.sys file.

b.) Changes to the config.sys file does not take effect until you reboot your system.

6. Verify FILE SHARING is DISABLED

a.) If you are running on a network that allows file sharing, be sure to DISABLE this option.

Common Problems with XLI

Listed below are some common problems we have seen using the XLI print drivers.

1. Error printing! Retry??
2. Vertical lines appear on image
3. Garbage in the image, especially when cropping the image
4. Can not initialize printer - **(check your paper supply, this message appears when trying to print and the printer is out of paper.)**

The most common reasons why problems are encountered with the XLI print driver are

1. Printer is not ready, ie.) power is off or cables are not attached correctly
2. The software is incorrectly configured
3. Memory or IRQ conflict with LaserPix board
4. Other driver conflicts in your system

Before throwing your XLI board out the window, verify that 1-4 is not what is causing your grief.

How To Be Sure That 1-4 Is Not Causing Your Problems

1. **Printer is not ready -**
   a.) Check printer power, make sure it is turned on
   b.) Check cables, make sure all cables are connected properly and securely on both the host computer and the printer, also make sure cables are connected to the correct ports.
   One way to verify your serial/parallel connection is correct, is to print to your printer with the standard driver (not the XLI driver). If this is successful, you can be confident that your serial/parallel connection is okay. If printing in Windows does not work, try printing from DOS. If printing from DOS works fine, but printing from WINDOWS fails, check that your printer is configured and setup correctly. (Use the Control Panel).
   c.) Check the laserpix board (in the host computer) that it is seated securely
   d.) Check the XLI interface board (in the printer) that it is seated securely
   e.) And of course, check your paper supply

2. **Software configuration-**
   a.) Verify that software settings match hardware settings.
   1. Turn your machine off
   2. Pull the laserpix board out of your machine
   3. Look at interrupt jumper, write down the interrupt where jumper is set
   4. Look at memory address dials, write down the address
   5. Put board back in machine (securely).
   6. Turn machine back on
   7. Bring up windows, Select Control Panel - Printers
   8. In Printer Setup, select Options.....until you see HW Settings Select HW Settings
   9. Verify that the memory address and the IRQ displayed on the screen matches the values you wrote down (from the board).
   10. If these do not match, reconfigure the software by changing the appropriate values in the edit box.
   11. Exit Windows - (changes do not take place until Windows is ReStarted)
   12. Verify changes made by repeating steps 7-9.

3. **Memory and/or IRQ conflicts -**

It is a good idea to know what upper memory addresses and what Interrupts are currently being used in your system. A summary sheet which lists this information is helpful to have around.
Microsoft has a utility (MSD) that lists some of this information.
Most add-in boards come with a setup and/or a utility that list this information.
ie.) GPIB, NETWORK cards, video cards, etc.......
Another, (tedious way) to get this information is to boot your system and write down the information as it appears on your screen.
(use Pause or <ctrl s> to stop screen)

a.) Verify that your system has no memory and/or IRQ conflicts
b.) If you find conflicts , resolve them

4. **Other driver conflicts**

If after verifying all of the above still gives you problems, STRIP your config.sys and autoexec.bat or PULL all boards out of your machine. Get your system down to the BARE minimum.
If printing with the XLI driver begins to work with no problems, slowly add things back to your system. This should help in determining what is causing the problem.
It has been found that if you run WorkGroup For Windows 3.1 with **ODINSUP.COM** (used with NOVELL and WorkGroup For Windows, for WGW network capabilities) causes problems when printing with XLI driver.

5. **Known Problems -** (the following are currently being researched and/or confirmed)

a.) Conflict with ODINSUP.COM , this is a driver used when WFW31 and Lan WorkPlace For Dos (TCIP) are used together. Without ODINSUP.COM WFW31 network capabilities do not work.

Mitsubishi P68 Printer Settings

Rear of printer:

1) Mode 1 DSQ pin configuration: All pins are OFF except pin # 2

2) Mode 2 DSQ pin configuration: All pins are OFF except pin # 1

3) RGB switch is in the ANALOG position

4) Monitor cable attaches to the “RGB Out” serial port.

5) VGA cable attaches from the “RGB In” serial port to the 256 gray level graphics card serial port on the computer.

Graphic Card Configuration:

1) Graphics card should be configured as “Non-Interlaced 800 x 600”

Front panel of printer:

1) Direction: Normal

2) Scan: Frame

3) Contrast: Normal

Recommended function settings   **DISPLAYED IN WINDOWS 3.1**

(Function settings for each monitor may vary slightly. Use the settings below as a starting point. Refer to the users guide for more information).

Screen 1:	RGB. A			
			USER	**STD**
	B-LEV	-7	0	
	W-LEV	-10	0	
	A G C		OFF	OFF
	DITHER		OFF	OFF
Screen 2:	RGB. A			
			USER	
	H-WID	831		
	H-TRC	-10		
	H-RES	640		
	H-STR	56		
	V-LIN		628	
	V-STR	24		
	V-RES	600		
	H-SIZ		100	
	V-SIZ		74	

Printer Paper:

Mitsubishi Thermal Paper for Video Copy Processor
Model K65H (Bio-Rad Catalog Number 170-7581)

Appendix I: Enhancing Image Quality With Filtering And Transformation

Transformation Tool

The guidelines to enhance image quality with the Transformation Tool are as follows:

a. Only use Nonlinear Transformation on images with high signal. Low signal images should always be transformed with linear transformation.

b. Set the minimum for the transformation to the minimum value in the image or to a value which is somewhat higher than this minimum. This lightens the background.

c. Set the maximum for the transformation to the maximum value in the image or to a value which is somewhat higher than this maximum. I would not set it to a value much higher than the maximum in the image unless visual sensitivity of low signal image objects is not a major priority.

d. The transformation tool can also be used to lighten the background in the displayed image beyond what was suggested in b. above. This does not affect the raw data used for quantitation, but it will give the visual effect of thresholding. For purposes of visual appeal, this can be very helpful. The best way to do this is to use the histogram as a guide to where to set the minimum value for transformation. If the histogram has a large, narrow peak as the low end, one could choose a a value at the top of this peak or one which would cut off most of the peak without cutting into the signal. There is no way around the fact that this will always be a judgment call; however, the histogram can be very helpful in getting an estimate for the minimum value to use in the transformation.

e. Every time the raw data is filtered in any way, retransform the image. Filtering will produce new values for the minimum in the image and will likely have a dramatic effect on the histogram at the low end. Transforming after filtering to further lighted the background can significantly improve an image.

f. Keep in mind that use of the Transformation Tool to lighten background or otherwise improve an image is one of the operations which also affects the 8-bit TIFF image which is exported to an external package such as Photoshop. This can give such external packages a head start in the direction of publication quality images.

Contrast, Brightness, and Gamma Tool

This tool operates on the 8-bit display image and can significantly improve an image as it is displayed on the computer screen. The user can develop an intuitive feel for this tool fairly quickly.

a. CONTRAST. Increasing the contrast can be used to lighten up the background of an image and to make the distinction between background (light) and signal (dark) sharper.

b. BRIGHTNESS. Increasing the brightness can also lighten the background of an image by sort of bleaching it out. However, it may also have the negative effect of making the signal, such as bands, appear lighter than is desired. For two partially resolved bands or spots, increasing the brightness can improve the visible separation between these bands. In a high signal image, it is possible that reducing the brightness could reveal bands which were not seen before.

c. GAMMA. This gives a nonlinear relationship between the stored 8-bit image and the displayed 8-bit image. It is very analogous to nonlinear transformation. Adjusting it in one direction will accentuate the low signal and the background and tend to mute the high signal. This is less useful with low signal images, because it accentuates the noise in or near the background of the image.

Adjusting it in the other direction will cause the background to be softened and for the high signal to be accentuated.

Filtering

There are basically two types of filters: 1) softening (smoothing) and 2) sharpening. For noise reduction, smoothing filters are appropriate and sharpening filters are to be avoided. Sharpening filters will aggravate noise and could produce artifacts by making certain regions of noise appear to be signal. Softening filters can reduce noise; however, all of them achieve this by performing some sort of spatial averaging or at least something closely related to this (as in the median filter). Strong smoothing will always produce a blurred image. If this effect is too strong, it will reduce the perceived image quality even though it may reduce noise.

A. Smoothing Filters: Included in this class of filters are the median, weighted mean, ORPA, and the adaptive noise reduction filters. All of these have the potential for blurring the image beyond what is desirable. However, each of these also has a particular way of reducing noise.

1. Median Filter. This is mostly useful for low signal images or regions of an image where the noise distribution is asymmetric, that is, it has enough very high outliers to be annoying visually. The median filter clips these outliers off and can dramatically reduce this kind of noise. This is particularly useful when quantitation is a key objective with low signal. When the background and the signal are low and the noise distribution is very asymmetric (has spikes) the median filter will do a much better job of conserving quantitative relationship than will any of the other smoothing filters which depend on averaging.

   Unfortunately, this filter also produces some bizarre visual effects. Their is a certain mottling effect which can be quite irritating. Also, if this filter is used with a large filter window, it can lead to significant blurring. With smaller windows, this filter is very good at preserving the edges of well defined objects such as bands or dots. This preserves the sharpness of bands and dots better than most other noise reduction filters.

2. Weighted Mean Smoothing Filter (Hat Filter, Gaussian Filter). This is the classing noise reduction filter. The bigger the filter window, the more smoothing which is achieved. There is a fairly direct tradeoff between the beneficial effects of smoothing to reduce noise and the detrimental effects of blurring which reduces perceived image quality. The only way to know what size of a filter to use, is by trial and error for a particular image.

3. Out Range Pixel Averaging (ORPA). This filter is directed at eliminating outliers somewhat like the median filter. However, it only acts on outliers above a certain threshold and then replaces them with an averaged value. So it is much like the averaging filter on some images. It can give substantial blurring, but it is very useful in regions with large isolated spikes. It is also very useful to use just after using the Weighted Noise Reduction Filter. This is because this latter filter leaves a few large isolated spikes.

4. Weighted Noise Reduction (ANR). The ANR filter is specialized to operate mostly on background noise and some low signal while at the same time doing very little modification of the edges of image object such as dots and bands. It also avoids doing much with noise which is on top of high signal. For some images, this does a very good job of reducing noise while at the same time preserving much of the sharpness of the image. The combination of ANR followed by ORPA is a good general approach to images in which the low signal and background will be scrutinized. This also has the advantage that it can be used on raw data from which quantitative data will be extracted after filtering. It does a good job of preserving quantitative relationships.

B. Sharpening Filters: Included in these are the "Sharpen" filter, the "Unsharp Mask," other high pass filters and one could include the Weighted Noise Reduction filter in this as well. None of these filters usually does a good job of preserving quantitative relationships; however, they can improve the visual image quality of some images. Usually, the images have to be low in noise before these filters can be applied successfully. An exception to this is the Weighted Noise Reduction which rescales the images in a way which can substantially improve the way some images look. This has been applied with great success to images of chemiluminescent samples.

One form of image processing which also sharpens images, when it works, is deconvolution. The advantage to deconvolution is that it can greatly improve quantitative relationships as well as improve image quality dramatically. However, it only works on images which have been blurred in a way which is understood in detail. This understanding is used to work backward toward what the image was before it was blurred. The main problem with deconvolution is that it does not work well with images which have much noise in them. It is for this reason that we have not used this technique much in the Molecular Imager software.

> 1. Sharpen. This has a parameter which can be adjusted to control the amount of sharpening. The only way to really tell if this is producing adequate results is to test it by trial and error.
>
> 2. Weighted Noise Reduction. This filter also incorporates a parameter called the boost factor. This allows the user to accentuate the background and lower signal after the filter has been applied. Again, trial and error is required, preferably with a filter preview window, to obtain good results with this filter.

In general, and except for the Weighted Noise Reduction filter, it is better to sharpen image objects with the Transformation Tool or the Contrast, Brightness, Gamma Tool than to use the various sharpening tools which are sensitive to noise.

C. Superfluous Filters: These include things like Laplacian, the Sobel filters, and DW-MTM. The first two of these are useful in object detection, but not to improve image quality.

Appendix J: Installing Lens & Adjusting The CCD Camera Focus When Zooming

This procedure should only be done when focus is completely off when zooming from high to low magnification. Refer to drawing on the following page.

Tools Required: Standard small screwdriver, resolution target (provided with camera), and .060" hex tool (provided with camera).

Purpose of Procedure: Installation of the lens on camera and adjustment of the flange back so that correct focus is held when zooming from high to low magnification.

1. Loosen lens mount set screw on camera body with hex tool.

2. Install zoom lens by screwing lens into camera body.

3. When lens is completely seated in the camera body, continue to rotate lens clockwise to line up zoom, focus, and iris index line with the face of the camera body. (This will be the face with the two (2) 1/4-20 tapped holes). Resistance will be felt as the lens is rotated to line up the index line. The lens will slip through a maximum of 350 degrees.

4. Tighten the lens mount setscrew with the .060" hex tool

5. Screw the lens diopter and other filters onto the zoom lens.

6. Mount the camera/bracket assembly onto the hood. Place the resolution target into the hood. Connect the camera-lens assembly so that a live image can be viewed on the computer monitor.

7. Remove the tracking adjustment hole cover. Loosen the tracking adjustment locking screw with the screwdriver. CAUTION: Do not allow dust or other particulates to accumulate in the camera body during this adjustment procedure.

8. Open up the iris on the lens until maximum illumination (completely open). Source light may need to be reduced in order to prevent saturation of CCD.

9. Rotate the zoom ring to one extreme so that the highest magnification of the resolution target is achieved. Adjust the focus ring for best focus. **Do not adjust the tracking adjustment screw at this zoom.**

10. Rotate the zoom ring to the other extreme so that the lowest magnification of the resolution target is achieved. Adjust the tracking adjustment screw for best focus.

11. Repeat steps 9 and 10 above until rotation of zoom ring requires no adjustment for focus (Generally, this is an iterative procedure and will converge in no more that two steps).

12. Tighten the tracking adjustment screw and replace the tracking adjustment access hole cover.

13. The zoom lens should now be properly installed onto the CCD camera with the focus adjusted properly for high and low magnification.

Appendix K: Installing Hardware Protection Keys "Dongles"

Molecular Analyst is protected by a hardware key. It is necessary to have a key connected either to the computer on which Molecular Analyst is installed, to a computer that has been designated as a network server, or to any client computer within a WINDOWS NT or NOVELL Netware network.

Single User Key

A single user key allows access to Molecular Analyst by one computer at a time. The key must be attached to the computer on which Molecular Analyst will be accessed. It is not required for Molecular Analyst to be installed on the computer that has the key attached. Molecular Analyst can reside anywhere on a network.

1. Plug the single user key into the parallel port in the back of your computer. (note: if a printer is connected to the parallel port, plug the key to the parallel port then attach the printer cable to the key.)

2. Run MA

Network Key

A multiple user key allows 5 users to access Molecular Analyst concurrently. The multiple user key (also referred to as a network key), can be connected to either a server or any client computer within a WINDOWS NT or NOVELL Netware network. There are many different ways to install the network key on any given system. For successful installation of the network key, it is required that the HASP server (the machine on which the network key is installed) and all its client workstations use the same communications protocol. For example, if the HASP server uses the NETBIOS protocol then all its clients must also use the NETBIOS protocol. If the HASP server uses the IPX protocol then all its clients must also use the IPX protocol. Servers may specify the protocol by loading the network server program with either "-NETBIOS" or "-IPX" option. Clients may specify the protocol by setting the NETHASPPROTOCOL environment variable to NETBIOS or IPX. See the list below to determine which installation option meets your needs.
Note: Please be aware that the HASP server will be servicing client stations throughout the network. It is therefore required that the HASP server remains powered up while Molecular Analyst is executing. If the HASP server goes down, access to Molecular Analyst will denied.

NOVELL Netware Server (recommended)

If you wish to provide multiple access to Molecular Analyst across the network, this is considered to be the easiest and the most practical way to install the network key. Installing the network key on a NOVELL Netware server requires using the IPX communication protocol. Follow the steps below.

Installing the network key on a NOVELL Netware Server - Server Software (haspserv.nlm)

1. Attach the key to the parallel port of the server.
2. Copy HASPSERV.NLM file to the SYS:SYSTEM directory.
3. To load the HASP server, type:
   load clib (probably already loaded)
   l**oad haspserv.nlm**

Alternatively, these lines can be added to the AUTOEXEC.NCF file which resides n the SYS:SYSTEM directory.

HASPSERV.NLM may be unloaded by typing the following command:
unload haspserv

Client Settings - (HASP server is a NOVELL Netware server)
All clients accessing Molecular Analyst with the network key attached to a NOVELL Netware server must use the IPX communication protocol. To be assured that each client is using the IPX communication protocol make sure,
1. IPX/SPX is installed on the client machine and
2.that the environment variable NETHASPPROTOCOL is set to IPX on each client machine. This can be set by adding the following line to your autoexec.bat file: SET NETHASPPROTOCOL=IPX

Windows 3.1, Windows 3.11, Windows For Workgroups 3.11, Windows 95 and Windows NT Workstations

If you wish to install the network key on a workstation, follow the steps below. Again, the requirements for a succesful installation requires both the HASP client(s) and the HASP server must be running the same communication protocol (IPX or NETBIOS).

1. Run nhsrvwin.exe using File Manager or Program Manager
Note: The server software will automatically choose the appropriate protocol. A window will come up and display the selected protocol. If you wish to override the selected protocol, run nhsrvwin.exe with the specified protocol. It is recommended to let the server software select the protocol. If it does not select IPX or NETBIOS, then it is recommended to override the protocol with IPX or NETBIOS.
If you choose to use IPX protocol then run "nhsrvwin.exe -ipx"
If you choose to use NETBIOS protocol then run "nhsrvwin.exe -netbios"

Client Settings - (HASP server is a Workstation)
All clients accessing Molecular Analyst with the network key attached to a workstation must use the same protocol that the server is using. To be assured that each client is using a consistent protocol with the server make sure,
1. The protocol is installed on the client machine and
2.that the environment variable NETHASPPROTOCOL is set to the selected protocol on each client machine. This can be set by adding the following line to your autoexec.bat file:
If the server is using the IPX protocol:
SET NETHASPPROTOCOL=IPX
If the server is using the NETBIOS protocol:
SET NETHASPPROTOCOL=NETBIOS

WinNT Server

If you wish to install the network key on a WINDOWS NT server you will need to install device drivers using the hinstall.exe with the /i option. This will automatically install the device drivers. Once the device drivers are loaded then run the nhsrvwnt.exe server program.
Note: The server software will automatically choose the appropriate protocol. A window will come up and display the selected protocol. If you wish to override the selected protocol, run nhsrvwnt.exe with the specified protocol. It is recommended to let the server software select the protocol. If it does not select IPX or NETBIOS, then it is recommended to override the protocol with IPX or NETBIOS.
If you choose to use IPX protocol then run "nhsrvwnt.exe -ipx"

If you choose to use NETBIOS protocol then run "nhsrvwnt.exe -netbios"

Client Settings - (NT server)
All clients accessing Molecular Analyst with the network key attached to a WINDOWS NT server must use the same protocol that the WINDOWS NT server is using. To be assured that each client is using a consistent protocol with the HASP server make sure,
1. The protocol is installed on the client machine and
2.that the environment variable NETHASPPROTOCOL is set to the selected protocol on each client machine. This can be set by adding the following line to your autoexec.bat file:
If the server is using the IPX protocol:
SET NETHASPPROTOCOL=IPX
If the server is using the NETBIOS protocol:
SET NETHASPPROTOCOL=NETBIOS

Local Networks and Internetworks under NOVELL

This section is relevant to users working under NOVELL networks, consisting of several interconnected networks. If you are not working under such a network, skip to the next chapter.

An internetwork is a series of two or more networks that are linked via internal or external routers. A local network is a single cabling scheme identified by a unique network number, to which one or more stations are attached.

When HASPSERV is loaded, by default it advertises its name to the entire internetwork and serves all of its networks and stations. You can, however, instruct HASPSERV to serve only a specific local network. If the default situation suits you, and you wish HASPSERV to serve the entire internetwork, the following sections are of no relevance to you, and you may skip this section.

The following sections describe how to load the various HASPSERV programs so as to serve a local network only.

Specifying Local Networks with HASPSERV. NLM

If the file server on which HASPSERV is loaded is a ìbridgeî, HASPSERV will serve only those stations which are part of the networks connected directly to this ìbridgeî. In addition to specifying that HASPSERV will serve a local network, you can specify with which network numbers HASPSERV will advertise its name. This is performed by loading HASPSERV.NLM with the NET switch as follows:

LOAD HASPSERV NET m [NET n...]
where m and n are network numbers in the internetwork.

Please note that although HASPSERV advertises its name with specific networks, it will serve all of the sttions in the entirre internetwork. HASPSERV will serve the specified networks only if the switch LOCALNET is used in conjunction with the NET switch.

Examples:

1. HASPSERV NET 111
In this example, HASPSERV advertises its name with network 111 and serves the entire internetwork.

2. HASPSERV NET 111 NET 222 LOCALNET
In this example, HASPSERV advertises its name with networks 111 and 222, and serves these two local
networks only.

Loading HASPSERV with the NET switch will take effect if you set the stationís environment variable, NETHASPPROTOCOL, as follows:

SET NETHASPPROTOCOL = LOCALNET

This ensures that the station will access only those HASPSERVs which advertised their names with the stationís network number. That is, the station will access a HASPSERV.NLM which was loaded with the switch NET n, where n equals the local network number of the station.

Specifying Local Networks with HASPSERV.EXE

To instruct HASPSERV.EXE to serve a local network, load it using the following switch:

HASPSERV -LOCALNET

If HASPSERV.EXE is loaded without using the LOCALNET switch, it will serve all of the networks and stations in the entire internetwork. You can, however, instruct a station to access only the HASPSERV.EXE that advertised its name with the stationís local network. This is done by setting the stationís environment variable, NETHASPPROTOCOL, as follows:

SET NETHASPPROTOCOL = LOCALNET